Notes on

A-Level Physics

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These notes review the H2 Physics (9749) and H3 Physics (9814) syllabus.

Updated October 17, 2026.

Preface

Acknowledgements

Much of the information within this document has been derived from the Senior High Physics Department at Dunman High School. Some pictoral illustrations are also taken from Serway & Jewett's *Physics for Scientists and Engineers* (9th Edition).

Thank you to my family, friends, and teachers for inspiring & supporting the writing of this document!

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This is not an official textbook. All errors are attributable to my own oversight.

No part of this document has been generated by an artificial intelligence model.

Data List

This data list is equivalent to that provided by the Singapore Examinations & Assessments Board for the 9749 syllabus.

Description	**Symbol**	**Value**
speed of light in free space	c	$= 3.00 \times 10^{8}\ \mathrm{m\,s^{-1}}$
permeability of free space	μ_0	$= 4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$
permittivity of free space	ε_0	$= 8.85 \times 10^{-12}\ \mathrm{F\,m^{-1}}$
elementary charge	e	$= 1.60 \times 10^{-19}\ \mathrm{C}$
the Planck constant	h	$= 6.63 \times 10^{-34}\ \mathrm{J\,s}$
unified atomic mass constant	u	$= 1.66 \times 10^{-27}\ \mathrm{kg}$
rest mass of electron	m_e	$= 9.11 \times 10^{-31}\ \mathrm{kg}$
rest mass of proton	m_p	$= 1.67 \times 10^{-27}\ \mathrm{kg}$
molar gas constant	R	$= 8.31\ \mathrm{J\,K^{-1}\,mol^{-1}}$
the Avogadro constant	N_A	$= 6.02 \times 10^{23}\ \mathrm{mol^{-1}}$
the Boltzmann constant	k	$= 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$
gravitational constant	G	$= 6.67 \times 10^{-11}\ \mathrm{N\,m^{2}\,kg^{-2}}$
acceleration of free fall	g	$= 9.81\ \mathrm{m\,s^{-2}}$

Contents

I Measurement 1

1 Measurement 2
1.1 Physical Quantities . . . 2
1.2 Scalars & Vectors . . . 3
1.3 Errors & Uncertainties . . . 3

II Newtonian Mechanics 6

2 Kinematics 7
2.1 Kinematic Quantities . . . 7
2.2 Graphical Methods . . . 8
2.3 Equations of Motion . . . 9
2.4 Free Fall . . . 10
2.5 Projectile Motion . . . 10

3 Dynamics 14
3.1 Linear Momentum . . . 14
3.2 Newton's Laws of Motion . . . 14
3.3 Mass & Inertia . . . 15
3.4 Impulse . . . 16
3.5 Free-Body Diagrams . . . 16
3.6 Conservation of Linear Momentum . . . 17
3.7 Collisions . . . 17

4 Forces 20
4.1 Conservative & Dissipative Forces . . . 20
4.2 Common Mechanical Forces . . . 20
4.3 Elastic Force . . . 21
4.4 Upthrust . . . 23
4.5 Static Equilibrium . . . 24

5 Work, Energy, & Power 27
5.1 Work . . . 27
5.2 Energy . . . 27
5.3 Conservation of Mechanical Energy . . . 30
5.4 Power . . . 30

6 Motion in a Circle 31
6.1 Angular Quantities . . . 31
6.2 Centripetal Acceleration . . . 32
6.3 Centripetal Force . . . 33
6.4 Applications of Uniform Circular Motion . . . 33
6.5 Applications of Non-Uniform Circular Motion . . . 36

7 Gravitational Fields 38
7.1 Gravitational Force . . . 38
7.2 Gravitational Field . . . 38
7.3 Gravitational Potential Energy . . . 39
7.4 Gravitational Potential . . . 40
7.5 Energy Considerations . . . 41
7.6 Kepler's Third Law . . . 42
7.7 Geostationary Orbits . . . 42

III Thermal Physics 44

8 Temperature and Ideal Gases 45
8.1 Zeroth Law of Thermodynamics . . . 45
8.2 Temperature Scales . . . 45
8.3 Ideal Gases . . . 46
8.4 Kinetic Theory of Gases . . . 47

9 First Law of Thermodynamics 51
9.1 Internal Energy . . . 51
9.2 Heat Capacity . . . 52
9.3 First Law of Thermodynamics . . . 53

IV Oscillations and Waves 55

10 Oscillations 56
10.1 Oscillatory Quantities . . . 56
10.2 Free Oscillations . . . 58
10.3 Damped Oscillations . . . 61
10.4 Forced Oscillations . . . 63

11 Wave Motion 64
11.1 Waves . . . 64
11.2 Intensity . . . 66
11.3 Polarisation . . . 67

12 Superposition 69
12.1 Superposition Principle . . . 69
12.2 Stationary Waves . . . 71
12.3 Diffraction . . . 73
12.4 Diffraction Grating . . . 76

V Electricity and Magnetism 77

13 Electric Fields 78
13.1 Electric Charge . . . 78
13.2 Electric Force . . . 78
13.3 Electric Field . . . 79
13.4 Motion of Charged Particles . . . 80

13.5 Electric Potential Energy . . . 80
13.6 Electric Potential . . . 81

14 Current of Electricity **83**
14.1 Electric Current . . . 83
14.2 Voltage . . . 84
14.3 Resistance . . . 84
14.4 Electrical Power . . . 87
14.5 Internal Resistance . . . 88

15 DC Circuits **91**
15.1 Components . . . 91
15.2 Kirchhoff's Laws . . . 91
15.3 Effective Resistance . . . 92
15.4 Potential Divider . . . 92
15.5 Potentiometers . . . 93

16 Electromagnetism **96**
16.1 Magnetic Fields . . . 96
16.2 Source of Magnetic Fields . . . 96
16.3 Magnetic Force . . . 98

17 Electromagnetic Induction **102**
17.1 Magnetic Flux . . . 102
17.2 Faraday's Law . . . 102
17.3 Lenz's Law . . . 103
17.4 Motional Electromotive Force . . . 103
17.5 A.C. Generators . . . 105
17.6 Eddy Currents . . . 105

18 Alternating Current **107**
18.1 AC Sources . . . 107
18.2 Transformers . . . 109
18.3 Diode Rectification . . . 110

VI Modern Physics **112**

19 Quantum Physics **113**
19.1 Photons . . . 113
19.2 Photoelectric Effect . . . 113
19.3 Wave-Particle Duality . . . 118
19.4 Energy Levels in Isolated Atoms . . . 119
19.5 Heisenberg's Uncertainty Principle . . . 124

20 Nuclear Physics **125**
20.1 Rutherford Scattering . . . 125
20.2 Notation . . . 126
20.3 Mass Defect & Binding Energy . . . 128
20.4 Binding Energy Per Nucleon . . . 130
20.5 Nuclear Reactions . . . 131

20.6 Decay Processes . . . 132
20.7 Radioactivity . . . 134

VII H3 Physics 136

21 Inertial Frames 137
21.1 Frames of Reference . . . 137
21.2 Center of Mass . . . 138

22 Rotational Motion 143
22.1 Angular Quantities . . . 143
22.2 Moment of Inertia . . . 146
22.3 Torque . . . 151
22.4 Angular Momentum . . . 154
22.5 Rotational Kinetic Energy . . . 156
22.6 Rolling Motion . . . 158

23 Planetary & Satellite Motion 160
23.1 Elliptical Orbits . . . 160
23.2 Vis-Viva Equation . . . 161
23.3 Kepler's Laws . . . 162
23.4 Newton's Shell Theorem . . . 164
23.5 Effective Radial Potential . . . 165
23.6 Hohmann Orbital Transfer . . . 168

24 Electric & Magnetic Fields 171
24.1 Properties of Electric Charges . . . 171
24.2 Electric Field . . . 172
24.3 Electric Flux . . . 178
24.4 Gauss's Law . . . 179
24.5 Conductors in Electrostatic Equilibrium . . . 184
24.6 Electric Potential . . . 185
24.7 Electric Dipole . . . 187
24.8 Magnetic Fields . . . 189
24.9 Magnetic Dipoles . . . 190

25 Capacitors & Inductors 193
25.1 Capacitors . . . 193
25.2 Dielectrics & Ferromagnetism . . . 194
25.3 Inductance . . . 194
25.4 Linear Analog Filters . . . 196

Part I

Measurement

1 Measurement

1.1 Physical Quantities

When dealing with physics, we utilise physical quantities as a method of understanding the objects and phenomena around us.

Definition 1.1: Quantities

A **physical quantity** is any scientifically measurable quantity (e.g. temperature, force, mass). All physical quantities consist of a **numerical value** and a **unit**.

A **base quantity** is one of the seven physical quantities of the S.I. system by which all other physical quantities are defined.

A **derived quantity** is a physical quantity that can be expressed in terms of products and quotients of base quantities.

The seven base quantities, along with their S.I. units, are shown below.

Base quantity	Base unit	Symbol
Mass	kilogram	kilogram
Length	metre	m
Time	second	s
Electric current	ampere	A
Thermodynamic temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

Base units can be used to check the **dimensional homogeneity** of a physical equation. If the units of each term are equal on both sides of the equation, the equation is said to be **homogenous** or dimensionally constant.

Quantities use prefixes to indicate decimal sub-multiples or multiples.

10^n	Prefix	Symbol	Name
10^{12}	tera	T	trillion
10^{9}	giga	G	billion
10^{6}	mega	M	million
10^{3}	kilo	k	thousand
10^{-1}	deci	d	tenth

10^n	Prefix	Symbol	Name
10^{-2}	centi	c	hundredth
10^{-3}	milli	m	thousandth
10^{-6}	micro	μ	millionth
10^{-9}	nano	n	billionth
10^{-12}	pico	p	trillionth

1.2 Scalars & Vectors

Definition 1.2: Scalars & Vectors

A scalar is a quantity has **magnitude** but **no direction**. It is completely specificed by its numerical value and unit.

A vector is a quantity that has **both** magnitude and direction. It must be specified with its value, unit, and direction.

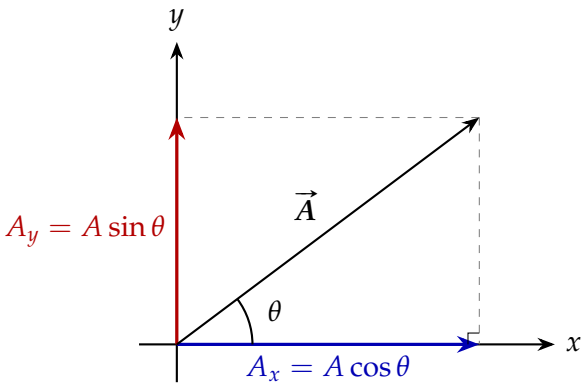
The change in a certain physical quantity $\vec{Q}$ can simply be taken as the subtraction between its final and initial state.

$$\Delta\vec{Q} = \vec{Q}_f - \vec{Q}_i$$

Since two vectors can be added together to give a resultant vector, this means that we can also "decompose" a vector into two constituent vectors.

This is done by **resolving a vector** into two mutually perpendicular and independent components. Often, this gives us the x-component and the y-component of a vector, which will be demonstrably useful in solving later problems.

In the graph below, we see that we have a vector $\vec{A}$. By projecting the length of this vector onto the x and y axes, we can visualise the independent components that make up this vector.



As a result, given a vector $\tilde{\mathbf{A}}$ with magnitude A inclined at an angle θ from the positive x-axis, its corresponding x and y components are given as:

$$A_x = A\cos\theta, \qquad A_y = A\sin\theta$$

The reason this is useful is because within a two-dimensional space, vectors can span an infinite number of directions, making calculations difficult at face value.

By resolving the vectors into its x-components and y-components, we effectively collapse the infinite spectrum of directions into just two perpendicular directions, hence enabling us to proceed forward with the problem.

1.3 Errors & Uncertainties

We can be tempted to take measurements as true, exact values. However, in practicality, the measurements that we take are subject to some order of uncertainty.

In this section, we make clear of the differences between uncertainty, error, precision, and accuracy, and how to deal with calculations under the compounding effects of uncertainties.

Definition 1.3: Uncertainty

Uncertainty is the range of values of a measurement in which the actual value of the measurement is expected to lie within, commonly arising from limitations of the observer, apparatus, or method.

In a scale reading, the **absolute uncertainty** is half of its smallest division, rounded off to 1 significant figure. In general, all readings can be recorded in the form $R \pm \Delta R$, where ΔR is the absolute uncertainty.

The uncertainty of a measured value can also be presented as a percent or a simple fraction. The **fractional uncertainty** of R is $(\Delta R)/R$, whereas its **percentage uncertainty** is $(\Delta R)/R \times 100\%$.

The absolute uncertainty (with units, 1 s.f.) indicates the **scale sensitivity** of the measuring instrument used. The percentage uncertainty (dimensionless, 2 s.f.) is useful to compare the **significance of the error**.

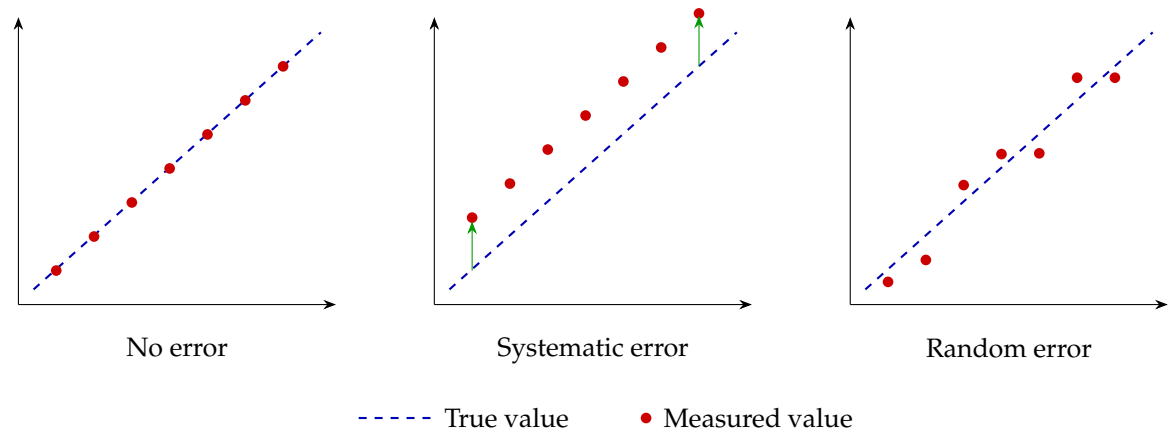
Definition 1.4: Error

Error is the difference between the measured value and the true value.

Systematic errors occur when the measured values produce errors of the same magnitude and sign. They cannot be eliminated by averaging, but can be reduced through careful design of experimental processes.

Random errors occur when the measured values produce errors of different magnitudes and signs. These readings are scattered about the **mean value** with no fixed pattern.

We illustrate the difference between systematic errors and random errors in the set of graphs below.



A purely systematic error will cause **fixed offsets** between the measured and true values, as represented by the green arrows. On the other hand, a purely random error will cause the measured values to exhibit **random deviations** from the true values.

1.3.1 Precision & Accuracy

Moving on, we differentiate between precision and accuracy.

Definition 1.5: Precision

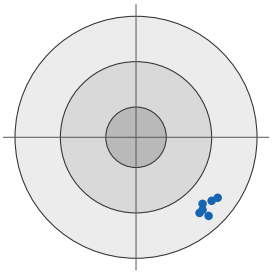
The precision of a measurement refers to how close the experimental values are to each other.

Definition 1.6: Accuracy

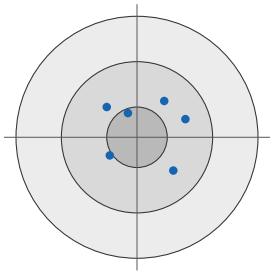
Accuracy is the closeness of a reading on an instrument to the true value of the quantity being measured.

In essence, precision describes the level of uncertainty in an instrument's scale. Good precision indicates small random errors. Good accuracy indicates small systematic errors.

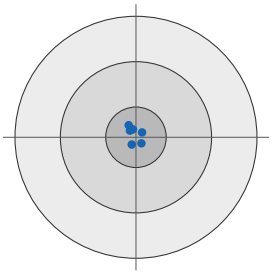
To draw a tangible example, precision and accuracy heavily affects scores in archery. This itself is equivalent to the earlier illustration depicting the plots between measured and true values.



Good precision, poor accuracy



Good accuracy, poor precision



Good accuracy, good precision

1.3.2 Propagation of Uncertainties

Consequential uncertainty is the overall estimate of uncertainty throughout an entire experiment. To statistically calculate consequential uncertainty, we consider **maximum uncertainty**, which occurs in the worst-case scenario.

Let A and B be measured independent quantities, with ΔA and ΔB being their corresponding uncertainties. The following rules will apply for a derived quantity Z.

$$Z = mA + nB \implies \Delta Z = |m|\Delta A + |n|\Delta B$$

$$Z = A^m \times B^n \implies \frac{\Delta Z}{Z} = |m|\frac{\Delta A}{A} + |n|\frac{\Delta B}{B}$$

When adding or substracting measurements, add their absolute uncertainties. When multiplying or dividing measurements, add their fractional uncertainties. If a quantity A is being divided by k, simply regard it as a multiplication of k^{-1}.

For complicated functions, it is acceptable to calculate the maximum uncertainty as the median value between the range of possible quantities.

$$\Delta Z = \frac{Z_{\max} - Z_{\min}}{2}$$

Part II

Newtonian Mechanics

2 Kinematics

In kinematics, we study the geometrical aspects of the motion of physical bodies. It is fundamental to understanding how objects move over time, without first diving too deeply into how forces work.

2.1 Kinematic Quantities

We begin with the quantity of linear length.

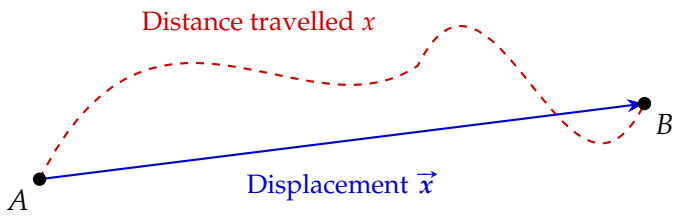
Definition 2.1: Distance

Distance, x, is the total length of path covered by a moving object irrespective of the direction of motion. It is a scalar quantity.

Definition 2.2: Displacement

Displacement, denoted by $\vec{x}$, is the linear distance of the position of a moving object from a given reference point. It is a vector quantity.

Before proceeding, we may want to make clear of the difference between distance and displacement. We will need to carry this distinction between a scalar and vector quantity with us.



In the figure above, a body travels along the dotted path. The total length of the dotted path is equal to the distance travelled x by the body. However, the displacement $\vec{x}$ of the body is defined as a vector linking its initial and final positions.

Let us move on to speed, the rate of change of distance, and its vector counterpart.

Definition 2.3: Speed

Speed, often denoted by u, is the distance travelled per unit time. It is a scalar quantity.

$$u = \frac{\mathrm{d}x}{\mathrm{d}t} \implies \langle u \rangle = \frac{\Delta x}{\Delta t}$$

Instantaneous speed is given by the time-based derivative of distance, whereas average speed is the change in distance over a time interval Δt.

Definition 2.4: Velocity

Velocity, denoted by $\vec{v}$, is the rate of change of displacement with time, acting in the direction of change of displacement. It is a vector quantity.

$$\vec{v} = \frac{\mathrm{d}\vec{x}}{\mathrm{d}t} \implies \langle \vec{v} \rangle = \frac{\Delta \vec{x}}{\Delta t}$$

Instantaneous velocity is given by the time-based derivative of displacement, whereas average speed is the change in displacement over a time interval Δt.

Now, we touch on acceleration, the rate of change of velocity.

Definition 2.5: Acceleration

Acceleration, denoted by $\tilde{\mathbf{a}}$, is the rate of change of velocity with time.

$$\vec{a} = \frac{\mathrm{d}\vec{v}}{\mathrm{d}t} \implies \langle \vec{a} \rangle = \frac{\Delta \vec{v}}{\Delta t}$$

Acceleration is a vector quantity that acts in the direction of the change of velocity. An object is said to be accelerating if its velocity is changing, either in magnitude or direction.

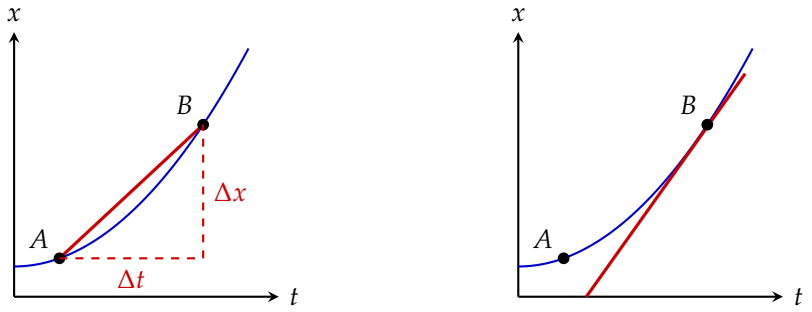
2.2 Graphical Methods

2.2.1 Displacement-Time Graphs

A displacement-time graph tracks the variation of a body's displacement on a time axis. Since $\vec{v} = \mathrm{d}\vec{x}/\mathrm{d}t$, the velocity of a body at any instant can be found using the displacement-time graph.

It is here that we may expand a bit on average and instantaneous quantities.

Suppose we are concerned with the motion of a body at two chronological points, A and B. The two graphs below illustrate how we can obtain the average and instantaneous velocities of the body from these points.



On the left graph, we are drawing a secant line between A and B, allowing us to determine the overall rise and run between these two points. The ratio $\Delta\vec{x}/\Delta t$ gives us the magnitude of the **average** velocity $\langle \vec{v} \rangle$.

On the right graph, we are drawing a tangent line to the graph directly at B. Taking the gradient of the tangent, $\mathrm{d}\vec{x}/\mathrm{d}t$, gives us the **instantaneous** velocity at B.

2.2.2 Velocity-Time Graphs

A velocity-time graph tracks the velocity of an object on a time axis. The method for obtaining the average and instantaneous accelerations is similar to that previously discussed. What is more interesting is that we are able to obtain the displacement of the object from a velocity-time graph.

Consider the following relation.

$$\vec{v} = \frac{\mathrm{d}\vec{x}}{\mathrm{d}t} \implies \vec{x} = \int \vec{v}\,\mathrm{d}t$$

This means that for a body that travels an overall displacement $\Delta\vec{x}$ between times t_1 and t_2, it is equivalent to the area under a velocity-time graph for that time duration.

$$\Delta\vec{x} = \int_{t_1}^{t_2} \vec{v}\,\mathrm{d}t$$

In other words, the displacement is the time integral of velocity $\vec{v}$. Similarly, the average velocity of an object can be found by taking the time integral of a body's acceleration. We will continue to see the use of such graphica methods in further chapters.

2.3 Equations of Motion

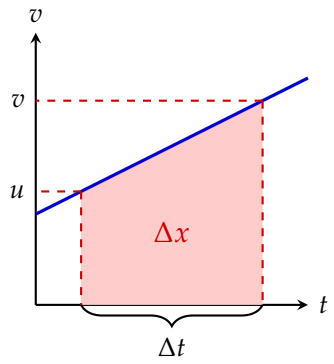
Proposition 2.1: Kinematic Equations of Motion

Consider a particle moving in a straight line. Let u be the initial velocity, v be the final velocity, t be the time taken, and x be the displacement along a straight line. If the particle moves with **constant** acceleration a, the following kinematic equations of motions hold.

1. $v = u + at$,
2. $x = \frac{1}{2}(u+v)t$,
3. $x = ut + \frac{1}{2}at^2$,
4. $x = vt - \frac{1}{2}at^2$,
5. $v^2 = u^2 + 2ax$

Proof. From the definition of acceleration, $v = u + at$ is a trivial result.

For constant acceleration, the velocity-time graph is that of a straight line. This means that displacement, which is given by the area under the velocity-time graph, is simply the area of a trapezium with side lengths u, v, and width t.



Hence, we show that $x = \frac{1}{2}(u+v)t$. The other equivalences can be shown by substituting u

and v from the first equation.

Finally, squaring v gives us the last equation.

$$\begin{aligned} v^2 = (u+at)^2 \implies v^2 &= u^2 + 2u \cdot at + (at)^2 \\ &= u^2 + 2a \cdot \left(ut + \frac{1}{2}at^2 \right) \\ &= u^2 + 2ax \end{aligned}$$

□

Note that while the vector typeface has been omitted to improve readability in the above equations, the quantities should still be interpreted as vectors. As a result, one must be conscious of the appropriate signs when substituting values.

For all rectilinear vector problems, directions are indicated using positive and negative signs. For example, if we define the upwards direction as positive, then anything pointing in the downwards direction must carry a negative sign.

2.4 Free Fall

In this section, we discuss scenarios where an object is released from a height and allowed to fall freely.

For many theoretical problems, we neglect air resistance, which is the drag force that resists an object's falling motion. Nonetheless, we will still need to understand how kinematic motion changes when accounting for air resistance.

Where air resistance is negligible, the only force acting on the ball is gravity. The following equation suffices.

$$v = u + at = gt$$

In reality, air resistance tends to increase approximately with the square of the object's speed at high velocities, and increase proportionally with the object's speed at low velocities.

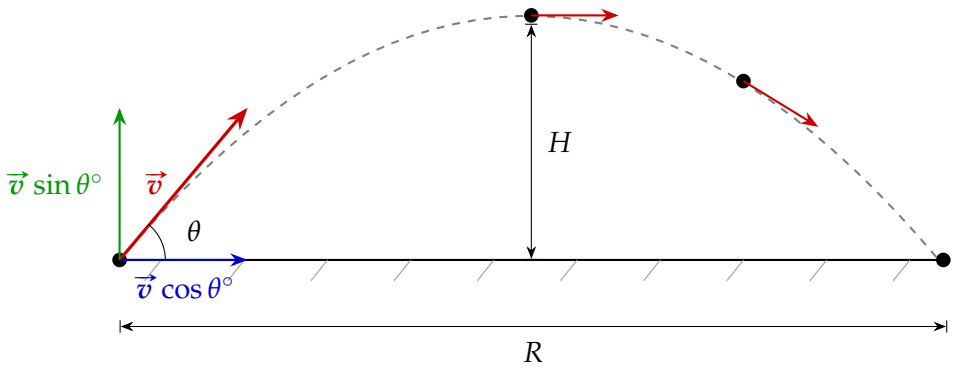
The larger the velocity, the greater the air resistance. The actual acceleration is the combined gravitational acceleration and the slowdown due to air resistance.

For bodies falling from great heights, the air resistance can be so large that it completely counteracts the gravitational force, causing the body to fall at a constant **terminal velocity**.

2.5 Projectile Motion

Thus far, we have immersed ourselves in the mathematics of kinematics. It is now time to apply this knowledge to the motion of projectiles.

We restrict ourselves to discussing objects moving in a two-dimensional plane. Bodies undergoing projectile motion experience uniform velocity in one direction and a uniform acceleration in a perpendicular direction, giving rise to a **parabolic path**. These objects are often launched into the air at an angle θ from the horizontal, at a speed v.



The velocity of the projectile at any time is given by the vector sum of v_x and v_y, the horizontal and vertical components of the object's velocity.

$$v = \sqrt{{v_x}^2 + {v_y}^2}$$

The angle of direction at any point can be ascertained using trigonometry.

$$\theta = \arctan\left(\frac{v_y}{v_x}\right)$$

2.5.1 Horizontal Motion

Horizontally, the body moves with a constant velocity. There is no resultant force, and therefore no horizontal acceleration. This means that the horizontal component of the object's velocity remains the same throughout its motion.

$$\begin{aligned} a_x = 0 &\implies v_x = u_x + a_x t = u_x \\ &\implies s_x + u_x t + \frac{1}{2} a_x t^2 = u_x t \end{aligned}$$

2.5.2 Vertical Motion

Vertically, the body moves under the influence of gravity with a constant acceleration of g, causing the velocity to change with time. For the following, we take upwards as positive.

$$\begin{aligned} a_y = g &\implies v_y = u_y - gt \\ &\implies s_y = u_y t - \frac{1}{2} g t^2 \\ &\implies {v_y}^2 + {u_y}^2 - 2 g s_y \end{aligned}$$

2.5.3 Useful Results

We present a series of formulas that allow us to ascertain the maximum height, total time of flight, and range of a projectile.

Proposition 2.2: Maximum Height

The **maximum height** reached by a body in projectile motion is given by

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

Proof. Note that at the maximum height, $v_y = 0$. Taking the time-independent equation,

$$\begin{aligned} {v_y}^2 = {u_y}^2 + 2gs_y &\implies {v_y}^2 (u \sin\theta)^2 - 2gH \\ &\implies H = \frac{u^2 \sin^2 \theta}{2g} \end{aligned}$$

□

Proposition 2.3: Total Time of Flight

The **total time of flight** of a body in projectile motion is given by

$$T = \frac{2u \sin \theta}{g}$$

Proof. To determine the time taking to reach the maximum height,

$$\begin{aligned} v_y = u_y + a_y t &\implies u \sin\theta - g t_{\text{up}} \\ &\implies \boxed{t_{\text{up}} = \frac{u \sin \theta}{g}} \end{aligned}$$

As the trajectory is a parabola, the time taken to travel upwards is equal to the time taken to travel downwards. Therefore,

$$\begin{aligned} t_{up} = t_{down} &\implies T = 2t_{\text{up}} \\ &\implies \boxed{T = \frac{2u \sin \theta}{g}} \end{aligned}$$

□

Proposition 2.4: Range

The **total horizontal distance** covered by the projectile, known as the **horizontal range** R, is given by

$$R = \frac{u^2 \sin 2\theta}{g}$$

Proof. Considering the vertical distance travelled over the total time of flight,

$$R = v_x T = (u\cos\theta)\frac{2u\sin\theta}{g}$$

$$\implies R = \frac{u^2 \sin 2\theta}{g}$$

□

Proposition 2.5: Maximum Range

The **maximum range** occurs when the projection angle is $\theta = 45°$, and is given by

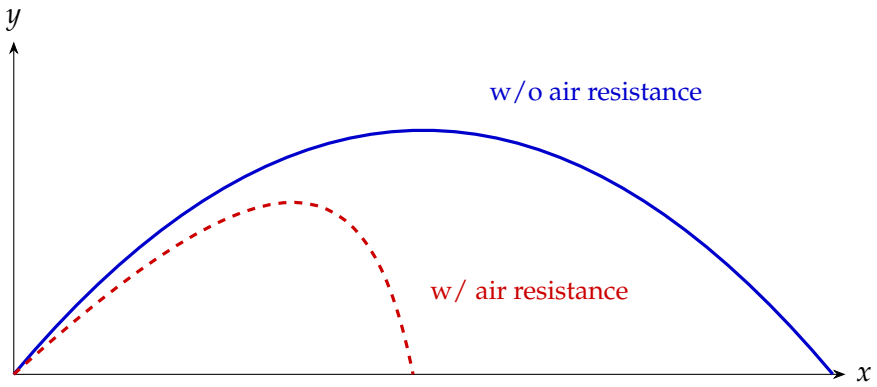
$$R_{\max} = \frac{u^2}{g}$$

Proof. R is maximum when $\sin 2\theta = 1 \implies \theta = 45°$. □

2.5.4 Projectile Motion with Air Resistance

Air resistance opposes both the horizontal and vertical components of the projectile's motion.

This causes its horizontal velocity to decrease to zero. During the object's upwards journey, the vertical velocity decreases at a decreasing rate. During its downwards journey, the vertical velocity increases at a decreasing rate.



This results in the projectile hitting the ground much faster, as compared to the scenario without air resistance.

3 Dynamics

> All soul is immortal; for that which is always in movement is immortal.
>
> — PLATO, *Phaedrus*

3.1 Linear Momentum

Definition 3.1: Linear Momentum

Linear momentum of a body is the product of its mass and velocity.

$$\vec{p} = m\vec{v}$$

where $\vec{p}$ is the linear momentum, m is the mass, and $\vec{v}$ is the velocity of the body.

Momentum is a vector quantity acting in the same direction as velocity. The S.I. unit of momentum is the newton second (N s).

3.2 Newton's Laws of Motion

Definition 3.2: Newton's First Law

Newton's first law of motion states that a body continues in its state of rest or uniform motion in a straight line, unless a resultant external force acts on it.

This law implies that

1. the **state of rest** requires no resultant force to maintain;
2. the **state of uniform velocity** also requires no resultant force to maintain.

Definition 3.3: Newton's Second Law

Newton's second law of motion states that the rate of change of momentum of a body is proportional to the resultant force acting on it, and occurs in the direction of the resultant force.

From the above definition, we see that

$$\sum \vec{F} \propto \frac{\mathrm{d}\vec{p}}{\mathrm{d}t} \implies \sum \vec{F} = k\frac{\mathrm{d}\vec{p}}{\mathrm{d}t}$$

where k is the constant of proportionality.

The S.I. unit force is the newton (N). One newton is defined as the force which produces an

acceleration of $1\,\mathrm{m\,s^{-2}}$ when applied to a mass of 1kg, thus conveniently setting $k = 1$. This gives

$$\sum \vec{F} = \frac{\mathrm{d}\vec{p}}{\mathrm{d}t} = \frac{\mathrm{d}(m\vec{v})}{\mathrm{d}t}$$
$$= m\frac{\mathrm{d}\vec{v}}{\mathrm{d}t} + \vec{v}\frac{\mathrm{d}m}{\mathrm{d}t}$$

When **mass** is constant,

$$\sum \vec{F} = m\frac{\mathrm{d}\vec{v}}{\mathrm{d}t} = m\vec{a}$$

When **velocity** is constant,

$$\sum \vec{F} = \vec{v}\frac{\mathrm{d}m}{\mathrm{d}t}$$

Hence, depending on the context, force can be represented in various ways.

Definition 3.4: Newton's Third Law

Newton's third law of motion states that if a body A exerts a force on a body B, then body B exerts an equal but opposite force on body A.

$$\vec{F}_{AB} = -\vec{F}_{BA}$$

where $\vec{F}_{AB}$ is the force that A exerts on B, and $\vec{F}_{BA}$ is the force that B exerts on A.

3.3 Mass & Inertia

Though we are already familiar with the concept of mass, we should analyse how it directly affects motion.

Definition 3.5: Mass

Mass is the amount of matter in a body. It is a scalar quantity with S.I. unit kg.

Consequently, mass is a measure of the inertia of a body, which is a body's reluctance to start or stop moving.

Going back to Newton's second law, we see that for a constant mass and constant net force acting on an object, its acceleration is inversely proportional to its mass.

$$\sum \vec{F} = m\vec{a} \implies m \propto (\vec{a})^{-1}$$

We can hence conclude that the larger the mass of a body, the larger its inertia.

3.4 Impulse

Definition 3.6: Impulse

The **impulse of a force** is defined as the time integral of a force during which the force is acting on an object.

$$\vec{J} = \int_{t_i}^{t_f} \sum \vec{F} \, \mathrm{d}t$$

Definition 3.7: Impulse-Momentum Theorem

The **impulse-momentum theorem** states that the impulse of a force acting on an object is equal to the **change** in the momentum of the object.

$$\int_{t_i}^{t_f} \sum \vec{F} \, \mathrm{d}t = \Delta \vec{p} = \vec{p}_f - \vec{p}_i$$

Proof. From Newton's Second Law,

$$\begin{aligned} \vec{F} = \frac{\mathrm{d}\vec{p}}{\mathrm{d}t} &\implies \vec{F} \, \mathrm{d}t = \mathrm{d}\vec{p} \\ &\implies \int_{t_i}^{t_f} \vec{F} \, \mathrm{d}t = \int_{\vec{p}_i}^{\vec{p}_f} \mathrm{d}\vec{p} = \vec{p}_f - \vec{p}_i = \Delta \vec{p} \end{aligned}$$

□

The extent to which the momentum of a body is changed by a given net force depends on the magnitude of the net force, and the duration for which the force acts on the body.

3.5 Free-Body Diagrams

When considering problems, it is helpful to draw a **free-body diagram** with the following procedure:

1. Isolate the system in question.
2. Identify all external forces acting on the system and draw them as vectors as appropriate points on the system.

Then, follow the general procedure for solving.

1. Consider **various systems** and their free-body diagrams. To choose systems, examine the forces that you need to solve. If they do not include internal forces (e.g. friction, normal forces between surfaces), consider **paired objects** as a **whole system**. Otherwise, segregate the objects.
2. Write the $\sum \vec{F} = m\vec{a}$ equations for all considered systems.
3. If necessary, identify certain relationships between variables to formulate additional equations.

3.6 Conservation of Linear Momentum

Definition 3.8: Principle of Conservation of Linear Momentum

The principle of conservation of linear momentum states that the total momentum of a system **remains constant**, provided that **no** net external resultant force acts on the system.

Proof. This is a direct consequence of Newton's laws of motion.

Consider a mass m_1 travelling with initial velocity $\vec{u}_1$ that collides with another mass m_2 travelling with initial velocity $\vec{u}_2$. During the collision, the two masses are in contact and exert an average force $\langle \vec{F} \rangle$ on each other. Let the duration of impact be Δt.

By Newton's 2nd law, the average force exerted on each object is equal to its change in momentum.

$$\langle \vec{F}_{1\to 2} \rangle = \frac{\Delta \vec{p}_2}{\Delta t} = \frac{m_2 \vec{v}_2 - m_2 \vec{u}_2}{\Delta t}$$
$$\langle \vec{F}_{2\to 1} \rangle = \frac{\Delta \vec{p}_1}{\Delta t} = \frac{m_1 \vec{v}_1 - m_1 \vec{u}_1}{\Delta t}$$

By Newton's 3rd law, the forces are equal and opposite.

$$\begin{aligned} \langle \vec{F}_{1\to 2} \rangle = -\langle \vec{F}_{2\to 1} \rangle &\implies \frac{m_2 \vec{v}_2 - m_2 \vec{u}_2}{\Delta t} = -\frac{m_1 \vec{v}_1 - m_1 \vec{u}_1}{\Delta t} \\ &\implies m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2 \\ &\implies \vec{p}_i = \vec{p}_f \end{aligned}$$

Therefore, total linear momentum is conserved. □

3.7 Collisions

With the previous principle in mind, we may examine two main types of one-dimensional collisions between bodies, namely **elastic collisions** and **inelastic collisions**.

In elastic collisions, the total kinetic energy of the system is conserved throughout. Contrariwise, inelastic collisions, total kinetic energy is not conserved.

It is important to be conscious about the direction of the velocities. The velocities should not be substituted with the magnitudes only, but the sign of said quantities ought to be accurate too.

3.7.1 Elastic Collisions

Proposition 3.1: Relative Velocities in Elastic Collision

For a two-body elastic collisions, the **relative velocity of approach is equal to the relative velocity of separation**, such that

$$\vec{u}_1 - \vec{u}_2 = \vec{v}_2 - \vec{v}_1$$

where u_1 and u_2 are the initial velocities of the bodies, and v_1 and v_2 are the final velocities of the bodies.

Proof. Consider two bodies of masses m_1 and m_2, colliding elastically with each other. Their initial velocities are $\vec{u}_1$ and $\vec{u}_2$, with final velocities $\vec{v}_1$ and $\vec{v}_2$ respectively.

Since kinetic energy is conserved for elastic collisions,

$$\begin{aligned}\Delta K = 0 \implies & \frac{1}{2}m_1{u_1}^2 + \frac{1}{2}m_2{u_2}^2 = \frac{1}{2}m_1{v_1}^2 + \frac{1}{2}m_2{v_2}^2 \\ \implies & m_1(\vec{u}_1 + \vec{v}_1)\cdot(\vec{u}_1 - \vec{v}_1) = m_2(\vec{u}_2 + \vec{v}_2)\cdot(\vec{u}_2 - \vec{v}_2)\end{aligned}$$

From the conservation of linear momentum,

$$m_1\vec{u}_1 + m_2\vec{u}_2 = m_1\vec{v}_1 + m_2\vec{v}_2 \implies m_1(\vec{u}_1 - \vec{v}_1) = m_2(\vec{u}_2 - \vec{v}_2)$$

Hence, we may remove the equivalent term on both sides of our equation.

$$\begin{aligned}m_1(\vec{u}_1 + \vec{v}_1)(\vec{u}_1 - \vec{v}_1) = m_2(\vec{u}_2 + \vec{v}_2)(\vec{u}_2 - \vec{v}_2) \implies & \vec{u}_1 + \vec{v}_1 = \vec{u}_2 + \vec{v}_2 \\ \implies & \vec{u}_1 - \vec{u}_2 = \vec{v}_2 - \vec{v}_1\end{aligned}$$

□

Consequently, we may derive the closed-form expressions for the final velocities for each body.

$$\vec{v}_1 = \frac{m_1 - m_2}{m_1 + m_2}\vec{u}_1 + \frac{2m_2}{m_1 + m_2}\vec{u}_2 \qquad \vec{v}_2 = \frac{2m_1}{m_1 + m_2}\vec{u}_1 + \frac{m_2 - m_1}{m_1 + m_2}\vec{u}_2$$

From these results, we may highlight expected behaviour in three special, notable situations.

If **two identical masses** collide, they exchange velocities after the collision.

$$\vec{v}_1 = \vec{u}_2 \quad \text{and} \quad \vec{v}_2 = \vec{u}_1$$

If an **extremely big** mass, m_1, collides with a **very small mass at rest**, m_2 (i.e. $m_1 >> m_2$ and $v_2 = 0$), the big mass continues with a speed close to its initial speed, while the small mass bounces off with a speed about twice the initial speed of m_1.

$$\vec{v}_1 \approx \vec{u}_1 \quad \text{and} \quad \vec{v}_2 \approx 2\vec{u}_1$$

If a **very small** mass, m_1, collides with a **very big mass at rest** (i.e. $m_2 >> m_1$ and $v_2 = 0$), the small mass rebounds with a speed close to its initial speed, and the big mass remains almost stationary.

$$\vec{v}_1 \approx -\vec{u}_1 \quad \text{and} \quad \vec{v}_2 \approx 0$$

3.7.2 Inelastic Collisions

In realistic collisions, change in kinetic energy take place although the momentum of the system is still conserved. This loss in kinetic energy is due to the increase in internal energy and heat dissipated to the surroundings.

For **inelastic collisions**, total momentum is conserved.

$$m_1 \vec{u}_1 + m_2 \vec{u}_2 = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

However, total kinetic energy is not conserved.

$$\frac{1}{2} m_1 \vec{u}_1{}^2 + \frac{1}{2} m_2 \vec{u}_2{}^2 \neq \frac{1}{2} m_1 \vec{v}_1{}^2 + \frac{1}{2} m_2 \vec{v}_2{}^2$$

For **perfectly inelastic collisions**, the two bodies will coalesce post-collision, moving off together with a common velocity.

$$\vec{v}_1 = \vec{v}_2$$

4 Forces

4.1 Conservative & Dissipative Forces

A force is defined as the rate of change of momentum of a free-moving object. The direction of the force is in the direction of the change of momentum.

A force is conservative if the work it does on an object moving between two points is **independent** of the path the object takes.

Conversely, a force is dissipative if the work it does on an object depends on the path taken by the object between its endpoints.

For instance, gravity is a conservative force, while friction is a dissipative force.

4.2 Common Mechanical Forces

4.2.1 Weight

Definition 4.1: Weight

Weight is defined as the force acting on a body due to gravity. For any given non-massless body,

$$\vec{W} = m\vec{g}$$

where $\vec{W}$ is the weight of the body, m is the mass of the body, and $\vec{g}$ is the acceleration due to gravity.

The mass of an object is constant irrespective of its position, but its weight is a **force** with a magnitude dependent on the value of g.

An object's weight on the Moon is a fraction of that on Earth, due to the Moon's weaker force of gravity. Nonetheless, it retains its inertia, as the mass is the same. Note that this is a simplification.

4.2.2 Normal Contact Force

The normal force is the **contact** force that acts perpendicular to the surface between two objects. When we are standing on the ground, the normal force on us due to the ground opposes the gravitational force and prevents us from accelerating to the Earth's core.

4.2.3 Friction

Definition 4.2: Frictional Force

Frictional force is the force exerted by one body on another body when two bodies slide over one another.

It is caused by irregularities in the surfaces in mutual contact and depends on the surfaces in contact as well as how much they are pressed against each other. It is the **component of contact force** along surface of contact.

Between any two surfaces in contact, there will always be a normal force (perpendicular to the interface) and a frictional force (parallel to the interface).



Static friction
(object stationary)

Kinetic friction
(object moving)

Static friction, $\vec{\mathbf{f}}_s$, is always directed to oppose the **relative impending motion**. For small applied forces, the magnitude of the static frictional force is larger than the applied force, causing the object to remain stationary.

$$f_s \leq \mu_s n$$

where μ_s is the coefficient of the static frictional force.

Kinetic friction, $\tilde{\mathbf{f}}_k$, is always directed to oppose the **relative motion**. When the magnitude of applied force exceeds that of static friction, the object accelerates with kinetic friction opposing the motion.

$$f_k = \mu_k n$$

where μ_k is the coefficient of the kinetic frictional force.

4.2.4 Viscous Drag

Definition 4.3: Drag

Viscous drag is the frictional force experienced either by an object as it moves through a fluid or by a fluid as it moves over a surface.

The magnitude of drag is proportional to the speed of the object.

4.2.5 Tension

Tension is the force exerted by a rope when pulled on. Every piece of the rope feels a tension force in both directions, except for the endpoints, which feel a tension on one side and a separate force by the object on the other.

Note that in most cases, ropes are assumed to be **massless**, hence the **tension must be the same at all points**. Otherwise, this would yield a net force, consequently resulting in infinite acceleration.

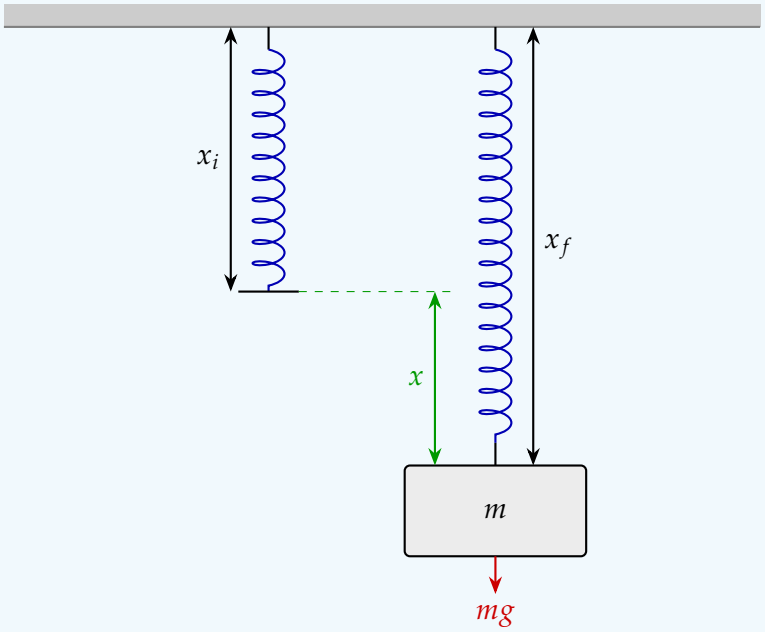
4.3 Elastic Force

The force caused by an elastic object, whether tensive or compressive, follows Hooke's law given that it has not been deformed.

Definition 4.4: Hooke's Law

Hooke's law states that the change in length x of a material is directly proportional to the resultant force F applied, provided the limit of proportionality has not been exceeded.

For the given diagram, where a spring of unstretched length x_i is stretched to a length x_f,



$$F = kx = k(x_f - x_i)$$

where

- F is the force applied to the material
- k is the proportionality constant
- x_i is the unstretched length of material
- x_f is the final length of material
- x is the extension of the material, where $x = x_f - x_i$

Assuming that a material obeys Hooke's law, the force applied is **linearly proportional** to the extension of the material. This law applies to springs, as well as metals in the form of wires.

For springs connected in series,

$$\frac{1}{k_{\text{eq}}} = \frac{1}{k_1} + \frac{1}{k_2} + \cdots = \sum_i \frac{1}{k_i}$$

For springs connected in parallel,

$$k_{\text{eq}} = k_1 + k_2 + \cdots = \sum_i k_i$$

4.4 Upthrust

Definition 4.5: Upthrust

Upthrust is the **net upward force** exerted by the surrounding fluid when a body is **submerged fully or partially** in a fluid.

It is the resultant force due to the difference in pressure exerted by the fluid at the top and bottom surfaces of the body. It is equal in magnitude to the weight of fluid displaced by the body.

Definition 4.6: Archimedes's Principle

Archimedes's Principle states that for any object immersed partially or fully in a fluid, the upthrust is equal in magnitude and opposite in direction to the weight of fluid displaced by the object.

$$U = W_{\text{fluid displaced}} = m_f g = \rho_f g V_{\text{fluid displaced}}$$

Proof. Consider a solid cylinder of height h and cross-sectional area A, submerged in a fluid of density ρ.

Recall that pressure is force over area.

$$p = \frac{F}{A}$$

Hence, the pressure of the top surface is given by

$$\begin{aligned} p_{\text{top}} = \frac{\text{weight of fluid}}{\text{cross-sectional area}} + p_{\text{atm}} &= \frac{m_{\text{top}} g}{A} + p_{\text{atm}} \\ &= \frac{\rho_f A h_1 g}{A} + p_{\text{atm}} \\ &= \rho_f h_1 g + p_{\text{atm}} \end{aligned}$$

Similarly, the pressure on the bottom surface is given by

$$p_{\text{bottom}} = \rho_f h_2 g + p_{\text{atm}}$$

Hence, the **upthrust** (resultant upward force) on the cylinder due to pressure difference between the top and bottom surfaces is

$$\begin{aligned} U = F_2 - F_1 &= \rho_f g (h_2 - h_1) A \\ &= \rho_f g h A \\ &= \rho_f g V_{\text{fluid displaced}} \\ &= m_f g \end{aligned}$$

This gives us Archimedes's principle. □

Definition 4.7: Principle of Floatation

The principle of floatation states that when an object is **floating in equilibrium** in a fluid, the **weight** of the object is equal to the weight of **fluid displaced** by the object.

$$W_{\text{object}} = W_{\text{fluid displaced}}$$

An object floats because the **upthrust** acting on it is equal in magnitude and opposite in direction to the weight of the object. The object sinks when the upthrust acting on it is less than its weight.

A ship made of steel can float because its internal hollow volume displaces a large amount of water, producing sufficient upthrust to keep the ship floating.

4.5 Static Equilibrium

Definition 4.8: Equilibrium

When a body is in equilibrium, there is **no resultant force** acting on the body in any direction and **no resultant moment** acting on the body about any point.

Paraphrasing, the conditions for any body in equilibrium are that

1. the resultant force on it must be zero in any direction;
2. and the resultant torque on it must be zero about any axis of rotation.

4.5.1 Center of Gravity

Definition 4.9: Center of Gravity

The center of gravity of a body is the point at which the whole weight of the body seems to act upon.

Every particle of mass is attracted towards the centre of the earth by force of gravity.

Instead of considering weight as being applied upon the multiplicity of particles within the body, the weight may instead be taken to be **acting on a single point**, whic is at its center of gravity.

The center of gravity of a homogeneous, symmetric body must lie on its axis of symmetry.

4.5.2 Translational Equilibrium

When the resultant force on a body is zero in any direction, there is no acceleration of its centre of mass and the body is said to be in translational equilibrium.

If the forces can be resolved into components in two chosen perpendicular directions (such as the x and y axes), then

$$\sum F_x = 0 \text{ and } \sum F_y = 0$$

Similarly, since the resultant force acting on a body is zero, the **vector sum** of forces acting on the body must be zero in any direction as well. The vector diagram showing the addition of all forces acting on the body will be a closed polygon.

4.5.3 Rotational Equilibrium

When the resultant torque of a body is zero in any axis of rotation, there is no angular acceleration of the object and the body is said to be in rotational equilibrium. We introduce the notion of moments of a force, as well as the torque of a couple to formalise rotational equilibrium.

Definition 4.10: Moment of a Force

The moment of a force about a point τ is the product of the magnitude of the force F and the perpendicular distance $r_\perp$ of the force from that point.

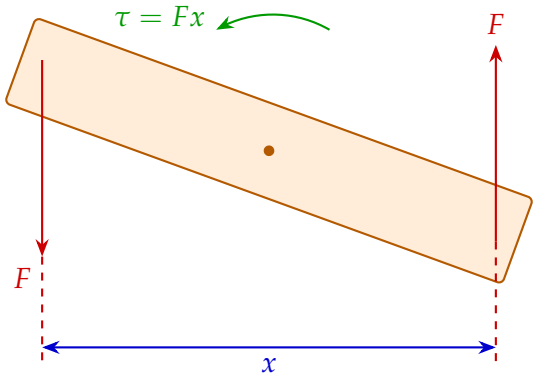
$$\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{F}} \times \vec{\boldsymbol{r}} \implies \tau = F(r \sin\theta)$$

Definition 4.11: Couple

A couple consists of a pair of forces of **equal magnitude** but acting in **opposite directions** whose lines of action are **parallel but separate**.

Definition 4.12: Torque of a Couple

The torque of a couple is the **product** of one of the forces and the perpendicular distance between the forces.



As shown from the figure above, the magnitude of the torque of the couple, τ, is given by

$$\tau = \frac{1}{2}Fx + \frac{1}{2}Fx = Fx$$

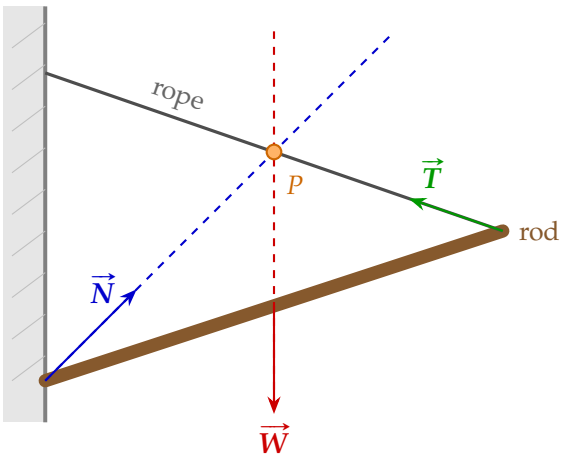
Definition 4.13: Principle of Moments

The principle of moments states that when a system is in equilibrium, the sum of clockwise moments about any axis must be equal to the sum of anticlockwise moments about the same axis.

When the resultant torque of a body is zero about any axis of rotation, it further implies that the **sum of moments of all forces** acting on the body would be zero about any axis.

Hence, the principle of moments may be used to deal with bodies in rotational equilibrium.

For systems in equilibrium, the lines of action of the non-parallel forces **pass through a common point**.



Lines of action of $\vec{N}$, $\vec{W}$ and $\vec{T}$ all pass through point P.

This is evident by considering the translational and rotational conditions for static equilibrium, and can be useful for identifying unknown forces in complicated setups.

5 Work, Energy, & Power

> All the rivers run into the sea; yet the sea is not full; unto the place from whence the rivers come, thither they return again.
>
> — KING SOLOMON, *Ecclesiastes 1:17*

5.1 Work

Definition 5.1: Work

Work, W, on an object by a **constant force** $\vec{F}$ is the product of the force and displacement $\vec{s}$ in the direction of the force.

$$W = \vec{F} \cdot \vec{s} = (F \cos\theta)s$$

where F is the magnitude of force, s is the magnitude of displacement, and θ is the angle between the force and the displacement vectors.

Work is a scalar quantity with S.I. unit joule (J), representing the transfer of energy to a system.

For several forces acting on an object, the **net work done** is the sum of the work done by each force separately, or the dot product between the net force and the displacement.

$$\begin{aligned}\sum W &= W_1 + W_2 + \cdots + W_n \\ &= \sum \vec{F} \cdot \vec{s} = \left(\sum F \cos\theta\right) s\end{aligned}$$

5.1.1 Work Done by Variable Forces

For a body displaced by a variable force, the net work done is the **integral of the force** with respect to displacement s.

$$\sum W = \int_{s_i}^{s_f} F \, \mathrm{d}s$$

In other words, the total work done on the body is equal to the **area under the force-displacement graph**.

5.2 Energy

Definition 5.2: Energy

Energy is the capacity to **do work**.

Energy is a scalar quantity with S.I. unit joule (J).

When work is done on a system, energy is transferred to the system. Likewise, when work is done by a system, energy is transferred away from the system.

Definition 5.3: Principle of Conservation of Energy

The principle of the conservation of energy states that energy can neither be destroyed nor created. It can be transformed from one form to another and transferred from one body to another.

5.2.1 Kinetic Energy

Definition 5.4: Kinetic Energy

Kinetic energy K is the energy that a body possesses by virtue of its motion.

$$K = \frac{1}{2}mv^2$$

where K is the kinetic energy of a body of mass m moving with speed v.

Definition 5.5: Work-Energy Theorem

The work-energy theorem states that the work done by a resultant external force on a body is equal to the change in the kinetic energy of the body.

$$\sum W = \Delta K$$

Proof. Consider an object of mass m moving through a displacement to the right under a net force $\sum \vec{F}$, also directed to the right.

The net work done is

$$\sum W = \int_{x_i}^{x_f} \sum F \, \mathrm{d}x$$

Using Newton's 2nd law, we substitute for the magnitude of the net force $\sum F = ma$, where m is constant,

$$\sum W = \int_{x_i}^{x_f} ma \, \mathrm{d}x = \int_{x_i}^{x_f} m\frac{\mathrm{d}v}{\mathrm{d}t} \, \mathrm{d}x = \int_{v_i}^{v_f} m\frac{\mathrm{d}x}{\mathrm{d}t} \, \mathrm{d}v = \int_{v_i}^{v_f} mv \, \mathrm{d}v$$

$$= \frac{1}{2}m{v_f}^2 - \frac{1}{2}m{v_i}^2$$

The common quantity $\frac{1}{2}mv^2$ is the kinetic energy, K. As such, we can rewrite the equation as

$$\sum W = K_f - K_i = \Delta K$$

Hence, the net work done is equal to the change in kinetic energy, provided that the system only undergoes translational motion, where speed is the only change. □

5.2.2 Potential Energy

Definition 5.6: Potential Energy

Potential energy U is the ability to do work as a result of the position, shape, or state of an object.

For a **conservative force** F, it is related to the potential energy U such that

$$F = -\frac{\mathrm{d}U}{\mathrm{d}x}$$

5.2.3 Gravitational Potential Energy

Definition 5.7: Gravitational Potential Energy

The gravitational potential energy of an object is its capacity to do work as a result of its position in a gravitaitonal field, and is given by

$$U_g = mgh$$

Proof. Consider an object elevated through a height h, at a constant speed, by a constant force F equal and opposite to the weight of the object.

Since there is no acceleration, the force F must be equal to the weight mg. The gravitational potential energy is thus

$$U_g = F \cdot s = mgh$$

□

5.2.4 Elastic Potential Energy

Definition 5.8: Elastic Potential Energy

The elastic potential energy of an elastic object is its capacity to do work because of a change of shape in the object, and is given by

$$U = \frac{1}{2}kx^2 = \frac{1}{2}Fx = \frac{F^2}{2k}$$

Proof. With reference to Hooke's law, an external force F acting on a wire and causing it to extend performs work given by

$$W = \int F \,\mathrm{d}x = \int kx \,\mathrm{d}x = \frac{1}{2}kx^2$$

This work is stored as elastic potential energy in the wire. □

5.3 Conservation of Mechanical Energy

Definition 5.9: Mechanical Energy

Mechanical energy of a system is the sum of the kinetic energy and potential energy in the system.

Definition 5.10: Principle of Conservation of Mechanical Energy

The principle of conservation of mechanical energy states that in the absence of any net external force, the total mechanical energy of the system is conserved.

$$\Delta K + \Delta U = 0$$

This principle illustrates that while the kinetic and potential energies of a system may be converted or transformed into one another, the **sum of the energies is always conserved**.

However, when moving bodies experience dissipative forces, the total mechanical energy is **not conserved**. The work done by dissipative forces is equal to the loss in mechanical energy, that is,

$$K_i + P_i + W_d = K_f + P_f$$

The work done by the dissipative forces W_d is negative, as the forces always act in opposite direction to the displacement.

5.4 Power

Definition 5.11: Power

Power is defined as work done per unit time, given as

$$P = \frac{\mathrm{d}W}{\mathrm{d}t} = \frac{\mathrm{d}(\vec{F} \cdot \vec{s})}{\mathrm{d}t} = \vec{F} \cdot \left(\frac{\mathrm{d}\vec{s}}{\mathrm{d}t}\right) = \vec{F} \cdot \vec{v}$$

for a constant force $\vec{F}$ acting on an object moving at a velocity $\vec{v}$.

Power is a scalar quantity with S.I. unit watt. The average power $\langle P \rangle$ is given by

$$\langle P \rangle = \frac{\Delta W}{\Delta t}$$

5.4.1 Efficiency

Definition 5.12: Efficiency

The efficiency of a machine in its conversion of energy from one form to another is given as

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy input}} \times 100\%$$

Due to inevitable work done by dissipative forces, the energy conversion can never be 100%.

6 Motion in a Circle

Definition 6.1: Radian

A radian is the angle subtended at the centre of a circle by an arc of a length equal to the radius. One radian is equal to $180^\circ / \pi$.

Consider a section of a circle. Let the radius of the circle be r, the arc length of the section be s, and the angle spanned by the section be θ.

The angle θ (in radians) is then the ratio of the distance s along the circular arc subtended by θ divided by the radius r.

$$\theta = \frac{s}{r} \implies s = r\theta$$

6.1 Angular Quantities

Definition 6.2: Angular Displacement

The angular displacement of a body, denoted θ, is equal to the angle through which the body has travelled in a circular arc.

Angular displacement has S.I. unit radian.

Definition 6.3: Angular Velocity

The angular velocity of a body, denoted $\vec{\omega}$, is the rate of change of its angular displacement with respect to time.

$$\vec{\omega} = \frac{\mathrm{d}\vec{\theta}}{\mathrm{d}t} \implies \langle\vec{\omega}\rangle = \frac{\Delta\vec{\theta}}{\Delta t}$$

For a body moving with constant angular velocity, it is said to be undergoing uniform circular motion.

If a body undergiong uniform circular motion takes T seconds to complete a full revolution (e.g. it has a revolutionary period of T seconds), we obtain the following formula,

$$\omega = \frac{\theta}{t} = \frac{2\pi}{T} = 2\pi f$$

where f represents the frequency of the circular motion.

Proposition 6.1: Linear vs. Angular Velocity

The lienar and angular velocities of an object moving in an arc of fixed radius r are related by the following equation.

$$\vec{v} = r\vec{\omega}$$

Proof. From $s = r\theta$, the linear speed of an object can be determined by considering the dis-

placement derivative.

$$\frac{\mathrm{d}s}{\mathrm{d}t} = \frac{\mathrm{d}(r\theta)}{\mathrm{d}t} = r\frac{\mathrm{d}\theta}{\mathrm{d}t} + \theta\frac{\mathrm{d}r}{\mathrm{d}t}$$

If r is a constant, then

$$\frac{\mathrm{d}s}{\mathrm{d}t} = r\frac{\mathrm{d}\theta}{\mathrm{d}t}$$

Converting these derivatives to their representative quantities, we obtain an equation relating linear speed and angular velocity.

$$v = r\omega$$

□

The direction of the linear velocity is tangential to the circular path.

For an object rotating about an axis, every point on the object has the same angular velocity. The tangential speed of any point is proportional to its distance from the axis of rotation.

6.2 Centripetal Acceleration

Definition 6.4: Centripetal Acceleration

For a body in uniform circular motion, its acceleration a_c is termed centripetal acceleration and is always perpendicular to the velocity of the object and points towards the centre of a circle.

$$a_c = \frac{v^2}{r} = v\omega = r\omega^2$$

Proof. Consider a body moving in a circular path of radius r with constant angular velocity ω. As it moves within an infinitesimally brief time interval dt, the magnitude of its linear velocity does not change, and the body moves an infinitesimal distance of arc length $r\,d\theta$.

Hence, the instantaneous entripetal acceleration of the body during this time interval is the derivative of linear velocity with respect to time, given by

$$a_c = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{\mathrm{d}(r\omega)}{\mathrm{d}t} = \omega\frac{\mathrm{d}r}{\mathrm{d}t} + r\frac{\mathrm{d}\omega}{\mathrm{d}t}$$

Since $\mathrm{d}\omega / \mathrm{d}t$ is equal to zero, we obtain the following relations.

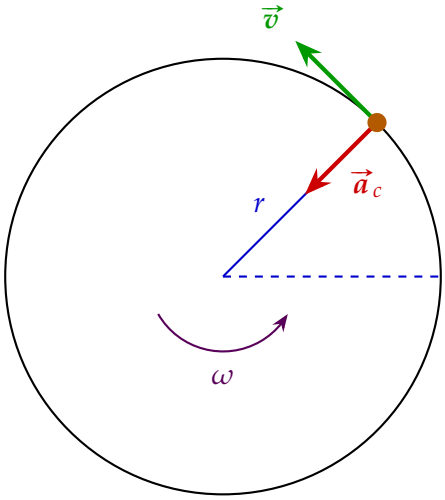
$$a_c = v\omega = r\omega^2 = \frac{v^2}{r}$$

□

Using trigonometric principles, it is possible to show that the direction of centripetal acceleration of the body undergoing uniform circular motion is towards the centre of the circular path.

A body's centripetal acceleration is always directed perpendicularly to its tangential velocity.

The following figure depicts a body moving in uniform circular motion, with its tangential velocity $\vec{v}$, angular velocity ω, and centripetal acceleration $\vec{a}_c$ labelled.



6.3 Centripetal Force

Definition 6.5: Centripetal Force

The centripetal force of a body is the force providing the centripetal acceleration to enable a body to travel along a circular path.

$$F = mr\omega^2 = m\frac{v^2}{r} = mv\omega$$

For uniform circular motion, the centripetal force is constant in magnitude and changing in direction. As the centripetal force always acts perpendicularly to motion, it only changes the direction of velocity, but not its speed. Due to its direction, centripetal force does no work.

Note that the centripetal force **is not an extra force** by itself. It represents the **net force** experienced by the body due to forces like tension and friction, and should not be drawn separately on free-body diagrams.

If the centripetal force producing the centripetal acceleration vanishes, the object does not continue to move in its circular path; instead, it will move along a tangential path.

6.4 Applications of Uniform Circular Motion

In this section, we analyse certain situations that involve uniform circular motion.

To solve a problem involving circular motion, we generally follow the procedure below. This is similar to how we would naturally approach problems involving Newtonian physics in previous chapters.

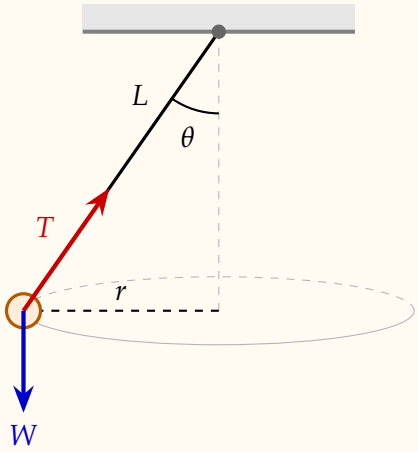
1. Draw a free-body diagram for the problem.
2. Identify the center of the circular motion.
3. Choose a suitable coordinate system.
4. Resolve the forces, if necessary.
5. Apply Newton's Second Law to solve the problem, while ensuring that r is the radius of the circular path.

6.4.1 Conical Pendulum

We begin with a simple example involving a conical pendulum.

Example 6.1. Serway & Jewett (Example 6.1) (modified)

A small ball of mass 2.0kg is suspended from a string of length 3.0m. The ball revolves with constant speed v in a horizontal circle of radius $r = 1.0\text{m}$ as shown in the figure.



Find v in terms of the geometry in the figure.

Let θ be the angle between the string and the vertical. By trigonometry,

$$\theta = \arcsin(r/L) \approx 0.340$$

Alongside the string's tension, the ball experiences a net vertical weight, resulting in a net horizontal centripetal force.

Resolving the forces using Newton's 2nd law,

$$\sum F_x = T\sin\theta = ma_c = mv^2/r$$
$$\sum F_y = T\cos\theta = mg$$

Dividing $\sum F_x$ by $\sum F_y$,

$$\tan\theta = \frac{v^2}{gr} \implies v = \sqrt{gr\tan\theta} \approx \boxed{1.86\text{m}\,\text{s}^{-1}}$$

Therefore, the ball travels at a linear speed of $v = 1.86\text{m}\,\text{s}^{-1}$.

6.4.2 Turning Vehicle

Rather ubiquitously, when focusing on cirucular motion, we encounter situations that involve a turning vehicle.

When a vehicle turns, the force that enables the car to remain in its circular path is its static friction, which is directed towards the centre of the path. In other words, **friction provides the centripetal force** of turning vehicles.

Example 6.2. DHS'24 Lecture Notes (Example 6.4)

A car of mass $m = 750\text{kg}$ is taking a corner with a flat horizontal surface and a radius of $r = 8\text{m}$. If the lateral friction forces cannot exceed a tenth of the normal contact force acting on the vehicle, what is the maximum speed at which it can take the bend?

To begin, the frictional force f by the road on the tyres is what provides the centripetal force.

$$f = m\frac{v^2}{r}$$

Furthermore, the only two vertical forces acting on the car are its weight and normal contact force. Since the car is in vertical equilibrium, both forces are equal to each other.

$$N = W = mg$$

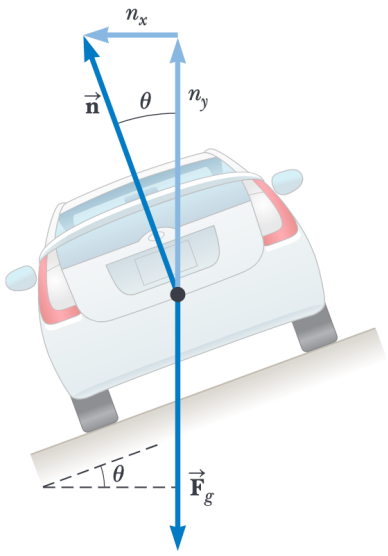
For maximum speed, where $f_{\text{max}} = mg/10$,

$$f_{\text{max}} = m\frac{{v_{\text{max}}}^2}{r} \implies \frac{mgr}{10} = m{v_{\text{max}}}^2$$
$$\implies v_{\text{max}} = \sqrt{\frac{gr}{10}} \approx \boxed{2.80\text{m s}^{-1}}$$

Therefore, the maximum speed at which the car can take the bend is $v = 2.80\text{m s}^{-1}$.

6.4.3 Banking Angle

As the tyres of a car can only provide a maximum friction, a car that is travelling too fast may not be able to get around the bend. To reduce skidding, roads can be **sloped** by a banking angle so that the normal contact force $\vec{n}$ of the road acting on the car provides a horizontal component.



For a particular speed, this force will produce the exact centripetal force required. Otherwise, some lateral frictional force is needed.

Example 6.3. Various sources.

Consider a car of mass m travelling on a slope with a banking angle θ between the road surface and the horizontal.

Find an expression for the optimal banking angle θ of the slope, such that the car is able to keep moving around the bend.

The horizontal component of the normal contact force N from the road provides the centripetal force required to keep the car moving around the bend.

Resolving the forces using Newton's 2nd law,

$$\sum F_x = N\sin\theta = m\frac{v^2}{r}$$
$$\sum F_y = N\cos\theta = mg$$

Dividing $\sum F_x$ by $\sum F_y$,

$$\frac{N\sin\theta}{N\cos\theta} = \frac{mv^2}{mgr} \implies \tan\theta = \frac{v^2}{gr}$$

Hence, the optimal angle for the slope to be banked is given by

$$\theta = \arctan\left(\frac{v^2}{gr}\right)$$

This gives the general formula for the optimal banking angle.

Proposition 6.2: Banking Angle

For a turning vehicle travelling at a speed v in a circular arc with radius r, the optimal banking angle is

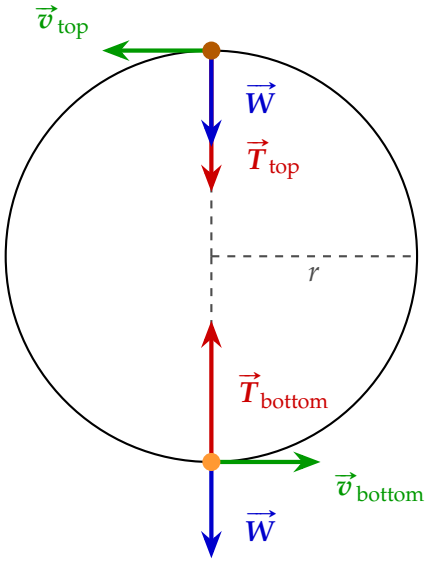
$$\theta = \arctan\left(\frac{v^2}{gr}\right)$$

6.5 Applications of Non-Uniform Circular Motion

Non-uniform circular motion occurs when the angular velocity of a body changes, while still moving in a circular path. This happens particularly as a body moves through a vertical circular path.

Consider a body of mass m moving in a vertical circle of radius r on the end of a string. The tension of the string is a constant contributor to the centripetal force. What differs how the weight of the body affects its centripetal force, and consequently its angular velocity.

Vertical circular motion is non-uniform because the speed decreases as the body moves upwards, where kinetic energy is gradually converted into gravitational potential energy.



From the figure above, at the topmost point, the centripetal force is given by the sum of the tension of the string and the weight of the body.

$$\vec{F}_{\text{top}} = \vec{T}_{\text{top}} + \vec{W} \implies m\frac{v^2}{r} = N + mg$$

At the bottommost point, the centripetal force is given by the difference between the normal contact force and the weight of the body.

$$\vec{F}_{\text{bottom}} = \vec{T}_{\text{bottom}} - \vec{W} \implies m\frac{v^2}{r} = N - mg$$

This is the dynamic-based analysis of the scenario, and it is also worthwhile to examine the energy changes of the body. Due to the conservation of mechanical energy, the increase in gravitational potential energy is equal to the decrease in kinetic energy as the body moves upwards.

$$\begin{aligned} \Delta K + \Delta U = 0 &\implies \frac{1}{2}m{v_{\text{bottom}}}^2 - \frac{1}{2}m{v_{\text{top}}}^2 = mg(2r) \\ &\implies {v_{\text{bottom}}}^2 - {v_{\text{top}}}^2 = 4gr \end{aligned}$$

Clearly, $v_{\text{bottom}} > v_{\text{top}}$, showing that the body is much faster as it crosses the bottom as compared to the top of the circular path.

7 Gravitational Fields

Here vigour fail'd the tow'ring fantasy:
But yet the will roll'd onward, like a wheel
In even motion, by the Love impell'd,
That moves the sun in heav'n and all the stars.

— DANTE, *Paradiso*

7.1 Gravitational Force

Two masses in proximity will exert an attractive force on each other. This force is termed gravity.

Definition 7.1: Newton's Law of Gravitation

Newton's law of gravitation states that the gravitational force F between two point masses m_1 and m_2 is directly proportional to the product of their masses and inversely proportional to the square of the separation r between their centres.

$$\vec{F} = -\frac{Gm_1m_2}{r^2}\hat{\mathbf{r}} \implies F = \frac{Gm_1m_2}{r^2}$$

where $G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$ is the gravitational constant of proportionality.

Point masses are theoretical bodies with non-zero mass but no volume. If two objects are placed sufficiently far apart, they can be approximated as point masses.

By Newton's 3rd law, the gravitational forces between two masses are equal and opposite, always acting along the line joining the two point masses. The negative sign in the gravitational force vector indicates its attractive nature.

7.2 Gravitational Field

Definition 7.2:

A gravitational field is a region of space within which a mass experiences a gravitational force.

Every mass creates a gravitational field, in which other masses will experience a gravitational force caused by said mass.

7.2.1 Gravitational Field Strength

Definition 7.3: Gravitational Field Strength

The gravitational field strength g at a point of distance r from point mass M is defined as the **gravitational force per unit mass** exerted on a small test mass placed at that point.

$$\vec{g} = -\frac{GM}{r^2}\hat{\mathbf{r}}$$

Proof. Consider two point masses M and m separated by a distance r.

Using Newton's law of gravitation, the magnitude of the gravitational force F acting on m due to M is

$$F = \frac{GMm}{r^2}$$

By its definition, gravitational field strength is equal to force per unit mass. The magnitude of g at the position of mass m is hence the following.

$$g = \frac{F}{m} = \frac{GM}{r^2}$$

□

The gravitational field strength is a vector quantity with unit $\mathrm{N\,kg^{-1}}$. A negative sign is used in the vector to show that the field strength points towards decreasing gravitational potential energy.

7.2.2 Field Lines

A gravitational field can be visualised through the illustration of field lines. The direction of a field at a point is **tangential** to the field line at that point.

The field line density at a point corresponds to the strength of the field at that point. Denser arrangements of field lines indicate greater gravitational field strength, and vice versa.

7.3 Gravitational Potential Energy

Definition 7.4: Gravitational Potential Energy

The gravitational potential energy of a mass at a point is defined as the **work done on the mass in moving it from infinity to that point**.

$$U = -\frac{GMm}{r}$$

Proof. Recall that for a conservative force,

$$F = -\frac{\mathrm{d}U}{\mathrm{d}r}$$

Since gravity is a conservative force, we can use this relation. From the definition of gravita-

tional potential energy at a point r away from a mass,

$$U = \int_{\infty}^{r} -\vec{F}_g \, \mathrm{d}r = \int_{\infty}^{r} \frac{GMm}{r^2} \, \mathrm{d}r = -\frac{GMm}{r}$$

□

Note that the gravitational potential energy is negative. This is because the external force acting on the mass is in the **opposite direction** to the gravitational attraction and the displacement of the mass, causing the work done on the mass to be negative. As the gravitational potential energy at infinite is a maximum value and equivalent at zero, any other point must have a negative gravitational potential.

7.4 Gravitational Potential

Definition 7.5: Gravitational Potential

The **gravitational potential** at a point is defined as the work done per unit mass (by an external force) in bringing a small test mass from infinity to that point.

$$\phi = \frac{U}{m} = -\frac{GM}{r}$$

Gravitational potential is a scalar quantity with unit $\mathrm{J\,kg^{-1}}$.

Proposition 7.1: Gravitational Potential Gradient

The gravitational field strength g can be said to be numerically equal but negative of the gradient of gravitational potential ϕ with respect to r.

$$g = -\frac{\mathrm{d}\phi}{\mathrm{d}r}$$

Proof. Taking the derivative of ϕ,

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{GM}{r}\right) = -\frac{GM}{r^2}$$

Substituting $g = -GM/r^2$,

$$g = -\frac{\mathrm{d}\phi}{\mathrm{d}r}$$

□

7.5 Energy Considerations

7.5.1 Total Energy

Proposition 7.2: Energies of Orbiting Body

For any body of mass m orbiting around a larger body of mass M, its kinetic energy is given by

$$K = \frac{GMm}{2r}$$

and its total energy is given by

$$E = -\frac{GMm}{2r}$$

Proof. Since gravitational force provides centripetal force,

$$\sum F = ma \implies \frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$\implies \frac{1}{2}mv^2 = \frac{GMm}{2r}$$

Hence, for gravitation,

$$E = K + U = \frac{GMm}{2r} + \left(-\frac{GMm}{r}\right) = -\frac{GMm}{2r}$$

□

The above holds true for circular orbits. For elliptical orbits, simply take r to be the semi-major axis length.

7.5.2 Escape Velocity

Definition 7.6: Escape Velocity

The minimum velocity required for a body to escape the gravitational pull of a larger body of mass M is termed the escape velocity of M, and is given by

$$v = \sqrt{\frac{2GM}{r}}$$

Proof. As an object travels indefinitely away from a body, $r \to \infty$, hence $K \to 0$, $U \to 0$, and $E \to 0$. As such, at infinity, we take the kinetic energy and gravitational potential energy to be zero.

Since the object-Earth system is isolated, calculating the minimum escape velocity v,

$$\Delta U = -\Delta K \implies 0 - \left(-\frac{GMm}{r}\right) = -\left(0 - \frac{1}{2}mv^2\right)$$

$$\implies \frac{GMm}{r} = \frac{1}{2}mv^2$$

$$\implies v = \sqrt{\frac{2GM}{r}}$$

□

If the initial velocity is greater than the minimum escape velocity required, the object will have some residual kinetic energy as $r \to \infty$.

7.6 Kepler's Third Law

Definition 7.7: Kepler's Third Law

Kepler's Third Law states that the square of a body's orbital period is directly proportional to the cube of the radius of its orbit, such that

$$T^2 \propto r^3$$

for circular orbits.

Proof. The gravitational force acting on the Moon by the Earth provides the centripetal force, giving acceleration $a_c = r\omega^2$.

$$\sum F = ma \implies \frac{GMm}{r^2} = mr\omega^2$$

$$\implies \omega^2 = \frac{GM}{r^3}$$

$$\implies \left(\frac{2\pi}{T}\right)^2 = \frac{GM}{r^3}$$

$$\implies r^3 = \frac{GM}{4\pi^2}T^2$$

$$\implies r^3 \propto T^2$$

□

7.7 Geostationary Orbits

Definition 7.8:

A geostationary orbit is an equatorial orbit around the Earth made by a satellite with an orbital period of 24 hours, moving in the direction from west to east.

To repeat, for an satellite to be in geostationary orbit, it must satisfy the following conditions.

- It must move from **west to east**, so that it moves in the same sense of self-rotation of the Earth about its own axis.
- It must have a orbital period of **24 hours**, the same as Earth's rotational period about its own axis.
- It must be an **equatorial orbit**, so that its axis of rotation coincides with Earth's axis of rotation.

Satellites in geostationary orbits appear stationary to an observer on the Earth's surface.

Part III

Thermal Physics

8 Temperature and Ideal Gases

> Every mathematician knows it is impossible to understand an elementary course in thermodynamics.
>
> — Vladimir Igorevich Arnold

8.1 Zeroth Law of Thermodynamics

We open our inquiry on thermal physics with a plethora of definitions to fix some notions of heat and temperature.

Definition 8.1: Temperature

Temperature is a measure of the degree of hotness of an object. Thermal energy moves from objects of higher temperatures to objects of lower temperatures. Note that temperature does not measure an object's thermal energy.

Definition 8.2: Heat

Heat is energy that flows from a higher-temperature object to a lower-temperature object due to differences in temperature.

Definition 8.3: Thermal Equilibrium

When two objects in thermal contact has no net heat transfer between them, they are said to be in thermal equilibrium and are at the same temperature.

With this, we are ready to define the most basic law of thermodynamics.

Definition 8.4: Zeroth Law of Thermodynamics

The zeroth law of thermodynamics states that if two objects A and B are separately in thermal equilibrium with object C, then objects A and B must be in thermal equilibrium with each other.

This forms the basis for the measurement of temperature, as the object C above can function as a thermal measuring device known as a thermometer.

8.2 Temperature Scales

Thermometers measure temperature based on a physical property that changes with temperature.

8.2.1 Empirical Temperature Scale

An **empirical temperature scale** is a scale of temperature that is based on the variation with temperature of a property of a substance, given that the property varies linearly with temperature.

This property is termed a **thermometric property**. Examples of thermometric properties include

- the volume of a fixed mass of liquid;
- the resistance of a metal;
- the pressure of a fixed mass of gas at constant temperature; and
- the electromotive force produced between junctions of dissimilar metals at different temperatures.

To set up an empirical temperature scale,

1. Choose an appropriate thermometric property.
2. Select two fixed temperature points; an upper point and a lower point to calibrate the thermometer.
3. Calibrate the thermometer by placing the thermometer at the two points, and record the thermometric quantity, letting the value at the lower point be X_0 and that at the upper point be X_{100}.

With these two values, the temperature θ of any subsequent medium can be calculated using the following formula.

$$\theta = \frac{X_\theta - X_0}{X_{100} - X_0} \times 100$$

8.2.2 Absolute Temperature Scale

The **absolute temperature scale** (also **thermodynamic temperature scale**) does not depend on the thermometric property of any particular substance, and has absolute zero and the triple point of water as fixed points.

- **Absolute zero**, defined as $-273.15\,^\circ\text{C}$ or $0\,\text{K}$, is the temperature at which all substances have *minimum internal energy*.
- The **triple point of water** is the particular temperature and pressure ($273.16\,\text{K}$, $4.58\,\text{mmHg}$) at which the three states of water can co-exist in equilibrium.

8.3 Ideal Gases

One **mole** is the S.I. unit for an amount of substance.

Avogadro's number is the exact number equal to the number of particles in one mole of any substance, and is given by $N_A = 6.02 \times 10^{23}$.

Two more important constants are the **ideal gas constant**, $R = 8.31\,\text{J mol}^{-1}\,\text{K}^{-1}$, and the **Boltzmann constant**, $k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$. These two are related by the following equation.

$$k = \frac{R}{N_A}$$

Definition 8.5: Ideal Gas

An ideal gas is a hypothetical gas that obeys the equation of state

$$PV = nRT$$

perfectly at all pressures P, volumes V, amounts of substances n, and temperatures T.

The equation of state of an ideal gas can be rewritten using the number of gas molecules $N = nN_A$.

$$PV = NkT$$

8.4 Kinetic Theory of Gases

The kinetic theory of gases is a simplified classical model of the thermodynamic behaviour of gases, where the macroscopic variables of pressure, volume, and temperature, are related to a description of gases on a microscopic scale.

In short, it applies Newton's laws of motion in a **statistical manner** to a large collection of particles, providing a reasonable description of thermodynamics. The theory itself relies on six core assumptions.

1. All gases consist of a very large number of particles.
2. These particles behave as if they are hard, perfectly elastic, identical spheres.
3. The particles are in constant, random motion, and obey Newton's laws of motion.
4. There are no forces of attraction or repulsion between the particles, unless they are in collision with each other or with the walls of the container.
5. The total volume of the particles is negligible when compared to the volume of the container.
6. The time of collisions is negligible compared to the time between collisions.

With these assumptions, we are able to derive further variations of the equation of state by analysing the effect of random collisions by the gaseous particles on the container's walls.

We first familiarise ourselves with a method of obtaining a statistical average for molecular speeds. Since the average of all molecular velocities is zero, we cannot take an arithmetic mean.

Definition 8.6: Mean Square Speed

For a container containing N gas molecules each with molecular speed c_i, the mean square speed $\langle c^2 \rangle$ of the gas is given by the arithmetic mean of the square of each molecular speed.

$$\langle c^2 \rangle = \frac{{c_1}^2 + {c_2}^2 + \cdots + {c_N}^2}{N} = \frac{1}{N}\sum_i {c_i}^2$$

The root mean square speed c_{rms} of the molecules is

$$c_{\text{rms}} = \sqrt{\langle c^2 \rangle} = \sqrt{\frac{{c_1}^2 + {c_2}^2 + \cdots + {c_N}^2}{N}}$$

Using the kinetic theory of gases, we propose a new variation of the ideal gas law.

Proposition 8.1: Ideal Gas Law (Kinetic Theory)

For an ideal gas of pressure P, volume V, number of molecules N, mass m, and mean square speed $\langle c^2 \rangle$,

$$PV = \frac{1}{3} Nm\langle c^2 \rangle$$

Proof. Consider a cubical box of sides l, containing N molecules of a gas, where the mass of each molecule is m.

Suppose a molecule has horizontal velocity c_x. By considering the elastic collision of this molecule with a wall, the change in momentum of the molecule Δp due to the collision is

$$\Delta p = (-mc_x) - (mc_x) = -2mc_x$$

By the conservation of linear momentum, this means that the change in momentum of the wall itself is

$$\Delta p_{\text{wall}} = -\Delta p = 2mc_x$$

For the next collision to occur with the same wall, the same molecule would have travelled a horizontal distance of $2l$. Hence, the time Δt between collisions is given.

$$\Delta t = \frac{2l}{c_x}$$

Therefore, the rate of change of momentum **for the wall** due to one molecule is

$$\frac{\Delta p_{\text{wall}}}{\Delta t} = \frac{m{c_x}^2}{l}$$

Summing up this value for all molecules, we obtain the total force being exerted on the wall.

$$F_{\text{wall}} = \sum_{i=1}^{N} \frac{m(c_x)_i^{\,2}}{l} = \frac{m}{l} \sum_{i=1}^{N} (c_x)_i^{\,2}$$

Finding the mean square horizontal speed, we can use it to find expressions for F_{wall} and consequently the pressure exerted on the wall P_{wall}.

$$\langle {c_x}^2 \rangle = \frac{1}{N} \sum_{i=1}^{N} (c_x)_i^{\,2} \implies F_{\text{wall}} = \frac{Nm}{l} \langle {c_x}^2 \rangle$$

$$\implies P_{\text{wall}} = \frac{F_{\text{wall}}}{l^2} = \frac{Nm}{l^3} \langle {c_x}^2 \rangle = \frac{Nm}{V} \langle {c_x}^2 \rangle$$

Applying Pythagoras' theorem to the 3-dimensional velocity vector for any molecule leads to

$$c^2 = {c_x}^2 + {c_y}^2 + {c_z}^2 \implies \langle c^2 \rangle = \langle {c_x}^2 \rangle + \langle {c_y}^2 \rangle + \langle {c_z}^2 \rangle$$

However, we assume that the motion of particles is completely random with no bias for direction, such that $\langle {c_x}^2 \rangle = \langle {c_y}^2 \rangle = \langle {c_z}^2 \rangle$. Therefore,

$$\langle {c_x}^2 \rangle = \frac{1}{3} \langle c^2 \rangle$$

Using this expression, we can find the total force and pressure exerted on the wall, giving us the proposition.

$$P = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle \implies PV = \frac{1}{3}Nm\langle c^2 \rangle$$

□

Using mean square speed, we can also find an expression for the average molecular kinetic energy of an ideal gas.

Proposition 8.2: Molecular Kinetic Energy

The average **molecular** kinetic energy of an ideal gas is given as

$$\langle K \rangle = \frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

Proof. The following representation of the ideal gas law are equal to each other.

$$PV = \frac{1}{3}Nm\langle c^2 \rangle = NkT$$

Isolating the kinetic energy term from the equivalence,

$$\left(\frac{1}{2}m\langle c^2 \rangle\right) N = \frac{3}{2}NkT \implies \frac{1}{2}m\langle c^2 \rangle = \frac{3}{2}kT$$

□

The term above can be summed up to obtain the average kinetic energy of a gas with N molecules.

$$\langle K_{\text{gas}} \rangle = \sum_{i=1}^{N} \frac{3}{2}kT = \frac{3}{2}NkT = \frac{3}{2}nRT$$

Proposition 8.3: Representations of RMS Speed

The following equations, involving density ρ, molar mass M_m, or mass m of a gas, are all equivalent to the root mean square speed of the gas.

$$c_{\text{rms}} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3RT}{M_m}} = \sqrt{\frac{3kT}{m}}$$

Proof. For a pressure P, volume V, amount of moles of gas n, amount of molecules of gas N, mass per molecule m, total mass of gas $M = Nm$, we have the following equations.

Given density ρ,

$$PV = \frac{1}{3}Nm\langle c^2 \rangle \implies PV = \frac{1}{3}M\langle c^2 \rangle$$
$$\implies c_{\text{rms}} = \sqrt{\langle c^2 \rangle} = \sqrt{\frac{3PV}{M}} = \sqrt{\frac{3P}{\rho}}$$

Given molar mass M_m,

$$PV = nRT \implies \frac{1}{3}M\langle c^2\rangle = \frac{M}{M_m}RT$$
$$\implies c_{\text{rms}} = \sqrt{\langle c^2\rangle} = \sqrt{\frac{3RT}{M_m}}$$

Given molecular mass m,

$$PV = NkT \implies \frac{1}{3}Nm\langle c^2\rangle = NkT$$
$$\implies c_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

□

9 First Law of Thermodynamics

9.1 Internal Energy

All systems possess energy by virtue of its microscopic components. This includes kinetic energy due to the random movements of its particles, as well as potential energy due to the intermolecular forces of attraction within the system.

Definition 9.1: Internal Energy

The **internal energy** of a system is the sum of the random distribution of the kinetic and potential energy associated with the molecules in a system.

$$U = \sum K + \sum P$$

In this section, we use U to denote internal energy and P to denote molecular potential energy.

Proposition 9.1: Internal Energy of an Ideal Gas

The internal energy of an **ideal gas** is equivalent to its kinetic energy.

$$U_{\text{ideal}} = \sum \langle K \rangle = \frac{3}{2} NkT$$

Proof. By the kinetic theory of gases, an ideal gas has no intermolecular attraction and thus no random potential energy. This means that all of its internal energy is due to its random kinetic energy.

Recall that

$$\langle K \rangle = \frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$$

The internal energy U of an ideal gas with N molecules is thus

$$U = \sum K = N \langle K \rangle = \frac{3}{2} NkT$$

□

It is worth clarifying that the internal potential energy of a system P is due to the work done to move the molecules further apart against intermolecular attraction. This means that non-ideal gases have a higher internal potential energy than solids.

9.2 Heat Capacity

Definition 9.2: Heat Capacity

The heat capacity C of an object is the thermal energy Q required to cause a unit temperature change ΔT.

$$C = \frac{Q}{\Delta T} \implies Q = C\Delta T$$

The unit of C is $\mathrm{J\,K^{-1}}$.

9.2.1 Specific Heat Capacity

Definition 9.3: Specific Heat Capacity

The specific heat capacity c of a substance is defined as the thermal energy Q per unit mass m required to cause a unit temperature change ΔT.

$$c = \frac{Q}{m\Delta T} \implies Q = mc\Delta T$$

The unit of c is $\mathrm{J\,K^{-1}\,kg^{-1}}$.

9.2.2 Specific Latent Heat

Definition 9.4: Specific Latent Heat

The specific latent heat L of a substance is defined as the thermal energy Q per unit mass m required to change its state at constant temperature.

$$L = \frac{Q}{M} \implies Q = mL$$

For melting and freezing, it is latent heat of fusion, and latent heat of solidification for the reverse. They occur at the substance's melting point, with constant pressure.

For boiling and condensation, it is latent heat of vaporisation, and latent heat of condensation for the reverse. They occur at the substance's boiling point, with constant pressure.

In both melting and boiling, the heat supplied is utilised by the molecules to do work against intermolecular attractions, thereby increasing the internal potential energy of the system. However, when boiling, work must also be done against atmospheric pressure, meaning that the specific latent heat of vaporisation is **higher** than the specific latent heat of fusion.

9.3 First Law of Thermodynamics

Definition 9.5: First Law of Thermodynamics

The first law of thermodynamics states that the increase in the internal energy of a system is the sum of the heat Q supplied **to** the system and the work done W **on** the system.

$$\Delta U = Q + W$$

The quantities Q and W are positive if they are done **on** the system. Otherwise, Q would be negative if heat is removed **from** the system, and W would be negative if work is done **by** the system.

Proposition 9.2: Work Done on Gas

The work done W on a gas as its volume changes from V_i to V_f is given by the following integral.

$$W = -\int_{V_i}^{V_f} P \, \mathrm{d}V$$

Proof. Recall the definition for work done W.

$$W = \int_{s_i}^{s_f} F \, \mathrm{d}s$$

As pressure is equal to force over area, this means that force is equal to the product of pressure P and area A. Since the external force is directed in the opposite direction to the pressure exerted by the gas, this gives

$$W = \int_{s_i}^{s_f} -PA \, \mathrm{d}s = -\int_{V_i}^{V_f} P \, \mathrm{d}V$$

□

9.3.1 Thermodynamic Processes

There are multiple ways in which a thermodynamic process can occur.

Definition 9.6: Adiabatic Process

An adiabatic process is a fast process during which **no heat** is supplied to or removed from the system.

$$Q = 0 \implies \Delta U = W$$

Definition 9.7: Isobaric Process

An isobaric process is a process that occurs at constant pressure.

$$W = -P\Delta V$$

Definition 9.8: Isovolumetric Process

An **isovolumetric process** (also **isochoric process**) is a process that occurs at constant volume.

$$V_f = V_i \implies W = 0 \implies \Delta U = Q$$

Definition 9.9: Isothermal Process

An **isothermal process** is a process that occurs at constant temperature.

$$\Delta U = 0 \implies Q = -W$$

9.3.2 Pressure-Volume Graph

For a system with N molecules, the **state** of the system is defined by its pressure P, volume V, and thermodynamic temperature T, and can be represented by a point in the pressure-volume graph.

The four thermodynamic processes are shown below.

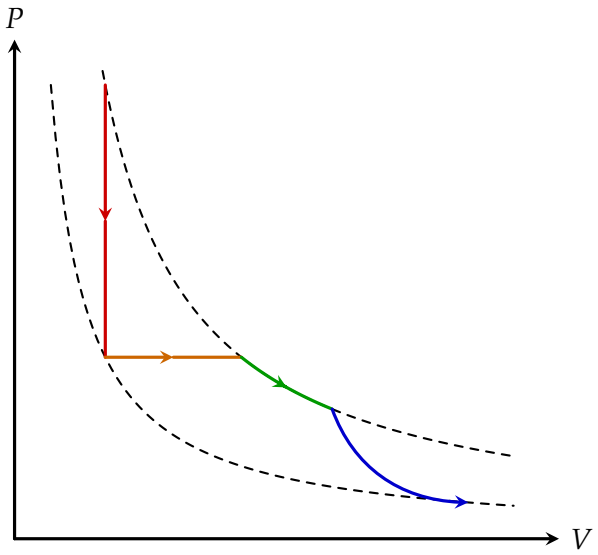


FIGURE 9.1. A pressure-volume graph depicting a isovolumetric, isobaric, isothermal, and adiabatic process.

In particular, isothermal processes trace out a line known as an **isotherm**. Since temperature T remains a constant, this means that PV is a constant as well, therefore creating a hyperbolic curve. These appear as dashed lines in Figure 9.1.

Part IV

Oscillations and Waves

10 Oscillations

> There is nothing staid, nothing settled, in this universe. All is rippling, all is dancing; all is quickness and triumph.
>
> — VIRGINIA WOOLF, *The Waves*

Until now, we have mainly considered motion that occurs once without any repeating factor. The only time we have come close is with the topic of circular motion. Yet, forms of repeating motion are commonplace in the study of mechanics.

In this chapter, we are concerned with the study of oscillatory motion, which comprises of repetitive and periodic variations in the displacement of a body.

10.1 Oscillatory Quantities

Definition 10.1: Oscillation

An **oscillation** is a periodic to-and-fro motion of an object between two limits.

An example of oscillatory motion appears in a pendulum.

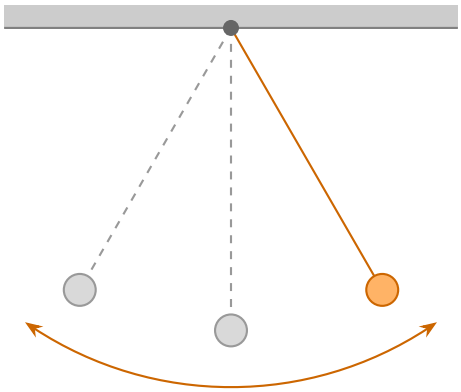


FIGURE 10.1. The trusty pendulum, swinging in an oscillatory motion.

As an ideal pendulum swings back and forth, there is a maximum distance that it travels in one direction. There is a constant time duration between each oscillation, as well as a certain speed at which it is angularly displaced. We solidify these concepts in the following definitions.

Definition 10.2: Displacement

The **displacement** x is the distance of the oscillating object from its **equilibrium position** in a stated direction.

Definition 10.3: Amplitude

The amplitude x_0 is the **maximum displacement** of the oscillating object from the equilibrium position.

Definition 10.4: Period

The period T is the time taken for one complete oscillation.

Definition 10.5: Frequency

The frequency f the number of complete to-and-fro cycles per unit time made by the oscillating object.

$$f = \frac{1}{T}$$

Definition 10.6: Angular Frequency

The angular frequency ω is the constant which characterises the particular simple harmonic oscillator and is related to its natural frequency.

$$\omega = 2\pi f = \frac{2\pi}{T}$$

Definition 10.7: Phase

The phase ϕ is an angle which gives a measure of the fraction of a cycle htat has been completed by an oscillating body.

10.1.1 Phase

We detour to expand more on the concept of phases. A phase is simply the fraction of a cycle that has been completed by some oscillating body, but converted to an angle.

If x is the current distance travelled by the oscillating body since completing the last oscillation and λ is the total distance travelled per oscillation, then

$$\phi = 2\pi \left(\frac{x}{\lambda}\right)$$

If t is the duration of time since completing the last oscillation and T is the period of the oscillating body, then

$$\phi = 2\pi \left(\frac{t}{T}\right)$$

To raise some examples, when one-fourth of a cycle is completed, the phase is $\pi/2$ radians. When one-half of a cycle is completed, the phase is π radians.

Definition 10.8: Phase Difference

The **phase difference** $\Delta\phi$ between two oscillating bodies is a measure of how much one oscillating body is out of step with the other.

The phase difference is itself an angle. Two bodies are **in phase** if their phase difference is zero. Two bodies are **in antiphase** if their phase difference is π.

10.2 Free Oscillations

Definition 10.9: Free Oscillation

An oscillating system undergoes **free oscillations** if it oscillates without any energy gain or loss, and if there is no external force acting on the system.

Under free oscillation, an object oscillates with constant amplitude, as there are no external forces acting on it.

Consequently, an oscillatory motion can be graphically represented by a sinusoidal curve. We can describe the displacement of an oscillating body with respect to time using the following equation.

$$x = x_0 \sin(\omega t + \phi) \text{ or } x_0 \cos(\omega t + \phi)$$

10.2.1 Simple Harmonic Motion

Free oscillations lead to a special type of periodic motion. We discuss this specific periodic motion below.

Definition 10.10: Simple Harmonic Motion

Simple harmonic motion is defined as the oscillatory motion of a particle whose acceleration $\vec{a}$ is directly proportional and in the opposite direction to its displacement $\vec{x}$ from the equilibrium position.

$$\vec{a} = -\omega^2 \vec{x}$$

Proof. We wish to prove that both $x = x_0 \sin(\omega t)$ and $x = x_0 \cos(\omega t)$ are particular solutions to the defining equation $a = -\omega^2 x$.

We begin with $x = x_0 \sin(\omega t)$. Differentiating displacement with respect to time twice,

$$\frac{\mathrm{d}x}{\mathrm{d}t} = w x_0 \cos(\omega t) \implies \frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = -\omega^2 x_0 \sin(\omega t)$$
$$\implies a = -\omega^2 x$$

We do the same for $x = x_0 \cos(\omega t)$.

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -w x_0 \sin(\omega t) \implies \frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = -\omega^2 x_0 \cos(\omega t)$$
$$\implies a = -\omega^2 x$$

Thus, simple harmonic motion holds for non-damped sinusoidal oscillatory motions. □

Proposition 10.1: Velocity in SHM

The velocity v of a body undergoing simple harmonic motion, with amplitude x_0 and displaced from the equilibrium position by x, can be expressed in the following equation.

$$v = \pm\omega\sqrt{({x_0}^2 - x^2)}$$

Proof. Observe that

$$v^2 = \omega^2 {x_0}^2 \cos^2 \omega t$$

Using the Pythagorean identity,

$$\begin{aligned} v^2 = {x_0}^2\omega^2 - {x_0}^2\omega^2 \sin^2 \omega t &\implies v^2 = \omega^2({x_0}^2 - x^2) \\ &\implies v = \pm\omega\sqrt{({x_0}^2 - x^2)} \end{aligned}$$

□

10.2.2 Energies under SHM

We now explore the formulations for kinetic, potential, and total energy for a body undergoing simple harmonic motion.

Proposition 10.2: Kinetic Energy of Oscillating Body

The kinetic energy K of an oscillating body is given by

$$K = \frac{1}{2}m\omega^2 {x_0}^2 \cos^2 \omega t = \frac{1}{2}m\omega^2({x_0}^2 - x^2)$$

Proof. Recall that

$$v = \omega x_0 \cos \omega t = \pm\omega\sqrt{{x_0}^2 - x^2}$$

Therefore,

$$\begin{aligned} K = \frac{1}{2}mv^2 &= \frac{1}{2}m\omega^2 {x_0}^2 \cos^2 \omega t \\ &= \frac{1}{2}m\omega^2({x_0}^2 - x^2) \end{aligned}$$

□

Proposition 10.3: Gravitational Potential Energy of Oscillating Body

The gravitational potential energy U of an oscillating body is given by

$$U = \frac{1}{2}m\omega^2 {x_0}^2 \sin^2 \omega t = \frac{1}{2}m\omega^2 x^2$$

Proof. Let us define the minimum value of U to be zero. Therefore, the total energy of the system is equal to the maximum kinetic energy of the system.

$$E = K_{\max} = \frac{1}{2}m\omega^2 {x_0}^2$$

Generally, due to the conservation of energy,

$$\begin{aligned} E = K + U &\implies U = E - K \\ &\implies U = \frac{1}{2}m\omega^2 {x_0}^2 - \frac{1}{2}m\omega^2 {x_0}^2 \cos^2 \omega t \\ &\implies U = \frac{1}{2}m\omega^2 {x_0}^2 \sin^2 \omega t \end{aligned}$$

Similarly, using the time-independent form of K,

$$U = \frac{1}{2}m\omega^2 x^2$$

□

Proposition 10.4: Total Energy of Oscillating Body

The total energy of an oscillating body without any external forces acting on it is constant and can be expressed as

$$E = \frac{1}{2}m\omega^2 {x_0}^2$$

Proof. See previous derivations for kinetic and potential energy. □

10.2.3 Graphical Illustrations

The quantities involved with simple harmonic oscillation vary in predictable and interlinked ways over time. We introduce some illustrations to better visualise the different relationships between these quantities.

In doing so, we hope to answer graph-based questions more easily when dealing with free oscillations and simple harmonic motion.

The velocity-displacement graph of a body undergoing simple harmonic motion takes the form of an ellipse, as seen in Figure 10.2.

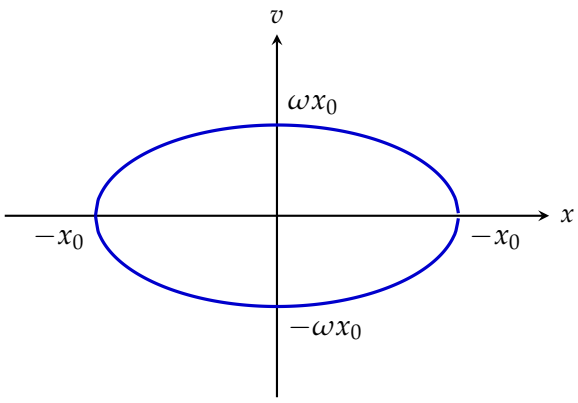


FIGURE 10.2. The velocity-displacement graph of an oscillating body undergoing simple harmonic motion.

As evident from the defining equation for simple harmonic motion, the acceleration-displacement graph is a straight line with a negative gradient. This is shown in Figure 10.3.

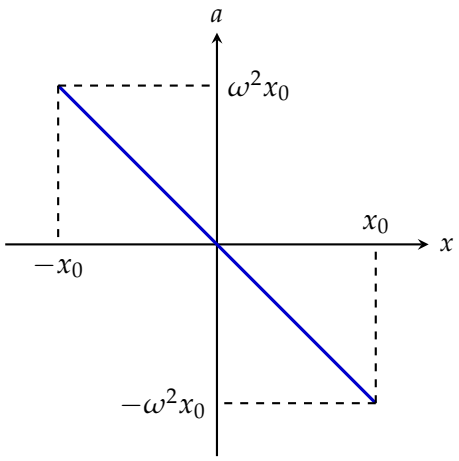


FIGURE 10.3. The acceleration-displacement graph of an oscillating body undergoing simple harmonic motion.

When we plot the variation of displacement, velocity, and acceleration with respect to time in Figure 10.4, we are able to notice the phase differences between each of these quantities.

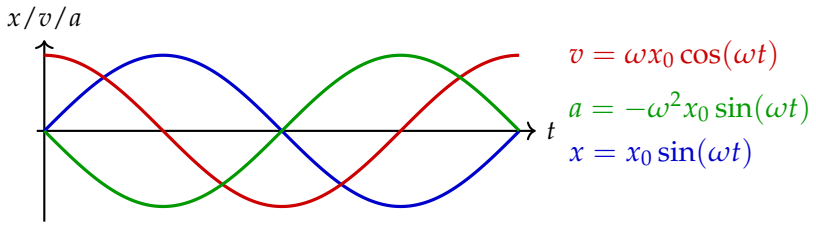


FIGURE 10.4. Time graphs of an oscillating body's displacement, velocity, and acceleration.

As for kinetic and potential energy, note that when a body passes by its equilibrium position, it will have a maximum kinetic energy and a minimum potential energy. This is shown in Figure 10.5.

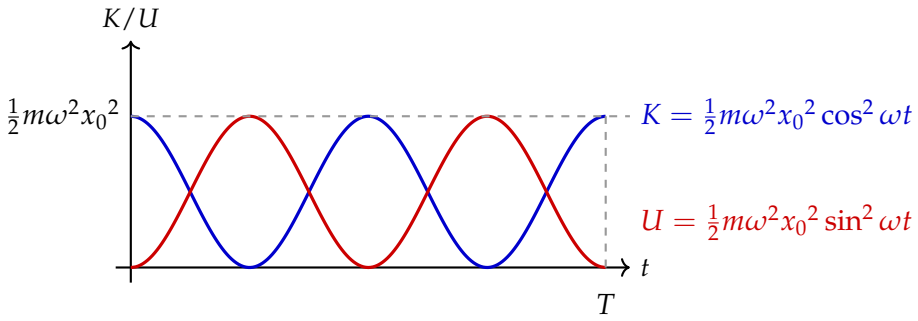


FIGURE 10.5. The change in kinetic energy and potential energy over time for a simple oscillating pendulum. They are always in antiphase.

10.3 Damped Oscillations

We have talked extensively about free oscillations. We now consider oscillations that experience energy loss, often due to the presence of drag forces.

Definition 10.11: Damping

Damping is a process where energy is taken from an oscillating system as a result of dissipative forces.

Definition 10.12: Damped Oscillation

Damped oscillation occurs when there is a **continuous dissipation of energy to the surroundings** such that the total energy in the system decreases with time, hence the amplitude of the motion progressively decreases with time.

There are three kinds of damping.

- Light damping occurs when the object oscillates with its amplitude decreasing exponentially over time, such as in a vertical spring-mass system.
- Critical damping occurs when the object does not oscillate, and displacement is brought to zero in the shortest possible time, such as in a car suspension system.
- Heavy damping occurs when the object does not oscillate, and the system takes a longer time to return to the equilibrium position.

The behaviour of each type of damping is visualised in Figure 10.6 below.

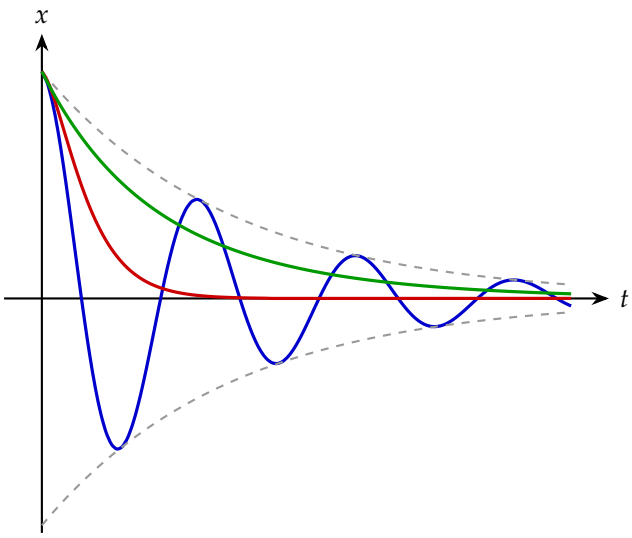


FIGURE 10.6. Displacement-time graphs for an oscillating body undergoing light damping, critical damping, and heavy damping. The dashed lines demarcate the exponential decay of amplitude for the lightly damped body.

10.3.1 Car Suspension Systems

One crucial application of critical damping appears in a car's mechanical system. The suspension system of a car allows passengers to be comfortable when moving along bumpy roads.

If the suspension system was only lightly damped, passengers would be thrown up and down due to the continuous oscillations. A good suspension system is one that critically damps the car's movements, thus ensuring a safe ride.

10.4 Forced Oscillations

We move on to the final type of oscillation.

Definition 10.13: Forced Oscillations

Forced oscillations are caused by the continual input of energy by external applied force to an oscillating system to compensate the loss due to damping in order to maintain the amplitude of the oscillation.

The system then oscillates at the frequency of the external periodic force.

10.4.1 Resonance

Definition 10.14: Natural Frequency

The natural frequency of a system is the rate at which the system tends to oscillate when disturbed and left to oscillate freely.

When the driving frequency f equals the natural frequency of a system f_0, the amplitude of the oscillation will be a maximum.

Definition 10.15: Resonance

Resonance occurs the driving frequency of the external driving force equals the natural frequency of the system, causing the resulting amplitude of the system to become a maximum.

Useful applications of include include radio and musical instruments, wheras undesirable effects of resonance include the shattering of glass and resonance in rigid structures.

10.4.2 Frequency Response Graph

Figure 10.7 below shows how the amplitude x_0 of a forced oscillation depends on the driving frequency f when the degree of damping is varied.

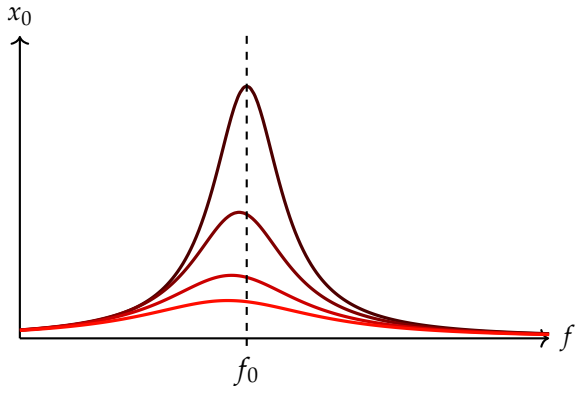


FIGURE 10.7. Amplitude-frequency graphs of varying driving frequencies. Brighter lines indicate greater damping, which results in the peak of the graph decreasing away from the natural frequency f_0.

11 Wave Motion

11.1 Waves

Apart from moving bodies, waves are also another phenomenon in which oscillations occur.

Definition 11.1: Progressive Wave

A **progressive wave** is a wave in which energy is carried from out point to another by means of vibrations or oscillations within the wave, without transporting matter.

Definition 11.2: Transverse Wave

A **transverse wave** is a wave with vibrations perpendicular to the direction of transfer of the wave's energy.

Definition 11.3: Longitudinal Wave

A **longitudinal wave** is a wave with vibrations parallel to the direction of transfer of the wave's energy.

Definition 11.4: Mechanical Wave

A **mechanical wave** is a wave that requires a medium for its transmission.

Definition 11.5: Electromagnetic Wave

A **electromagnetic wave** is a wave that consists of oscillating electromagnetic fields that are perpendicular to each other and to the direction of transfer of the wave's energy.

All electromagnetic waves travel in vacuum with the same speed of $c = 3.0 \times 10^8$ metres per second. They do not require a medium for transmission. The table below shows the categorisation between different electromagnetic waves.

Name	Wavelength	Uses
Radio waves	>10cm	Communications, radioastronomy
Microwaves	1mm to 10cm	Communications, cooking
Infrared	700nm to 1mm	Satellite surveying, TV controls
Visible light	400nm to 700nm	Sight, communication
Ultraviolet	1nm to 400nm	Food sterilisation, atomic structures
X-rays	1pm to 1nm	Diagnosis, radiotherapy, astronomy
Gamma rays	1fm to 1pm	Diagnosis, radiotherapy

11.1.1 Wave Characteristics

We introduce some definitions for waves, which are quite similar in spirit to the ones previously defined for oscillating bodies.

Definition 11.6: Displacement

Displacement is the distance in a specific direction of a point on the wave from its equilibrium position.

Definition 11.7: Amplitude

The amplitude is the maximum displacement of any point on the wave from its equilibrium position.

Definition 11.8: Period

The period T is the time taken for one complete oscillation of a point in a wave.

Definition 11.9: Frequency

The frequency f is the number of oscillations per unit time of a point on a wave.

$$f = \frac{1}{T}$$

Definition 11.10: Wavelength

The wavelength λ is the **minimum distance** between any two points of the wave with the **same phase** at the same instant.

Definition 11.11: Wave Speed

The wave speed v is the speed at which energy is transmitted by a wave.

$$v = \frac{\lambda}{T} = f\lambda$$

Definition 11.12: Phase

The phase ϕ is an angle which gives a measure of the fraction of a cycle htat has been completed by a wave.

$$\phi = 2\pi\left(\frac{x}{\lambda}\right) = 2\pi\left(\frac{t}{T}\right)$$

11.2 Intensity

Definition 11.13: Intensity

The intensity I of a wave is the power transmitted P per unit area A perpendicular to the wave velocity.

$$I = \frac{P}{A}$$

Proposition 11.1: Proportionality of Intensity

The intensity I of a wave is directly proportional to the square of the wave's amplitude x_0, as well as the square of the wave's frequency f.

$$I \propto {x_0}^2 \text{ and } I \propto f^2$$

Proof. From simple harmonic motion, we understand that the total energy associated with an oscillatory motion is proportional to the square of its amplitude.

$$E = \frac{1}{2}m\omega^2{x_0}^2 = \frac{1}{2}m(2\pi f)^2{x_0}^2$$

Since $I \propto E$ and $E \propto f^2{x_0}^2$, then

$$I \propto f^2{x_0}^2$$

□

Proposition 11.2: Inverse Square Law

By the inverse square law, intensity is inversely proportional to the square of the distance travelled for three-dimensional transmission.

$$I \propto \frac{1}{r^2}$$

Proof. Consider a point source which emits energy with power P. Energy is transferred radially outward in three dimensions, carried on an expanding spherical surface.

At a distance r from the source,

$$I = \frac{P}{A} = \frac{P}{4\pi r^2} \implies I \propto \frac{1}{r^2}$$

This is known as the inverse square law, where the observed intensity of the wave is inversely proportional to the square of the distance from the source. □

Note that the inverse square law does not hold **for two-dimensional transmission**. For a point source that emits energy two-dimensionally, the wave propagation can be encapsulated within a cylinder of height $h = 2x_0$. At a distance r from the source,

$$I = \frac{P}{A} = \frac{P}{2\pi rh} \implies I \propto \frac{1}{r} \text{ (2D)}$$

11.3 Polarisation

An ordinary beam of light contains a large number of transverse waves. The oscillation of these waves are not initially confined to any particular direction, but instead they span across all possible angles.

It is possible to restrict or confine the direction that such waves oscillate in. For instance, we can create a filter that only picks out horizontally oscillating waves while blocking out vertically oscillating waves. This is the principle of **polarisation**.

Definition 11.14: Polarised Waves

If the oscillations in a transverse are confined to one direction only, in a plane normal to the direction of energy transfer, the wave is said to be **polarised** in that direction.

A piece of material known as a **polariser** achieves this effect. An ideal polariser passes all incident light parallel to its polarising axis, and blocks all light perpendicular to this axis.

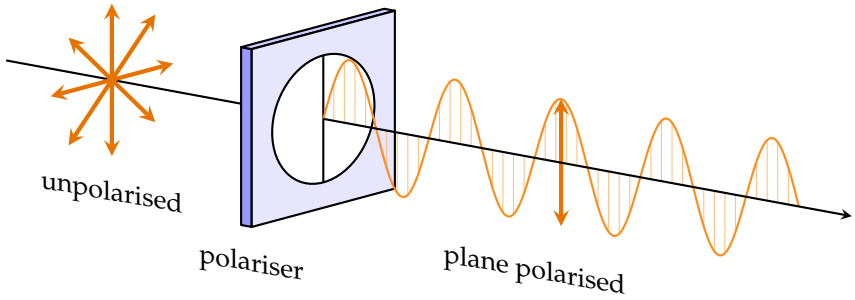


FIGURE 11.1. A beam of light becoming linearly polarised after passing through a sheet of polarising material.

Only transverse waves can be polarised. It is not possible to polarise sound waves for instance, as they are longitudinal and possess no oscillations in the plane normal to the direction of energy transfer of the wave.

Proposition 11.3: Intensity of Polarised Light

The intensity I of a beam of emerging polarised light from a beam of previously unpolarised light of intensity I_0 is

$$I = \frac{1}{2} I_0$$

Proof. The vertical and horizontal components of the oscillations across all the incident waves in a beam of unpolarised light will be equal on average. Therefore, polarisation along any axis will statistically result in the intensity of the light beam being halved. □

Definition 11.15: Malus's Law

Malus's law states that when an ideal polariser is placed in a polarised beam of light, the intensity I_1 of the light that passes through is

$$I_1 = I_0 \cos^2 \theta$$

where I_0 is the initial intensity of the light and θ is the angle between the light's initial polarisation direction and the axis of the polariser.

Proof. Let I_0 be the initial intensity of a beam of polarised light, and I_1 be the resulting intensity after the beam is incident on a polariser tilted at an angle θ.

As established beforehand, intensity is proportional to the square of amplitude. For some arbitrary constant k,

$$I_0 \propto A^2 \implies I_0 = kA^2$$

When two polarisers are placed along axes varied by an angle θ, only the parallel component of amplitude $A\cos\theta$ is transmitted past the polariser. This gives, for the same arbitrary constant k,

$$I_1 \propto (A\cos\theta)^2 \implies I_1 = kA\cos\theta$$

Taking I_0 / I_1, we obtain the following result.

$$\frac{I_0}{I_1} = \frac{kA^2}{kA^2\cos^2\theta} \implies I_1 = I_0\cos^2\theta$$

□

12 Superposition

12.1 Superposition Principle

Definition 12.1: Superposition Principle

The **superposition principle** states that when two or more waves of the same type meet at a point at the same time, the displacement of the resultant wave is the vector sum of the displacements of the individual waves at that point in time.

Two waves must be of the same type in order for the principle of superposition to apply.

Definition 12.2: Coherence

Two sources are said to be **coherent** if the waves produced by the two sources have a constant phase difference.

Definition 12.3: Interference

Interference occurs when two or more waves overlap to produce a resultant wave pattern.

12.1.1 Constructive & Destructive Interference

Definition 12.4: Constructive & Destructive Interference

Constructive interference occurs when two waves meet in phase. The resultant amplitude is the sum of the individual amplitudes of the waves.

Destructive interference occurs when two waves meet in antiphase. The resultant amplitude is zero.

Constructive and destructive interference can only occur between two coherent waves.

12.1.2 Path Difference

The distance along any path from source to receiver is called the **path length** r. The **path difference** between two waves is simply the absolute difference between their path lengths.

For two waves initially in phase, constructive interference occurs if

$$\Delta r = n\lambda \text{ or } \Delta\phi = 2n\pi$$

Likewise, destructive interference occurs if

$$\Delta r = \left(n + \frac{1}{2}\right)\lambda \text{ or } \Delta\phi = (2n+1)\pi$$

If the two waves start in antiphase instead with $\phi = \pi/2$, the above formulae will swap between constructive and destructive interference.

12.1.3 Two-Source Interference

We explore the simple scenario where waves emanating from two sources undergoes continuous interference with each other. Certainly, not all sources generate waves that can observably interfere with each other.

For two-source interference patterns **to be observable**, the waves must

- be coherent;
- have the same amplitude;
- be unpolarised or polarised in the same plane, for transverse waves.

For an example of two-source interference, let us examine the behaviour of interlapping sound waves as they emanate from two identical speakers in Figure 12.1, placed some distance away from each other.

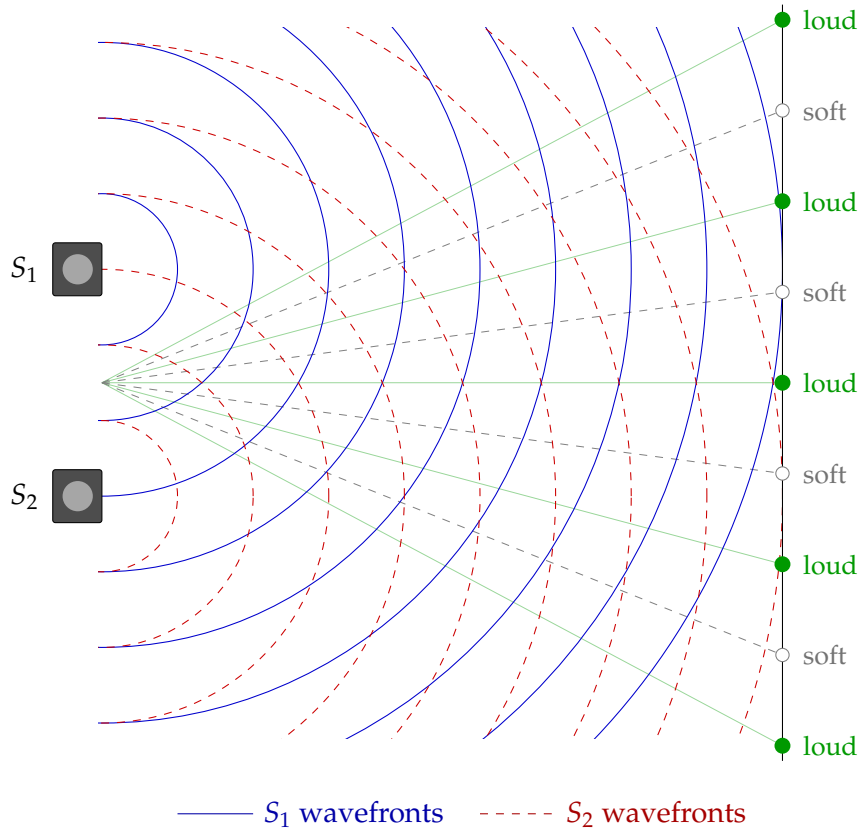


FIGURE 12.1. Two-source interference from speakers S_1 and S_2. Circular wavefronts overlap to produce alternating loud (constructive) and soft (destructive) fringes, marked by green and grey lines respectively.

We see that at the regions where the wavefronts intersect each other, the sound waves are in phase and therefore interfere constructively to produce a louder sound. The same logic applies for wavefronts that interfere destructively to produce softer sounds.

This is the essence of interference and superposition. We will revisit this two-source interference pattern later when we touch upon the topic of diffraction. For now, let us move on to superposed phenomena in waves.

12.2 Stationary Waves

Definition 12.5: Stationary Wave

A **stationary wave** (or **standing wave**) is formed when two progressive waves of the same type of equal amplitude, equal frequency, equal speed travelling in opposite directions meet and undergo superposition with each other.

On any stationary wave, there are two types of points of interest.

Definition 12.6: Node

A **node** is a point on a stationary wave vibrating with zero amplitude.

Definition 12.7: Antinode

An **antinode** is a point on a stationary wave vibrating with maximum amplitude.

Nodes occur at the point where the two constituent waves meet in antiphase, while antinodes occur when they meet in phase. The distance between adjacent nodes is $\lambda/2$.

In the following sections, we analyse the behaviour of stationary waves in strings and air columns, where they are subject to boundary conditions.

12.2.1 Stretched Strings

A common example of this boundary condition occurs in stretched strings with two fixed ends. When in such a position, the oscillatory pattern of the string can take on a discrete number of variations.

These discrete patterns are called **normal modes**, also termed as **harmonics**.

Proposition 12.1: Harmonics of Stretched Strings

Suppose a standing wave is formed over a stretched string of length L. Then, for the nth harmonic, the frequency of the standing wave is given by

$$\lambda_n = n\frac{v}{2L},\ n = 1,\ 2,\ 3,\ \cdots$$

where v is the speed of sound.

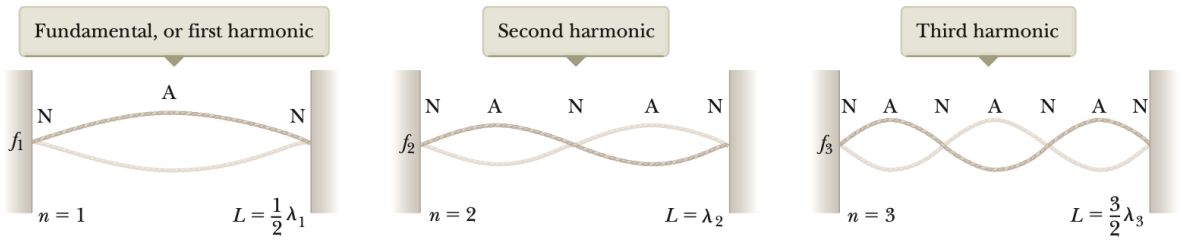


FIGURE 12.2. Harmonics of stationary waves in a stretched string.

12.2.2 Air Columns

Boundary conditions also present themselves when waves are created in air columns. A closed end of a pipe is always a displacement node.

Proposition 12.2: Harmonics of Air Columns

In a pipe closed at **one end**, the frequencies form a harmonic series that includes only **odd** integer multiples of the fundamental frequency.

$$\lambda_n = n\frac{v}{4L},\ n = 1,\ 3,\ 5,\ \cdots$$

In a pipe closed at **both ends**, the wavelengths form a harmonic series that includes all integer multiples of the fundamental frequency.

$$\lambda_n = n\frac{v}{4L},\ n = 1,\ 2,\ 3,\ \cdots$$

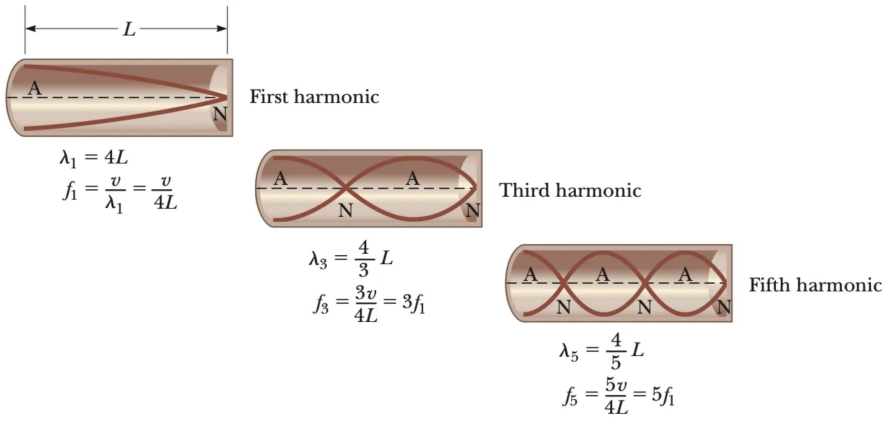


FIGURE 12.3. Harmonics of a closed-end air column.

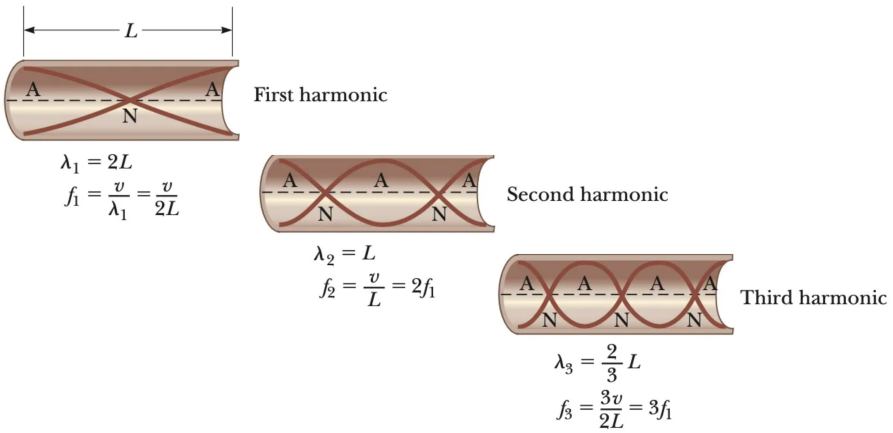


FIGURE 12.4. Harmonics of a open-end air column.

12.3 Diffraction

When light passes through an extremely narrow slit, commonly of a width comparable to the light beam's wavelength, the light spreads out in all directions upon passing through the slit.

Definition 12.8: Diffraction

Diffraction is the spreading of a wave as it passes through a gap or around an obstacle.

Diffraction is most significant when the width of the slit is of the same order of magnitude as the wavelength of the light passing through.

12.3.1 Single-Slit Diffraction

Previously, we have seen the maxima and minima fringes when it comes to regular two-source interference **without diffraction**. However, when diffraction enters the equation, fringes can appear even if there is only one source!

To demonstrate this, we imagine incoming light passing through a narrow slit of a width b, and projected onto a screen that is far away. What ends up appearing on the screen is a series of bright and dark fringes, termed a **Fraunhofer diffraction pattern**, as shown in Figure 12.5.

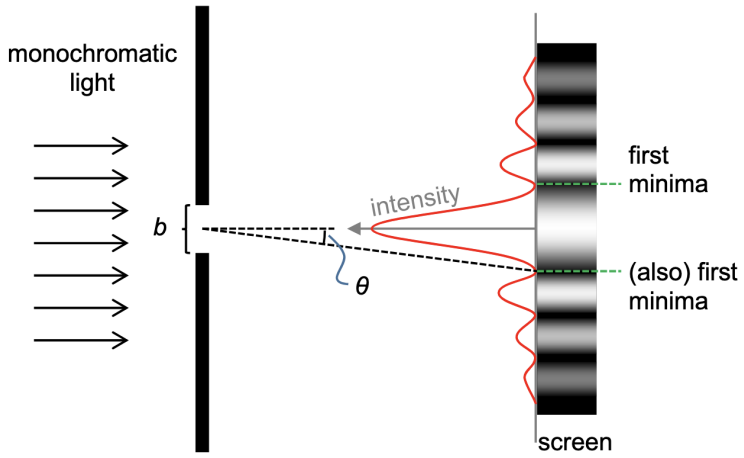


FIGURE 12.5. Resultant pattern of a light beam diffracting through a single slit.

The general reason why this occurs is due to the principle that each portion of the slit acts as its own source of light waves, causing multitudinous occurrences of interference that give way to the resultant diffraction pattern. In other words, the diffraction pattern is actually just a complicated interference pattern.

We now mathematically characterise the appearances of dark fringes in the single-slit Fraunhofer diffraction pattern.

Proposition 12.3: Minima of Single-Slit Diffraction

For a single-slit diffraction pattern caused by a slit with width b, dark fringes occur at angles θ from the center of the slit to the screen, as given by the following equation.

$$\sin\theta = n\frac{\lambda}{b},\ n \in \mathbb{Z}$$

Proof. Split the slit of width b into two halves. Note that all the light waves are in phase as they leave the slit.

Let us consider the behaviour of five numbered rays, as they travel at an angle θ with respect to the horizontal. Each of these waves are separated by a distance $b/4$.

Consider rays 1 and 3. As they travel towards and hit the faraway screen, ray 1 will have travelled an extra distance of $(b/2)\sin\theta$. This is also true for rays 2 and 4, and rays 3 and 5.

Recall that if the path difference between two waves of the same phase is exactly half a wavelength, they will undergo destructive interference, thereby leaving a dark fringe. This means that waves from the upper half of the slit will interfere destructively with those from the lower half of the slit when the following condition is true.

$$\Delta r = \frac{b}{2}\sin\theta = \frac{\lambda}{2}$$

Rearranging, we obtain

$$\sin\theta = \pm\frac{\lambda}{b}$$

When we divide the slit into four equal parts and utilise the same reasoning, we will obtain the following condition.

$$\sin\theta = \pm 2\frac{\lambda}{b}$$

It follows that the general condition for destructive interference is

$$\sin\theta = n\frac{\lambda}{b},\ n \in \mathbb{Z}$$

□

Take note that this equation fails with $n\lambda/b > 1$. For such values of n, no diffraction patterns can be observed.

12.3.2 Rayleigh Criterion

When two light sources placed far away from a slit project their images onto a screen, we expect to see two distinct images on the screen. However, there comes a certain point whereby if the two light sources are too close together, the images cannot be practically distinguised or resolved.

To determine whether two images are resolved, we utilise the following criterion.

Definition 12.9: Rayleigh Criterion

The Rayleigh criterion states that two patterns are just resolved when the central maximum of one patterns lies on the first minimum of the other pattern.

According to the previous section, the first minimum of the diffraction pattern from one source is given as

$$\sin\theta = \frac{\lambda}{b}$$

Using small-angle approximation, we can say that the limiting angle of resolution θ_{min} for a slit of width b is

$$\theta_{\text{min}} \approx \frac{\lambda}{b}$$

12.3.3 Double-Slit Diffraction

We now investigate the behaviour of light waves when we introduce a second slit. Recall that for double-slit diffraction patterns to occur, the light waves must be coherent and monochromatic. Coherence of light waves is often achieved by passing a light source through a single slit first, or by using a laser.

The resultant diffraction pattern will look like a regular two-source interference pattern, modulated by a single-slit diffraction pattern, as seen in Figure 12.6.

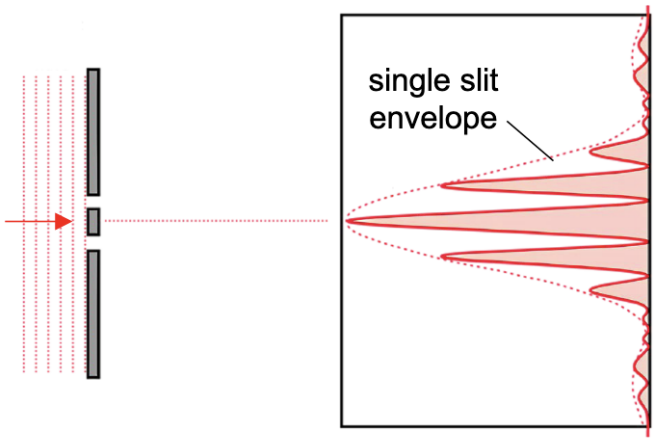


FIGURE 12.6. Resultant pattern of a light beam diffracting through two slits.

That said, just like the two-source interference pattern, separation between fringes will be a constant value. The relation between many variables involved in double-slit diffraction is summarised in the following.

Proposition 12.4: Fringe Separation in Double-Slit Diffraction

For a double-slit diffraction pattern with slit separation a, slit-to-screen distance L, and wavelength λ, the separation x between two fringes of the same type is given as

$$x = \frac{L\lambda}{a}$$

12.4 Diffraction Grating

Definition 12.10: Diffraction Grating

A **diffraction grating** consists of a large number of equally-spaced lines, each of which are capable of diffracting incident light.

In a diffraction grating with slit separation d, for the nth order maximum, the path difference is given by

$$d \sin \theta = n\lambda$$

Diffraction gratings are usually specified by the number of lines N per unit length.

$$d = \frac{\text{unit length}}{N}$$

Part V

Electricity and Magnetism

13 Electric Fields

I sing the body electric,
The armies of those I love engirth me and I engirth them,
They will not let me off till I go with them, respond to them,
Discorrupt them, charge them full with the charge of the soul.

— WALT WHITMAN, *Leaves of Grass*

13.1 Electric Charge

The electric force exists between objects with **electric charge**, commonly represented by Q. Charge has S.I. unit couloumb (C), which is equivalent to an ampere-second.

Charge comes in two varieties. There are **positive charges** due to the accumulation of protons, as well as **negative charges** due to that of electrons. Charges of the same sign repel one another, whereas charges of opposite signs attract one another.

The elementary charge $e = 1.60 \times 10^{-19}$ C is a constant describing the amount of charge a proton possesses. A proton has charge e, and an electron has charge $-e$.

13.2 Electric Force

Definition 13.1: Coulomb's Law

Coulomb's law states that the electric force between two point charges is directly proportional to the product of the charges, and inversely proportional to the square of the distance between them.

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r^2}$$

where q_1 and q_2 are the charges, r is their separation, and ε_0 is the permittivity of free space.

It may be easier to remember that $1/(4\pi\varepsilon_0) = 9.0 \times 10^9\ \mathrm{N\,m^2\,C^{-2}}$.

The direction of the force is along the line joining the two point charges. This relation is **only applicable** if

- the charges are point charges; or
- if the two charged bodies are separated far apart such that the size of each charged body is negligible compared to their separation.

13.3 Electric Field

Definition 13.2: Electric Field

An electric field is a region of space within which there is an electric force acting on a charged object placed in that field.

The direction of the electric field is the direction that an electric force would act upon a **positive** point charge.

13.3.1 Electric Field Strength

Definition 13.3: Electric Field Strength

The electric field strength E at a point in the field created by a charge Q is defined as the electric force per unit positive charge experienced by a charge placed at that point.

$$E = \frac{F}{q} = \frac{Q}{4\pi\varepsilon_o r^2}$$

13.3.2 Electric Field Lines

Just like a gravitational field, an electric field may be represented by electric field lines, which indicate the direction of the force a stationary positive point charge would experience if placed at that point.

Electric field lines must either originate on positive charges or come in from infinity, and either terminate on negative charges or extend out to infinity. Furthermore, the number of lines touching a charge must be proportional to the magnitude of the charge.

For an isolated positive charge, its field lines point radially outward. Conversely, for an isolated negative charge, its field lines point radially inward. We show this in Figure 13.1.

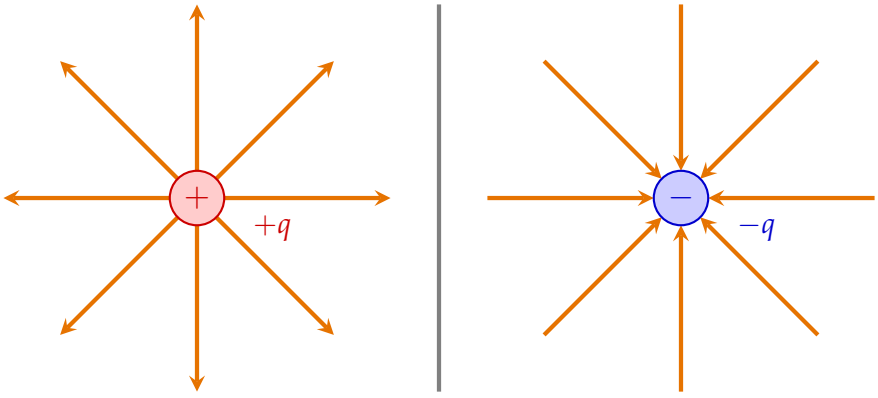


FIGURE 13.1. Electric field lines for isolated positive and negative charges, directed radially outward and inward respectively.

For two charges of equal magnitude, they cancel out. The number of field lines leaving the positive charge must be equal to the number of field lines terminating at the negative charge.

Similarly, for a positive charge $+2q$ and a negative charge $-q$, two field lines must leave the positive charge for every line that terminates on the negative charge.

13.3.3 Charged Parallel Plates

Consider a pair of charged parallel plates separated by a distance d with potential difference ΔV, as shown below.

The uniform electric field E between both plates has a field strength with magnitude

$$E = \frac{|\Delta V|}{d} \implies F = \frac{q|\Delta V|}{d}$$

Note that the direction of the electric field points from the plate with the higher electric potential to the plate with the lower electric potential.

13.4 Motion of Charged Particles

In a uniform electric field, the force on a charged particle is constant. The particle will consequently have a constant acceleration a as well. If the charged particle has a mass m, we have

$$\sum F = ma \implies qE = ma \implies \frac{q|\Delta V|}{d} = ma$$
$$\implies a = \frac{q|\Delta V|}{md}$$

Depending on the relative direction of the electric field to the particle's motion, the particle could either experience linear motion or projectile motion. In any case, its motion can be modelled using the kinematic equations of motion.

13.5 Electric Potential Energy

Definition 13.4: Electric Potential Energy

The electric potential energy U at a point is the work done in bringing a small positive test charge from infinity to that point.

Proposition 13.1: Electric Potential Energy of a Charge

For a system of two charges Q and q separated by a distance r, the electric potential energy of q due to Q is given by

$$U = \frac{1}{4\pi\varepsilon_0}\frac{Qq}{r}$$

In the general case, for a **system of several charges** containing $Q_1, Q_2, \ldots, Q_n$ located a distance $r_1, r_2, \ldots, r_n$ respectively from a charge q, the electric potential energy of q due to the system is given by

$$U = \frac{q}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{Q_i}{r_i}$$

Proof. Recall that for a conservative force,

$$F = -\frac{\mathrm{d}U}{\mathrm{d}r}$$

Since the electric force is a conservative force, we can use this relation. From the definition of electric potential energy at a point r away from a charge Q,

$$U = -\int_{\infty}^{r} F_e \, \mathrm{d}r = -\frac{1}{4\pi\varepsilon_0}\int_{\infty}^{r} \frac{Qq}{r^2} \, \mathrm{d}r = \frac{1}{4\pi\varepsilon_0}\frac{Qq}{r}$$

For a system of several charges containing $Q_1, Q_2, \ldots, Q_n$, the total electric field at each point is the vector sum of the fields due to the individual charges, and the total work done on a single point charge q during any displacement is the sum of the contributions from the individual charges.

$$U = \frac{q}{4\pi\varepsilon_0}\left(\frac{Q_1}{r_1} + \frac{Q_2}{r_2} + \cdots + \frac{Q_n}{r_n}\right) = \frac{q}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{Q_i}{r_i}$$

□

Proposition 13.2: Electric Potential Energy of a System

As for a system of several charges $Q_1, Q_2, \ldots, Q_n$, we can find the total electric potential energy by calculating U for every pair of charges. The generalised equation for the total electric potential energy of a system is

$$\sum U = \frac{1}{4\pi\varepsilon_0}\sum_{i<j}^{n}\frac{Q_iQ_j}{r_{ij}}$$

where r_{ij} is the separation between charges Q_i and Q_j.

13.6 Electric Potential

Definition 13.5: Electric Potential

The electrical potential V at a point is the work done per unit positive charge by an external agent to bring a small test charge q from infinity to that point, without producing any acceleration.

$$V = \frac{W}{q} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r}$$

where V is the electric potential at a distance r from the centre of Q.

Electric potential in electric fields is sometimes represented by equipotential lines, indicated by dashed lines. No work is done by an external force when a charge is moved along an equipotential line.

Proposition 13.3: Electric Potential Gradient

The electric field strength E can be said to be numerically equal but negative of the gradient of electric potential V with respect to r.

$$E = -\frac{\mathrm{d}V}{\mathrm{d}r}$$

Proof. Taking the derivative of V,

$$\frac{1}{4\pi\varepsilon_0}\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{Q}{r}\right) = -\frac{1}{4\pi\varepsilon_0}\frac{Q}{r^2}$$

Substituting $E = Q/(4\pi\varepsilon_0 r^2)$,

$$E = -\frac{\mathrm{d}V}{\mathrm{d}r}$$

□

The negative sign in the equation indicates the direction of the electric field, which points towards decreasing electric potential.

14 Current of Electricity

14.1 Electric Current

Definition 14.1: Electric Current

Electric current is the rate of flow of charge. It has the S.I. unit of ampere (A).

$$I = \frac{\mathrm{d}Q}{\mathrm{d}t} \implies \langle I \rangle = \frac{Q}{t}$$

Electric currents consist of charges in motion. When this motion takes place within a closed loop, the loop is termed an electric circuit.

In metallic conductors, current is caused by the flow of electrons. However, we often indicate the conventional current in circuitry diagrams, which is directed opposite to the flow of electrons and is meant to indicate the movement of 'positive charges'.

We can further improve our formulation of the electric current by considering a microscopic model of how electrons flow within metallic conductors.

Proposition 14.1: Microscopic Models of Current

Current is the rate of flow of charge, and is thus the number of charge carriers passing through per unit time multiplied by the charge per carrier.

$$I = nAvq$$

where n is the charge carrier density, A is the cross-sectional area of the conductor, v is the drift velocity, and q is the charge per carrier.

Proof. For a region of length l and cross-sectional area A, its total volume is $V = Al$. The total charge within this region is given by

$$Q = n \times q \times (Al)$$

Since current is the rate of flow of charge,

$$I = \frac{Q}{t} = \frac{nAql}{t} = nAvq$$

□

Some clarifications are in order. The term 'charge carrier' simply means some particle flowing through the circuit that possesses a charge. Most commonly, this refers to electrons.

The drift velocity of the charge carriers is due to the potential difference applied across the conductor. This allows them to travel at around $\approx 10^{-4}\ \mathrm{m\,s^{-1}}$, causing charge to be transferred.

14.2 Voltage

14.2.1 Potential Difference

Definition 14.2: Potential Difference

The potential difference or voltage V between two points in a circuit is defined as the work done W per unit charge Q to convert electrical energy into non-electrical energy when the charge passes from one point to the other.

$$V = \frac{W}{Q}$$

Potential difference applies to points within the circuit, where electrical energy is naturally converted into non-electrical energy.

14.2.2 Electromotive Force

Definition 14.3: Electromotive Force

The electromotive force or e.m.f. ε of a source can be defined as the work done W per unit charge Q when non-electrical energy is converted into electrical energy when the charge is moved around a complete circuit.

$$\varepsilon = \frac{W}{Q}$$

In order to maintain an electric current in an electrical circuit, a potential difference across the ends of the entire circuit is needed. To achieve this, the ends of the electrical circuit are connected to opposite terminals of a battery.

This potential difference maintained by the battery is the electromotive force. Unlike potential difference within a circuit, the electromotive force of a battery exists regardless of whether a current is flowing.

14.3 Resistance

Definition 14.4: Resistance

The resistance R of a component is defined as the ratio of the potential difference V across the component to the current I passing through it.

$$R = \frac{V}{I}$$

The resistance of a conductor is a measure of its opposition of flow of current through itself. It is an intrinsic property of a conductor.

14.3.1 Ohm's Law

Definition 14.5: Ohm's Law

Ohm's Law states that the current flowing through a conductor is directly proportional to the potential difference applied across the conductor, provided that physical conditions remain constant.

14.3.2 Current-Voltage Characteristics

The current-voltage graph of a material, obtained experimentally, allows the resistance of the material to be determined for various values of currents or voltages.

When the current passes through a material increases, the temperature of the material is likely to increase. When the latter occurs, two main changes occur.

1. The charge concentration n increases, which **reduces** the resistance of the material.
2. The thermal vibrations of the lattice atoms increases, which **increases** the resistance of the material.

Ohmic Conductor

If a conductor is maintained at a constant temperature, the charge concentration and the amplitude of atomic vibration remains unchanged.

Thus, the resistance remains a constant and the conductor is said to be ohmic. The current-voltage graph for an ohmic conductor is shown in Figure 14.1.

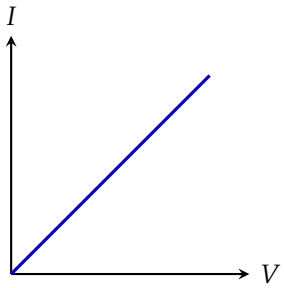


FIGURE 14.1. Current-voltage graph of an ohmic conductor.

Filament Lamp

A filament lamp heats up as time goes on. As a result, physical conditions are not kept constant and Ohm's law fails to apply.

As the temperature increases, charge concentration will not vary significantly, but the amplitude of atomic vibrations increases at a greater degree. Since the increase in atomic vibrations predominates over the increase in charge concentration, the overall effect is that resistance increases.

The resulting graph is shown in Figure 14.2, where large increases in voltage causes a less-than-proportionate increase in the current.

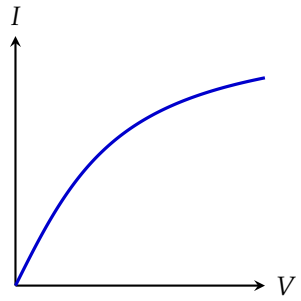


FIGURE 14.2. Current-voltage graph of a filament lamp. It starts out as a straight line, before waning in accordance with the increasing resistance.

Semiconductor Diode

A semiconductor diode is an electric component that allows electric current in one direction only, bearing low resistance in that direction and high resistance in the other. A current-voltage graph of a diode is shown in Figure 14.3.

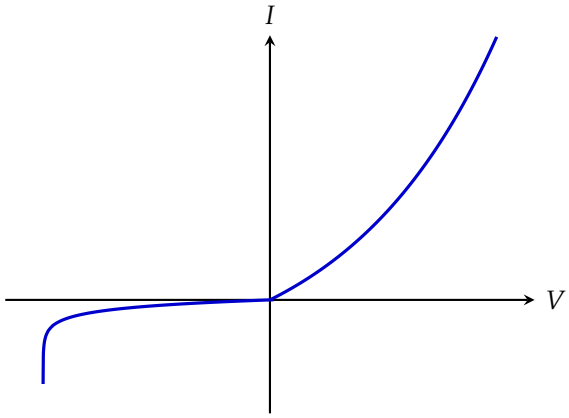


FIGURE 14.3. Current-voltage graph of a semiconductor diode.

Considering the region of low resistance, as voltage increases, the temperature of the diode increases. Electrons in the semiconductor material are more likely to have sufficient energy to escape from a particular atom if the temperature is higher, hence the charge concentration increases significantly. The rate of interaction of electrons with the vibrating atoms also increases. However, as the increase in charge concentration predominates over the increase in the rate of interaction, the overall effect is that resistance decreases.

If the forward voltage is greater than a threshold value, a large current will pass through the diode. For the reverse direction, no current flows through the diode until the breakdown voltage.

Thermistor

A thermistor is an electronic component with a resistance primarily dependent on its temperature. In particular, we analyse negative temperature coefficient (or NTC) thermistors where resistance is **inversely proportional** to temperature.

As the thermistor's temperature increases, the charge concentration of the thermistor increase, hence reducing the resistance. As such, its current-voltage characteristics is quite similar to that of a semiconductor diode, as seen in Figure 14.4.

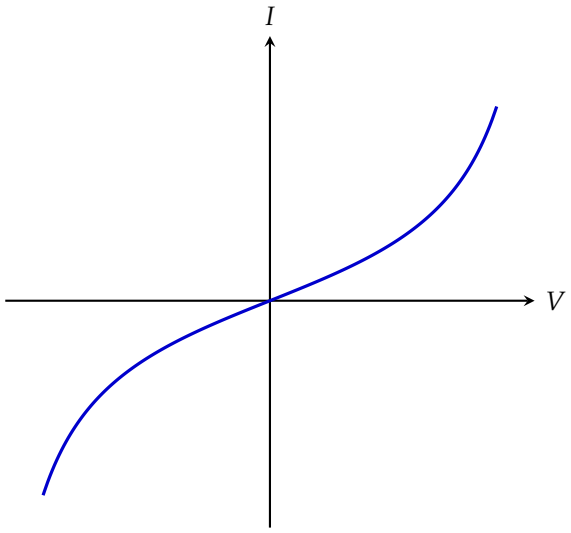


FIGURE 14.4. Current-voltage graph of an NTC thermistor.

14.3.3 Resistivity

Definition 14.6: Resistivity

For a conductor of resistivity ρ, length l, and cross-sectional area A, its resistance is given by

$$R = \rho \frac{l}{A}$$

Resistivity is mainly governed by the nature of the conductor's material, excepting some dependence on temperature. Metals generally have low resistivity values of around $10^{-8}\,\Omega\,\mathrm{m}$, whereas insulators have high resistivity values of around 10^{13} to $10^{16}\,\Omega\,\mathrm{m}$.

14.4 Electrical Power

Definition 14.7: Electrical Power

Electrical power P is the electrical energy dissipated by a conductor per unit time.

Proposition 14.2: Watt's Law

Watt's law provides the following relationships between power P, voltage V, and current I.

$$P = VI = I^2R = \frac{V^2}{R}$$

Proof. Electrical power P is defined as electrical energy dissipated per unit time. In other words, it is the amount of electrical energy converted into non-electrical energy per unit time.

$$P = \frac{W}{t}$$

Bringing in $V = W/Q$ and $I = Q/t$,

$$P = \frac{W}{Q} \times \frac{Q}{t} = IV$$

Since $V = IR$, we obtain the other two forms as well.

$$\begin{aligned} P = IV &\implies P = I(IR) = I^2R \\ &\implies P = \left(\frac{V}{R}\right) V = \frac{V^2}{R} \end{aligned}$$

□

Often times, devices will have a **power rating** labelled on it. The power rating is the rate at which energy is consumed by a device when the device is operating at the rated potential difference across it.

For instance, a device rated at 100 W and 220 V means that the device dissipates 100 W of power when a potential difference of 220 V is applied across its terminals.

14.5 Internal Resistance

In an ideal circuit, the potential difference V across the cell terminals is always equal to the electromotive force ε of the cell.

In practice, real battery cells always have some **internal resistance**. As such, the potential difference that is available to the other components within the circuit is often less than the electromotive force of the cell.

A real battery can be modelled as one ideal cell or a combination of ideal cells connected in series to a resistor r. This is demonstrated in Figure 14.5.

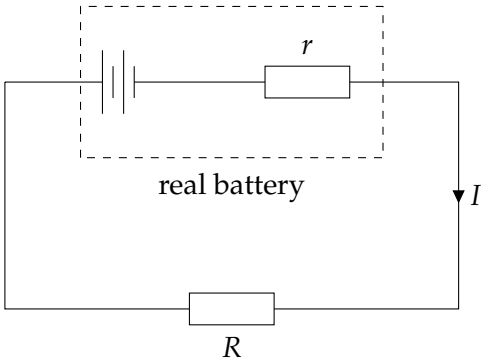


FIGURE 14.5. A battery possessing internal resistance may be modelled as an ideal cell coupled with a resistor, as shown by the dashed region.

Proposition 14.3: Terminal Potential Difference

The terminal potential difference V of a circuit is the potential difference made available to the rest of the circuit, and is given as

$$V = \varepsilon - Ir$$

where ε is the e.m.f. of the cell, I is the current flowing through the circuit, and r is the internal resistance of the cell.

Proof. Due to the conservation of energy, the sum of the power dissipated through the cell and through the rest of the circuit must be equal to the total power supplied to the circuit.

$$\begin{aligned} P_s = P_r + P_R &\implies I\varepsilon = I^2 r + I^2 R \\ &\implies \varepsilon = Ir + IR = Ir + V \\ &\implies V = \varepsilon - Ir \end{aligned}$$

□

14.5.1 Maximum Power Theorem

Definition 14.8: Maximum Power Theorem

The maximum power theorem states that maximum power is supplied to the external circuit when the resistance of the external circuit is equal to the internal resistance of the cell.

Proof. Consider a practical source with e.m.f. E and internal resistance r connected in series with an electrical device of resistance R.

The current passing through the circuit is given by

$$I = \frac{V}{R} = \frac{E}{R+r}$$

Thus, the power delivered to the device is

$$P = I^2 R = \left(\frac{E}{R+r}\right)^2 R$$

To find maximum P, we let the derivative be zero.

$$\begin{aligned} \frac{\mathrm{d}P}{\mathrm{d}R} = 0 &\implies \frac{E^2(R+r)^2 - 2E^2R(R+r)}{(R+r)^4} = 0 \\ &\implies E^2(R+r)^2 - 2E^2R(R+r) = 0 \\ &\implies R + r = 2R \\ &\implies \boxed{R = r} \end{aligned}$$

Hence, maximum power is dissipated through the circuit when the resistance of the external circuit is equal to the internal resistance of the cell. □

When operating an electrical device at maximum power, the output efficiency η of the source is

$$\eta = \frac{\text{useful energy}}{\text{total energy}} = \frac{I^2R}{I^2(R+r)} = \frac{R}{R+r} = \frac{R}{2R} = 50\%$$

Half the amount of power it generates is lost due to the internal resistance of the cell.

15 DC Circuits

15.1 Components

In dealing with circuits, we must familiarise ourselves with various electrical components. These include the **ammeter**, which measures current, and the **voltmeter**, which measures potential difference.

Other previously-discussed devices like cells, switches, resistors, and filament lamps are also considered electrical components. The following figure shows some common circuit symbols that pertain to the scope of our discussion.

Ammeters are connected in series, whereas voltmeters are connected in parallel. Ideal ammeters have zero internal resistance and ideal voltmeters have infinite internal resistance.

15.2 Kirchhoff's Laws

Kirchhoff's circuit laws are two laws that pertain to electrical current and potential difference, and are crucial in the field of circuit analysis.

15.2.1 Kirchhoff's Current Law

Definition 15.1: Kirchhoff's Current Law

Kirchhoff's current law states that the sum of currents $I_1, I_2, \ldots, I_n$ entering any junction in an electric circuit is always equal to the sum of current leaving that junction.

$$\sum I_{\text{in}} = \sum I_{\text{out}} \text{ (for a junction)}$$

Kirchhoff's current law is a direct result of the principle of conservation of charge. The total charge that enters a junction per unit time must be equal to the total charge that leaves the same junction per unit time.

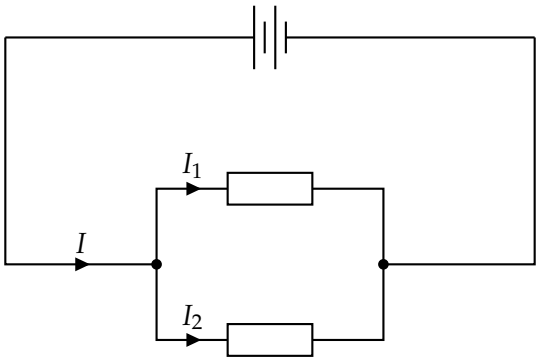


FIGURE 15.1. A simple circuit of two resistors arranged in parallel. As per Kirchhoff's current law, the three currents are related by $I = I_1 + I_2$.

15.2.2 Kirchhoff's Voltage Law

Definition 15.2: Kirchhoff's Voltage Law

Kirchhoff's voltage law states that in any closed loop in an electric circuit, the total electromotive force supplied ε equals to the total potential difference in that loop.

$$\varepsilon = \sum_{i=1}^{n} V_i \text{ (for a loop)}$$

This comes as a result of the principle of conservation of energy, which implies that the electrical energy produced by the source should be equal to the sum of electrical energy consumed by all the components.

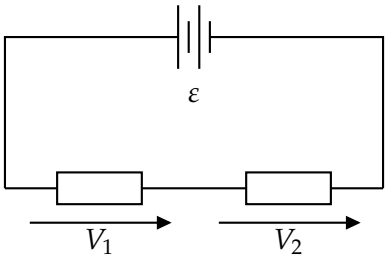


FIGURE 15.2. A simple circuit of two resistors arranged in series. As per Kirchhoff's voltage law, $\varepsilon = V_1 + V_2$.

15.3 Effective Resistance

The effective resistance of a circuit refers to the resistance of a single resistor that would draw the same current from a cell of the same electromotive force as the existing circuit.

For resistors connected in series,

$$R_{\text{eff}} = R_1 + R_2 + \cdots$$

The effective resistance of resistors in series is always larger than the largest resistor in the circuit.

For resistors connected in parallel,

$$R_{\text{eff}} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \cdots \right)^{-1}$$

The effective resistance of resistors in parallel is always smaller than the smallest resistor in the circuit.

15.4 Potential Divider

A potential divider is an arrangement of resistors that is used to obtain a fraction of the potential difference provided by a cell.

We demonstrate the concept of a potential divider by considering the circuit in Figure 15.3, which consists of two resistors of known resistances R_1 and R_2, connected to a cell with e.m.f. ε.

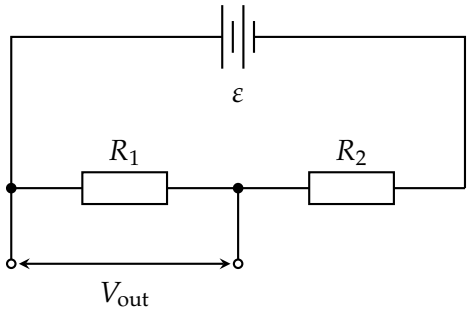


FIGURE 15.3. A simple set-up of a potential divider. The potential difference V_{out} is produced, and often connected to an external load.

The potential difference that we obtain is denoted here as V_{out}. In our simple two-resistor circuit, the potential difference is given as follows.

$$V_{\text{out}} = IR = \left(\frac{\epsilon}{R_{\text{eff}}}\right) R_1 = \left(\frac{E}{R_1 + R_2}\right) R_1$$

To create a potential divider that offers a continuously variable potential difference, the two resistors can be replaced with a singular rheostat.

To offer some easy clarifications on electric potential in D.C. circuits,

- two points have zero potential difference if they are connected directly by a zero-resistance wire;
- two points have a non-zero potential difference if they are separated by a component with resistance, unless there is no current passing through them.

15.5 Potentiometers

Real voltmeters do not have infinite internal resistance, causing small currents to flow through it, thus skewing measurements of voltage. A **potentiometer** can be used instead to compare voltages without the need for a current.

Let us first talk through the concept behind a generalised potentiometer, which is shown in **??**.

The circuit consists of an ideal cell with e.m.f. ε, a resistor R, and a resistance wire of length AB. Connected to the circuit is another circuit with unknown potential difference, a galvanometer, and a sliding contact placed along the resistance wire.

As the contact is slid across the wire, there will be a point where there is no current passing through the galvanometer, as the voltages of the primary circuit and the unknown circuit cancel each other out. This point is termed the **balance point**.

Denote the balance point as C. The length AC is then the **balance length**. We can say that the voltage across segment AC is equal to the voltage of the unknown circuit.

To find this voltage, we turn our attention to the resistance wire. We assume that the wire has a constant resistance per unit length. Then, we have

$$V_{\text{AC}} = \frac{R_{\text{AC}}}{R_{\text{AB}}} \times V_{\text{AB}} = \frac{L_{\text{AC}}}{L_{\text{AB}}} \times V_{\text{AB}} = \frac{V_{\text{AB}}}{L_{\text{AB}}} \times L_{\text{AC}}$$

The constant $k = V_{AB}/L_{AB}$ is known as the **potential per unit length** of the resistance wire. This process is the **potentiometer principle**.

15.5.1 Comparing Electromotive Force

Consider **??**, where we are comparing the balance points of two cells of unknown electromotive force. We wish to find the ratio of ε_1 to ε_2.

The first and second cells give a balance length of l_1 and l_2 respectively. From the previous section, we know that $\varepsilon_1 = kl_1$ and $\varepsilon_2 = kl_2$. This gives

$$\frac{E_1}{E_2} = \frac{l_1}{l_2} \implies E_1 = \frac{l_1}{l_2} E_2$$

15.5.2 Measuring Electromotive Force

Consider the setup as shown in **??**. We want to measure the electromotive force of the unknown cell.

A potentiometer has a resistance wire AB of length L and resistance R, powered by a cell of e.m.f. E, alongside a resistor of resistance R_s.

With an unknown e.m.f. E_1 in the branch circuit, the balance point is C.

The p.d. across wire AB is

$$V_{AB} = IR = \frac{R}{R + R_s} \times E = \frac{RE}{R + R_s}$$

Using the potentiometer principle,

$$E_1 = \frac{L_{AC}}{L_{AB}} \times V_{AB} = \frac{L_{AC}}{L} \left(\frac{RE}{R + R_s} \right)$$

15.5.3 Measuring Internal Resistance

To determine the internal resistance of a cell, consider the circuit in **??**.

Initially, switch S is **open**, meaning that there is no current passing through R_2. The balance point is C with balance length l_1.

From this alone, because there is **no current** passing through R, we can find the electromotive force ε of the unknown cell with ease.

$$E_1 = \frac{L_{AC}}{L_{AB}} \times V_{AB} = \frac{L_1}{L} \left(\frac{R_{AB} E}{R_{AB} + R_1} \right)$$

Now, we **close** switch S. This allows current to pass through R_2, creating a new balance point D with balance length l_2.

Since there is now current flowing through the bottom circuit, we can find the terminal potential difference V across the unknown cell.

$$V = \frac{L_2}{L} \left(\frac{R_{AB} E}{R_{AB} + R_1} \right)$$

The terminal potential difference V across the unknown cell will also be equal to the potential difference across R_2. Using Watt's law, we can find the current flowing through R_2.

$$V = IR \implies I = \frac{V_T}{R_2}$$

Using the equation $V = \varepsilon - Ir$, we can find the internal resistance r.

$$V = \varepsilon - Ir \implies r = \frac{\varepsilon - V}{I}$$

16 Electromagnetism

The world may be utterly crazy
And life may be labour in vain;
But I'd rather be silly than lazy,
and would not quit life for its pain.

— JAMES CLERK MAXWELL

16.1 Magnetic Fields

Definition 16.1: Magnetic Field

A **magnetic field** is a region of space within which a magnetic pole, a current-carrying conductor, or a moving charged particle may experience a force.

Magnetic fields can be visualised using magnetic field lines, which are similar to electric and gravitational field lines. A unique characteristic of magnetic fields is that a magnetic field line should come out of north poles and go into south poles.

16.1.1 Magnetic Flux Density

We now introduce a quantity that describes the strength of a magnetic field.

Definition 16.2: Magnetic Flux Density

The **magnetic flux density** $\vec{B}$ of a magnetic field is defined as the magnetic force F per unit length L per unit current I acting on a straight conductor placed normal to the field.

$$B = \frac{F}{IL\sin\theta}$$

The S.I. unit of magnetic flux density is the tesla (T).

16.2 Source of Magnetic Fields

16.2.1 Directional Hand Rules

Currents and moving charged particles are sources of magnetic fields. However, some visualisation is needed to determine the specific direction of the resultant magnetic fields.

Here, we show two simple rules that help with finding the direction of magnetic fields.

Ampere's Right-Hand Grip Rule

Ampere's right-hand grip rule provides a visualisation for the geometrical relationship between magnetic fields and current flow.

As seen in Figure 16.1, when gripping your right-hand around an imaginary straight wire with the thumb pointing towards the direction of current, the other four fingers provide the direction of the magnetic field. For coils or solenoids, when the four fingers are curled in the direction of the current, the thumb gives the direction of the magnetic field.

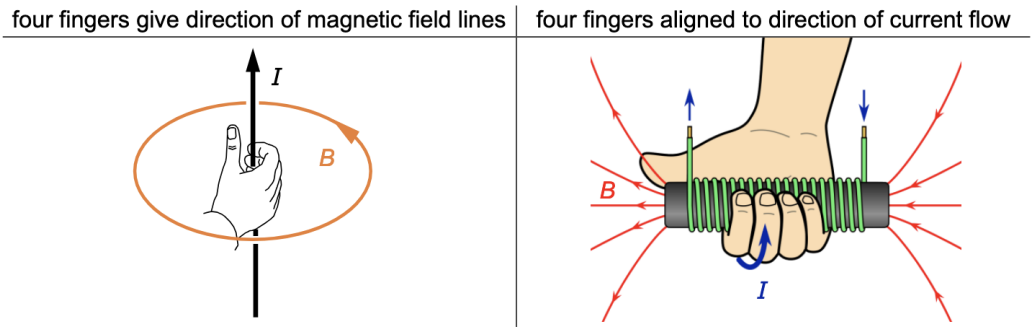


FIGURE 16.1. Two uses of Ampere's right-hand grip rule. For straight wires, the four curled fingers gives the direction of the magnetic field. In coiled wires, the thumb gives the direction of the magnetic field instead.

This rule is mostly used for straight wires, flat circular coils, and solenoids.

Fleming's Left-Hand Rule

Current-carrying conductors and moving charged particles in a magnetic field will experience a magnetic force. This force is always perpendicular to the direction of the current and the direction of the magnetic field.

Fleming's left-hand rule provides a handy visualisation for the directions of all three quantities, as seen in Figure 16.2.

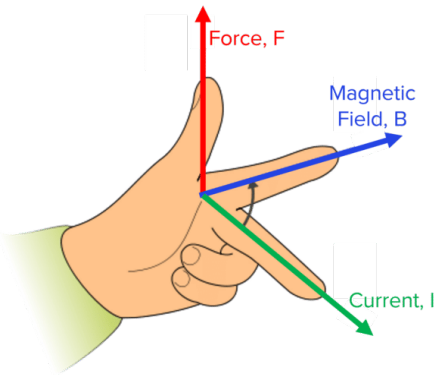


FIGURE 16.2. When the left hand is held in the orientation shown above, the thumb, index finger, and middle finger represent the force, magnetic field, and current respectively.

16.2.2 Coil Geometries

Closed-form equations for magnetic flux density can be difficult and complicated to define at times. However, there are three common geometries that have simple formulations for their magnetic fields, as shown below.

Proposition 16.1: Magnetic Field of Long Straight Wire

The magnetic flux density of a current-carrying long straight wire is given by

$$B = \frac{\mu_0 I}{2\pi d}$$

where I is the current through the wire and d is the perpendicular distance of the point in question from the wire.

Proposition 16.2: Magnetic Field of Flat Circular Coil

The magnetic flux density of a flat circular coil with N coil loops is given as

$$B = \frac{\mu_0 N I}{2r}$$

where I is the current through the coil and r is the radius of the coil.

Proposition 16.3: Magnetic Field of Solenoid

The magnetic flux density of a long solenoid is given by

$$B = \mu_0 n I$$

where I is the current through the solenoid and $n = N/L$ is the number of turns per unit length of the solenoid.

16.3 Magnetic Force

Definition 16.3: Magnetic Force

The magnetic force $\vec{F}$ acting on a current-carrying conductor with current I and length $\vec{L}$ placed at an angle θ to the magnetic field of flux density $\vec{B}$ can be expressed as

$$\vec{F} = \vec{B} \times I\vec{L} \implies F = BIL\sin\theta$$

where θ is the angle between the magnetic field and the current.

We must stress that the magnetic force is acting on the **electrons** flowing through the conductor. When the magnetic force is perpendicular to the direction of electron flow, the force on these electrons is transmitted to the wire upon collision.

The equation for the magnetic force acting on a charged particle is slightly different.

Proposition 16.4: Magnetic Force on Charged Particle

The magnetic force acting on a moving charged particle of charge q and speed v is given by

$$\vec{F} = \vec{B} \times q\vec{v} \implies F = Bqv\sin\theta$$

where θ is the angle between the magnetic field and the velocity of the particle.

Note that when using Fleming's left-hand rule for moving charged particles, the middle finger should point in the direction of motion of **positive charges**. This means that for negatively-charged particles, the middle finger should actually point in the opposite direction of the particle's movement.

16.3.1 Parallel Current-Carrying Conductors

If two long wires run parallel to, and near each other,

- if both their currents flow in the **same direction**, the wires will **attract** each other;
- if their currents flow in **opposite directions**, the wires will **repel** each other.

16.3.2 Motion of Charged Particles

When a charged particle is projected at a right angle into a magnetic field, notice that the magnetic force F must always be perpendicular to the particle's velocity v.

As the distance travelled in the direction of the force is zero, magnetic force does not do work on a charged particle.

Furthermore, since F is of constant magnitude and is always perpendicular to v, the charged particle undergoes circular motion, as seen in Figure 16.3.

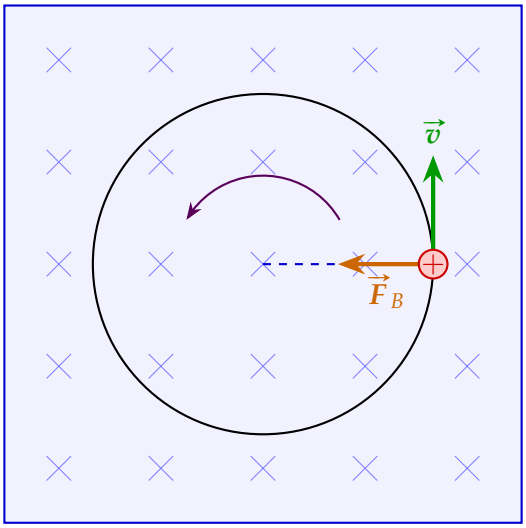


FIGURE 16.3. A positive charge moving with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ directed into the page. The magnetic force $\vec{F}_B$ always acts perpendicularly to the charge's motion, thus pulling it into circular motion.

Now, we will analyse the Newtonian mechanics behind the circular motion caused by magnetic forces. By Newton's second law,

$$\sum \vec{F} = \vec{F}_B = m\vec{a}$$

Since the particle undergoes circular motion, we can treat the magnetic force as a centripetal force providing a corresponding centripetal acceleration.

$$\vec{F}_B = Bq\vec{v} = \frac{m\vec{v}^2}{r}$$

From this equation, we can obtain various quantities.

$$r = \frac{mv}{qB} \implies \omega = \frac{qB}{m} \implies T = \frac{2\pi m}{qB}$$

16.3.3 Velocity Selector

A crossed-field **velocity selector** consists of a magnetic field and an electric field applied over the same region. This results in all undeflected particles passing through the region to have the same velocity. A schematic is shown in Figure 16.4.

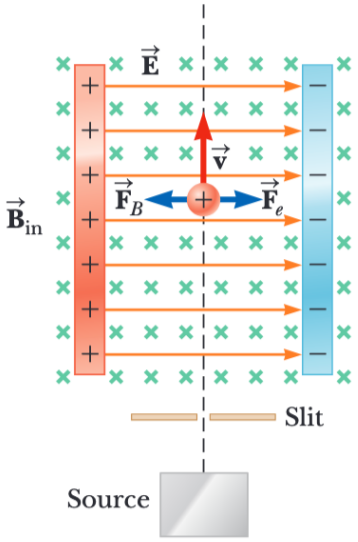


FIGURE 16.4. A particle passing through a velocity selector, experiencing a magnetic force and electric force in opposite directions.

As a charged particle enters the region, it will experience an electric force $\vec{F}_E$ and a magnetic force $\vec{F}_B$ simultaneously.

In order for the particle to not be deflected, it must be in equilibrium. In other words, $\vec{F}_E = \vec{F}_B$.

$$\vec{F}_E = \vec{F}_B \implies Bqv = qE$$

$$\implies v = \frac{E}{B}$$

Only particles having the speed $v = E/B$ will pass undeflected through the crossed-field region.

- If particles move with higher velocity v, the magnetic force becomes stronger than the electric force, causing leftwards deflection.

- If particles move with lower velocity v, the magnetic force becomes weaker than the electric force, causing rightwards deflection.

16.3.4 Mass Spectrometer

Some questions may include a **mass spectrometer**, where charged particles are first sent through a velocity selector, before being shot out into a region of uniform magnetic field that causes it to move in a semicircular path. This allows us to determine the mass per unit charge of any particle. A set-up is shown in Figure 16.5.

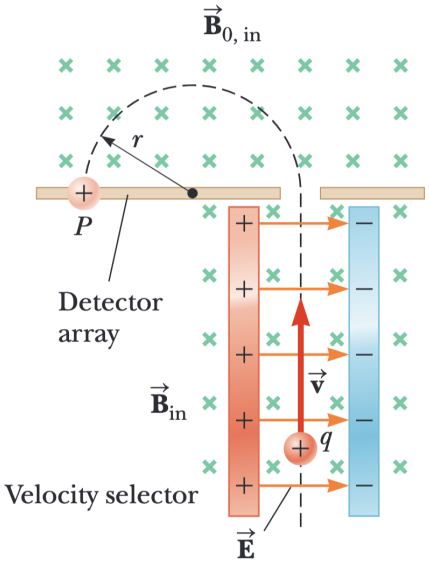


FIGURE 16.5. A mass spectrometer composed of a velocity selector with a continued magnetic field, providing a unique arc depending on the mass of the particle.

The velocity of the particles after passing through the selector is $v = E/B$.

Upon entering the new magnetic field $\vec{\boldsymbol{B}}_0$, the charges undergo circular motion, moving in a semicircular of radius r before hitting a flat plane.

$$r = \frac{mv}{qB} \implies \frac{m}{q} = \frac{rB_0}{v} = rB_0\left(\frac{B}{E}\right)$$

The above result provides us with a relation between the specific charge m/q, the radius r, as well as the magnitudes of each field strength.

17 Electromagnetic Induction

Just like how a change in an electric field can create a magnetic field out of thin air, changes in magnetic fields can also induce the creation of electric fields. This bidirectional and inextricable relationship forms the basis for electromagnetism.

17.1 Magnetic Flux

Definition 17.1: Magnetic Flux

Magnetic flux is defined as the product of an area and the component of the magnetic flux density perpendicular to that area.

$$\Phi = \mathbf{B} \cdot \mathbf{A} = BA\cos\theta$$

where Φ is the magnetic flux for a uniform magnetic flux density B at an angle θ to the normal of an area A.

The S.I. unit of magnetic flux is the weber (Wb).

Definition 17.2: Magnetic Flux Linkage

Magnetic flux linkage is the product of the magnetic flux Φ passing through the coil and the number of turns N on the coil.

$$N\Phi = NBA\cos\theta$$

The magnetic flux linkage is simply the total magnetic flux passing through an area A bounded by a coil with N turns.

17.2 Faraday's Law

Definition 17.3: Faraday's Law

Faraday's law of electromagnetic induction states that the magnitude of the induced e.m.f. is directly proportional to the rate of change of the magnetic flux linkage.

$$|\varepsilon| = \frac{\mathrm{d}(N\Phi)}{\mathrm{d}t} = \frac{\mathrm{d}(NBA\cos\theta)}{\mathrm{d}t}$$

17.2.1 Slidewire Generator

An example of Faraday's law in action appears in the slidewire generator, where a rod moving across a magnetic field can generate electricity.

Consider a metal rod of length l moving at a constant speed of v perpendicular to a magnetic field of flux density B. The rod is in contact with two other connected rods, forming a closed circuit as shown in Figure 17.1

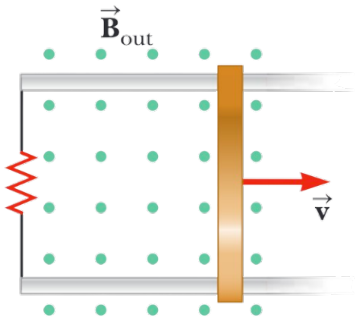


FIGURE 17.1. A diagram of a slidewire generator.

As the rod moves, the area bounded by the circuit increases, thereby corresponding to an increase in its magnetic flux linkage. Using Faraday's law, we can obtain an equation for the induced e.m.f.

$$\begin{aligned} |\varepsilon| = \frac{\Delta(NBA\cos\theta)}{\Delta t} &= B\frac{\Delta A}{\Delta t} \\ &= Blv \end{aligned}$$

17.3 Lenz's Law

Definition 17.4: Lenz's Law

Lenz's law states that the induced current is in a direction so as to produce effects which oppose the change in the magnetic flux.

$$\varepsilon = -\frac{\mathrm{d}(NBA\cos\theta)}{\mathrm{d}t}$$

Figure 17.2 shows the flow of induced current when a bar magnet moves towards and away from a conducting loop.

In other words, the induced current flows in such a way that it attempts to preserve the original magnetic flux. If the magnetic flux is decreasing, the induced current flows to increase it. If the magnetic flux is increasing, the induced current flows to decrease it.

Considering the slidewire generator shown in Example 17.1, as the rod moves outwards, the magnetic flux directed into the area enclosed by the circuit increases with time.

Due to Lenz's law, the induced current flows to oppose this change. It must produce a field directed out of the page. Hence, the induced current must be directed counterclockwise when the rod moves to the right.

17.4 Motional Electromotive Force

Motional electromotive force refers to the electromotive force induced in a conductor that moves through a constant magnetic field.

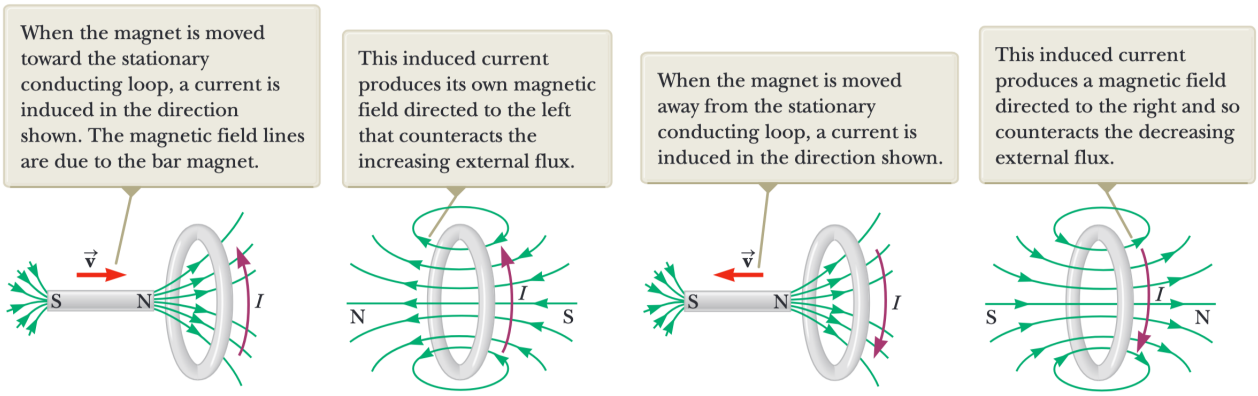


FIGURE 17.2. Step-by-step explanations on how the induced current in a circular loop changes as the magnetic flux linkage of the loop increases and decreases.

Consider a straight current-carrying conductor of length ℓ moving uniformly with velocity $\tilde{\mathbf{v}}$ in a direction perpendicular to a magnetic field $\tilde{\mathbf{B}}$, as seen in Figure 17.3.

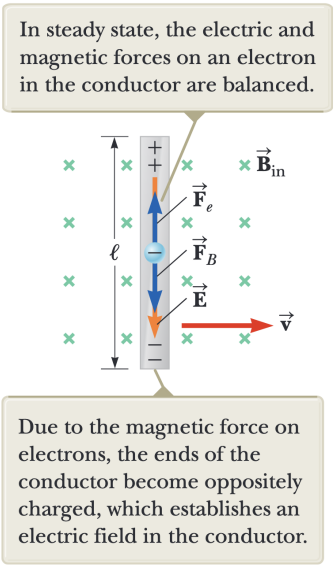


FIGURE 17.3. Electrons within a metal rod moving through a magnetic field.

The electrons in the conductor experience a force $\tilde{\mathbf{F}}_b = \tilde{\mathbf{B}} \times q\tilde{\mathbf{v}}$ directed along the length of the conductor. Electrons thus accumulate at one end of the conductor, creating an electric field $\tilde{\mathbf{E}}$ inside the conductor.

In order for the velocity to remain constant, the forces must balance.

$$qE = Bvq \implies E = vB$$

The magnitude of the electric field produced in the conductor is related to the potential difference across its ends, where $\Delta V = E\ell$. This gives

$$\Delta V = E\ell = B\ell v$$

Hence, a potential difference is maintained as long as the conductor continues to move across the field and "cut through" the magnetic flux.

17.5 A.C. Generators

The alternating-current generator is a loop of wire that rotates due to the magnetic force exerted upon it, consequently creating electricity due to the continual change in its magnetic flux. A set-up is shown in Figure 17.4.

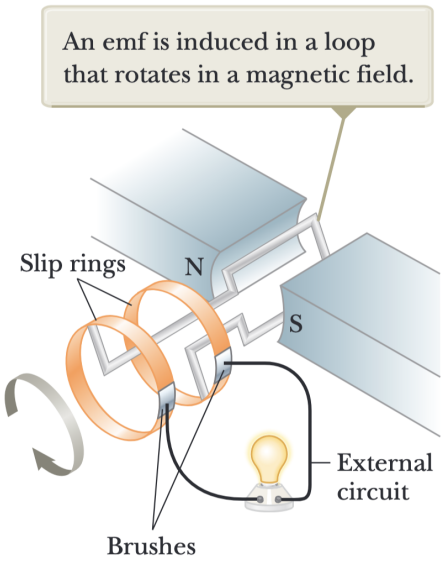


FIGURE 17.4. A simple A.C. generator, with a rectangular loop of wire placed between two magnetic poles.

As the loop rotates, its magnetic flux Φ changes with time, which induces an electromotive force and a current. Assuming that the generator rotates with constant angular speed ω,

$$\Phi = BA\cos\theta = BA\cos\omega t$$

The induced electromotive force in the coil is thus

$$\varepsilon = -\frac{\mathrm{d}(N\Phi)}{\mathrm{d}t} = -NBA\frac{\mathrm{d}}{\mathrm{d}t}(\cos\omega t)$$
$$= NBA\,\omega\sin\omega t$$

The electromotive force varies sinusoidally with time. The maximum electromotive force has the value $NBA\,\omega$, when ωt is either 0° or 180°, namely when $\tilde{\mathbf{B}}$ is perpendicular to the plane of the coil.

17.6 Eddy Currents

If a solid plate conductor is subjected to a changing magnetic flux, eddy currents will flow simultaneously along many different paths in swirls along the conductor's plane.

In Figure 17.5, as the plate enters the magnetic field, the plate cuts the magnetic flux, incuding an e.m.f. in the plate. However, the rate of cutting varies across the whole palte, so different electromotive forces are induced in different parts of the plate, forming eddy currents.

Directions of eddy currents can be derived using Lenz's law. For example, as the plate enters the field in the direction of $\tilde{\mathbf{v}}$ shown above, the field is going into the face of the plate, hence the eddy

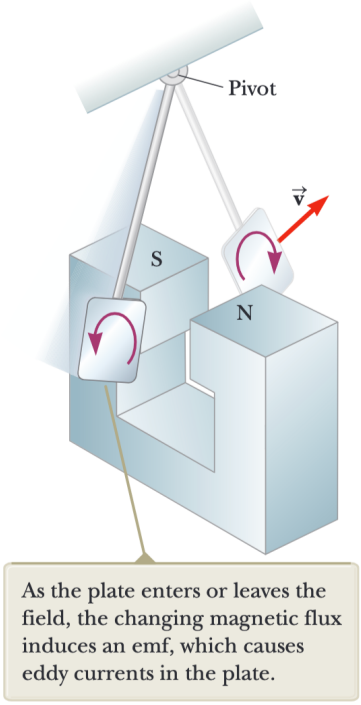


FIGURE 17.5. Eddy currents are created due to uneven changes in magnetic flux.

current must provide its own magnetic field out of the page. This causes the eddy current to flow counterclockwise.

Eddy currents can be reduced by eliminating paths for current flow, done by cutting slots in the plate or laminating the plate.

18 Alternating Current

18.1 AC Sources

As opposed to DC sources, AC sources provide a voltage that continuously alternates its direction and magnitude.

Definition 18.1: Period

The **period** T of an AC source is the time taken for the alternating current to complete one cycle.

Definition 18.2: Frequency

The **frequency** f of an AC source refers to the number of complete cycles per unit time.

Definition 18.3: Peak Value

The **peak value** I_0 of an AC source is the maximum absolute value of the alternating current or voltage in either direction of zero value in a periodic cycle.

Definition 18.4: Root-Mean-Square Current

The **root-mean-square current** I_{rms} is the value of the of the steady direct current which would dissipate heat at the same average rate in a given resistor.

$$I_{\text{rms}} = \sqrt{\langle I^2 \rangle} = \sqrt{\frac{1}{T}\int_0^T I^2 \, \mathrm{d}t}$$

Definition 18.5: Peak-to-Peak Value

The **peak-to-peak value** is the difference between the positive peak value and the negative peak value of the alternating current within a cycle.

18.1.1 Mean Power

Proposition 18.1: Mean Power

The **mean power** dissipated in a resistor of resistance R connected to an A.C. source with root-mean-square current I_{rms} and root-mean-square voltage V_{rms} is given as

$$\langle P \rangle = I_{\text{rms}} V_{\text{rms}} = I_{\text{rms}} R^2 = \frac{{V_{\text{rms}}}^2}{R}$$

Proof. This can be derived with a simple application of Watt's law. □

18.1.2 Sinusoidal AC Sources

Alternating sources commonly provide voltages that vary in a sinusoidal manner, graphed in Figure 18.1.

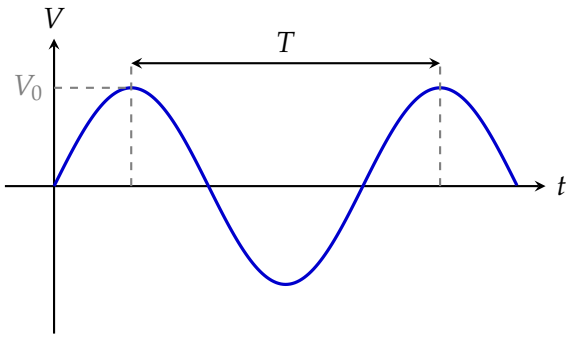


FIGURE 18.1. A sinusoidal alternating voltage.

The time-varying voltage from a sinusoidal source (and its respective current) is given by

$$V = V_0 \sin \omega t \implies I = I_0 \sin \omega t$$

where ω is the angular frequency of the A.C. source.

For a sinusoidal current, the mean current of the circuit will be zero. However, heat is dissipated when it flows through a resistor, implying that the mean current is not representative of the effective value of the alternating current.

Proposition 18.2: RMS Values for Sinusoidal AC

For a *sinusoidal* alternating current, the root-mean-square values for its current I_{rms} and V_{rms} share the same ratio of $1/\sqrt{2}$ with respect to their peak values I_0 and V_0. In other words,

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}}, \quad V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$

Proof. Working with the general formula for the root-mean-square current, for a sinusoidal alternating current,

$$\begin{aligned} I_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T I^2 \, \mathrm{d}t} &= \sqrt{\frac{1}{T}\int_0^T (I_0 \sin \omega t)^2 \, \mathrm{d}t} \\ &= \sqrt{\frac{{I_0}^2}{T}\int_0^T \sin^2 \omega t \, \mathrm{d}t} \\ &= \sqrt{\frac{{I_0}^2}{2T}\int_0^T (1 - \cos 2\omega t) \, \mathrm{d}t} \\ &= \sqrt{\frac{{I_0}^2}{2T}\left[t - \frac{\sin 2\omega t}{2\omega}\right]_0^T} \\ &= \sqrt{\frac{{I_0}^2}{2}} = \frac{I_0}{\sqrt{2}} \end{aligned}$$

The value for V_{rms} can be derived similarly. □

Proposition 18.3: Mean Power for Sinusoidal AC

For a *sinusoidal* alternating current, the mean power dissipated is half of the maximum power dissipated.

$$\langle P \rangle = \frac{1}{2} P_{\max}$$

Proof. For a sinusoidal alternating current, the instantaneous power dissipated in a resistor is

$$P = I^2 R = {I_0}^2 \sin^2(\omega t) R$$

The mean power is thus

$$\begin{aligned} \langle P \rangle = \langle I^2 \rangle R &= \langle {I_0}^2 \sin^2 \omega t \rangle R \\ &= \frac{1}{2} {I_0}^2 R \\ &= \frac{1}{2} P_{\max} \end{aligned}$$

□

18.2 Transformers

A simple iron-core **transformer** is illustrated in Figure 18.2, where two insulated conducting coils are wound on separate limbs of a soft iron core.

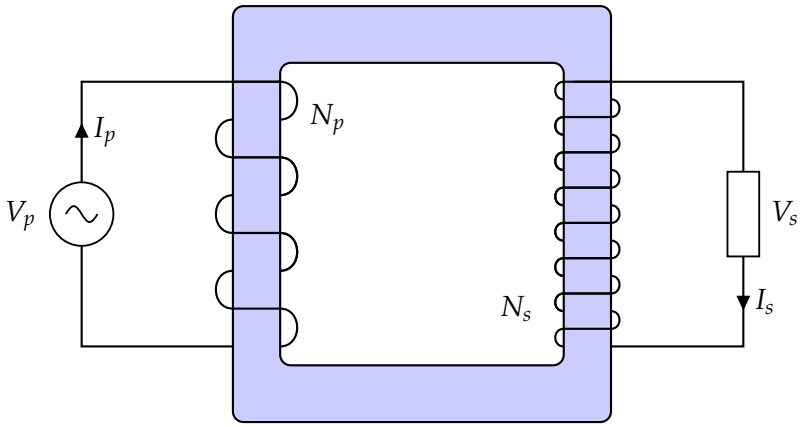


FIGURE 18.2. An iron-core transformer with a primary and secondary coil.

A transformer works by electromagnetic induction. The source creates an alternating current in the primary coil, which sets up an alternating magnetic flux in the iron core. From Faraday's law, this induces an alternating e.m.f. in each coil. The induced e.m.f. in the secondary coil gives rise to an alternating current, which delivers energy to the device to which the secondary is connected.

An ideal transformer is one in which there is no power loss, and the same magnetic flux passes through each turn of both the primary and secondary coils.

Proposition 18.4: Turn Ratio of Transformers

For an ideal transformer, the turn ratio is given by

$$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{I_p}{I_s}$$

where N_p, V_p, and I_p are the number of coils, voltage, and current in the primary coil respectively, and N_s, V_s, and I_s are that in the secondary coil.

Proof. From the laws of electromagnetic induction, we know that

$$V_s = -N_s \frac{\mathrm{d}\Phi}{\mathrm{d}t}, \quad V_p = -N_p \frac{\mathrm{d}\Phi}{\mathrm{d}t}$$

From these two equations,

$$\frac{N_s}{N_p} = \frac{V_s}{V_p}$$

In an ideal transformer without energy losses, the equivalence of power between the two coils must hold.

$$\begin{aligned} P_s = P_p &\implies I_s V_s = I_p V_p \\ &\implies \frac{V_s}{V_p} = \frac{I_p}{I_s} \end{aligned}$$

Therefore,

$$\frac{N_s}{N_p} = \frac{V_s}{V_p} = \frac{I_p}{I_s}$$

□

A transformer is **step-up** if $N_s > N_p$ and **step-down** if $N_s < N_p$.

18.2.1 Transmission of Electrical Power

The main benefit of using high alternating currents over direct currents primarily lies within long-distance power transmission, for alternating currents are more effective in minimising power loss with the use of transformers.

18.3 Diode Rectification

Recall that a diode is a device that conducts current much better in one direction than the other. Ideal diodes have zero resistance for one direction of current and infinite resistance for the other.

Rectification is the conversion of alternating current to direct current through the use of a diode.

18.3.1 Half-Wave Rectifier

Consider the circuit shown in Figure 18.3.

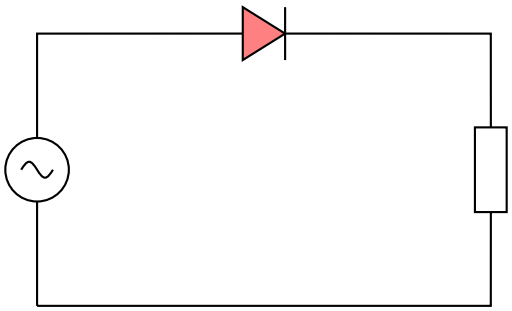


FIGURE 18.3. A circuit with an AC source connected to a diode, causing partial rectification.

The circuit is a half-wave rectifier, which only allows current to flow through the resistor during one-half of every alternating current cycle.

If the first half cycle acts in the forward-biased direction of the diode, current flows round the circuit, creating a potential difference. During the second half cycle, the diode is reverse-biased, where little or no current flows in the circuit and the potential difference across R is zero. This is repeated for each cycle of alternating current input.

This results in a **half-wave rectified** alternating current, as shown in Figure 18.4.

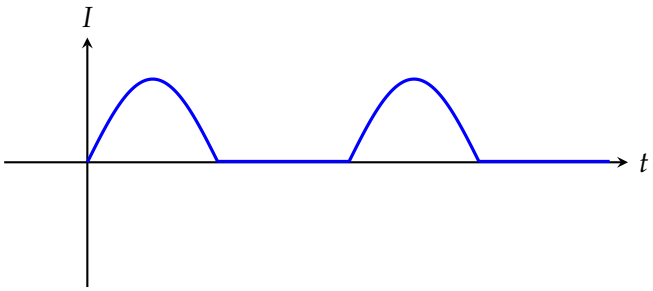


FIGURE 18.4. Current due to the half-wave diode rectification.

18.3.2 Rectifier Comparisons

The comparison of the root-mean-square current I_{rms}, mean power through a load of resistance $\langle R \rangle$, and the whole cycle average current $\langle I \rangle$ are shown below.

AC source	I_{rms}	$\langle P \rangle$	$\langle I \rangle$
Sinusoidal	$\frac{I_0}{\sqrt{2}}$	$\frac{{I_0}^2 R}{2}$	0
Half-wave rectified	$\frac{I_0}{2}$	$\frac{{I_0}^2 R}{4}$	$\frac{I_0}{\pi}$
Full-wave rectified	$\frac{I_0}{\sqrt{2}}$	$\frac{{I_0}^2 R}{2}$	$\frac{2\,I_0}{\pi}$
Square	I_0	${I_0}^2 R$	0

Part VI

Modern Physics

19 Quantum Physics

I hope that you will find someday that, after all, it isn't as horrible as it looks.

— Richard Feynman

19.1 Photons

Thus far, we have seen one underlying assumption within classical mechanics: that light is represented purely by electromagnetic waves.

However, scientists observed inexplicable phenomena on the atomic scale, obtaining results that were directly contradictory to the manifestation of light as continuous waves.

These scientists eventually considered the possibility that light acted as discrete particles that contained packets of energy, called photons, consequently giving way to new interpretations of physics that we now regard as quantum physics.

Definition 19.1: Photon

A photon is a discrete quantum of energy of electromagnetic radiation. The energy of one photon is directly proportional to the frequency f of electromagnetic radiation.

$$E = hf$$

where $h = 6.63 \times 10^{-34}$ J s is the Planck constant.

Definition 19.2: Electronvolt

In measuring photon energies, the electronvolt (eV) unit is much often used.

One electronvolt is the energy gained by one electron when it is accelerated through a potential difference of one volt.

19.2 Photoelectric Effect

Experiments showed that when light was shone on metal surfaces, electrons was emitted from the surfaces. This is the photoelectric effect, which specifically refers to the emission of electrons from a cold metal surface when electromagnetic radiation of a sufficiently high frequency falls on it.

In 1905, Albert Einstein came up with an explanation for the photoelectric effect by considering the particulate nature of light.

19.2.1 Einstein's Photoelectric Equation

Definition 19.3: Einstein's Photoelectric Equation

Einstein's photoelectric equation is given by the following equation.

$$hf = \Phi + \frac{1}{2}m{v_{\max}}^2$$

where the energy E provided by a single photon is equal to the work function energy Φ of the metal and the max kinetic energy of the electrons emitted by the metal surface.

Definition 19.4: Work Function Energy

The work function energy of a metal Φ is the **minimum energy** of a photon to cause the emission of electrons from the surface of said metal.

The work function energy depends on the nature of the material of the metal, as well as its surface condition. The existence of a work function energy also means that there must be a minimum frequency or a maximum wavelength for the photoelectric effect to occur.

Definition 19.5: Threshold Frequency

The threshold frequency f_0 is the **lowest frequency** of electromagnetic radiation that gives rise to the emission of electrons from the metal surface.

Definition 19.6: Threshold Wavelength

The threshold wavelength λ_0 is the **highest wavelength** of electromagnetic radiation that gives rise to the emission of electrons from the metal surface.

From the above definitions,

$$hf_0 = \frac{hc}{\lambda_0} = \Phi$$

Some characteristics of the photoelectric effect are that

1. a single photon can only interact and exchange its energy with a single electron, i.e. its interactions are one-to-one;
2. not all photons (of sufficient energy) get to interact with electrons;
3. the photoelectrons are emitted in all random directions with varying speeds.

19.2.2 Stopping Potential

Figure 19.1 shows an experimental setup that demonstrates the photoelectric effect.

A metal plate known as the **emitter** is connected to the negative terminal, whereas another metal plate known as the **collector** is connected to the positive terminal.

When monochromatic light of a sufficient frequency shines on the emitter, a current is detected by the ammeter due to the flow of photoelectrons across the emitter and collector.

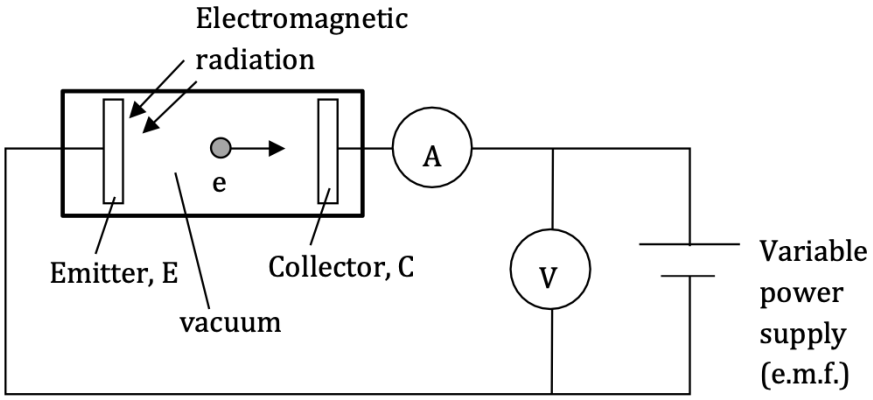


FIGURE 19.1. A circuit powered by photoelectrons being emitted due to electromagnetic radiation and hitting a collector.

Here comes the critical point: if the electric field points from collector to emitter, then all photoelectrons are accelerated towards the collector, contributing to the photocurrent based on its kinetic energy.

However, if we reverse the polarity of the cell, electrons can be prevented from reaching the collector due to the reversed electric field. At some potential, no electrons can reach the collector.

Definition 19.7: Stopping Potential

The **stopping potential**, V_s, is the minimum potential difference between the emitter and the collector that prevents the most energetic photoelectrons from reaching the collector.

$$eV_s = \frac{1}{2}m{v_{\max}}^2$$

We can also determine the maximum kinetic energy with which electrons leave the emitter. From the conversation of energy,

$$\begin{aligned}\Delta K + \Delta U = 0 &\implies (0 - K_{\max}) + (-e)(-V_s) = 0 \\ &\implies K_{\max} = \frac{1}{2}m{v_{\max}}^2 = eV_s\end{aligned}$$

From the photoelectric effect,

$$\frac{1}{2}m{v_{\max}}^2 = hf - \Phi \implies eV_s = hf - \Phi$$

Note that since intensity I does not appear in the equation, V_s is independent of intensity.

19.2.3 Wave-Model Contradictions

The above experiment, when perceived through the lens of Maxwell's picture of light as electromagnetic waves, presents three important contradictions.

1. We observe the **existence of a threshold frequency**.

- From the wave model, the intensity of an electromagnetic wave **depends on the amplitude, not frequency**. Thus, the photoelectric effect should not be affected by frequency in any way.
- Experimentally, the photocurrent is **dependent on light frequency**. Monochromatic light below a minimum threshold frequency yields no photocurrent, regardless of intensity.
- Due to the 1-to-1 photon-electron interaction, the photon energy must be greater than the work function energy of the metal in order to emit an photoelectron. Hence, if a photon has a smaller energy than the work function, the photoelectron will not be emitted. By $E = hf$, the frequency must be larger than a minimum threshold frequency.

2. There is an **instantaneous emission of photoelectrons**.
   - From the wave model, for low intensity light, a **long exposure time is needed** before the total energy absorbed by the surface can become equal to the work function energy.
   - Experimentally, there is an **instantaneous emission of photoelectrons** when the light is above a minimum frequency, no matter how low the intensity is.
   - Due to the 1-to-1 photon interaction, so long as the photon energy is larger than the work function of the metal, the electrons will be emitted instantaneously regardless of the intensity of the incident electromagnetic radiation.
3. The **stopping potential** is **independent of intensity** but **dependent on frequency**.
   - From the wave model, since energy delivered to the emitter is dependent on the light's intensity, **the stopping potential should increase with increasing light intensity**. We further expect that frequency should have nothing to do with the stopping potential.
   - Experimentally, **the stopping potential** (and the maximum kinetic energy of the emitted photoelectrons) **is independent of intensity but is dependent on frequency**.
   - Due to the 1-to-1 photon interaction, since photons have energy $E = hf$, increasing intensity will affect the rate of incidence of photons but not the photon energies themselves, causing the maximum kinetic energy of each photoelectron to remain the same. Stopping potential itself is given by the equation $E = \phi + eV_s$ which is independent of intensity.

It is only with Einstein's postulate regarding photons that these experimental results were adequately explained.

Furthermore, this is consistent with the observation that **a greater intensity creates greater photocurrent** as there is a greater number of photons per second absorbed, and thus a greater number of electrons emitted, resulting in a larger photocurrent.

19.2.4 Graphical Representations

We now take a look the graphical representations of the photoelectric effect.

Stopping Potential & Frequency

Rearranging Einstein's photoelectric equation, we can obtain a linearised equation between V_s and f.

$$\begin{aligned} hf = \Phi + K_{\max} &\implies hf = \Phi + eV_s \\ &\implies V_s = \frac{h}{e}f - \frac{\Phi}{e} \end{aligned}$$

A graph of V_s against f produces a straight line graph of gradient h/e, vertical intercept $-\Phi/e$, and horizontal intercept f_0, as shown in Figure 19.2.

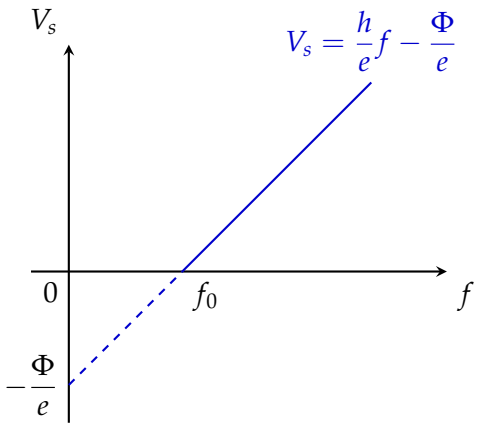
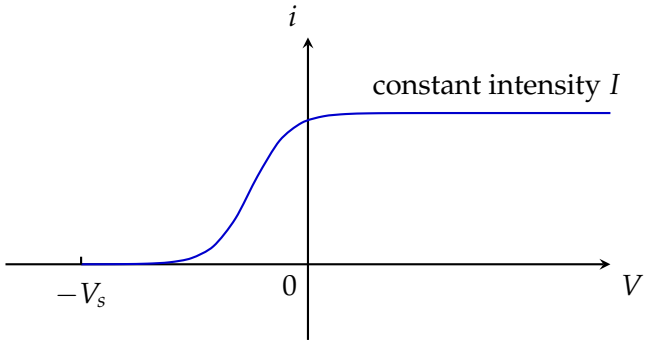


FIGURE 19.2. Stopping potential V_s shares a linear relationship with incident frequency f, resulting in a straight-line graph with vertical intercept $-\Phi/e$.

Photocurrent & Voltage

The graph of photocurrent i against V for the circuit in the photoelectric experiment for light of a given frequency f is show below.



For $V < 0$, as V increases from $-V_s$ towards zero, more photoelectrons reach the collector, resulting in an increase in the photocurrent.

At $V = 0$, there is zero potential, thus only photoelectrons with sufficient initial kinetic energy reach the collector.

For $V > 0$, as V becomes positive, it accelerates all emitted photoelectrons towards the collector. This current rapidly increases until it reaches a saturation value.

Beyond a certain positive voltage, the current reaches a **saturation value**, determined by the intensity of the incident light and the quantum efficiency of the emitter. Further increases in voltage do not increase the current.

19.3 Wave-Particle Duality

We understand that light can be modelled as both a particle and a wave. It is possible to then theorise that all forms of matter have both wave and particle properties. This is the concept of wave-particle duality.

In 1923, de Broglie postulated that a free particle with rest mass m, moving with non-relativistic speed v, should have a wavelength λ related to its momentum p.

Definition 19.8: de Broglie Wavelength

The de Broglie wavelength λ is the wavelength associated with a particle of mass m that is moving at a speed v, expressed as

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

where h is the Planck constant.

Specifically for electromagnetic radiation of a wavelength λ, it will have an associated momentum given by

$$p = \frac{h}{\lambda} = \frac{hf}{c}$$

Example 19.1. EJC'24 Lecture Notes (Ex. 19)

A lightweight spacecraft can be propelled via a photon sail by the Sun. As photons bounce off an ultra-thin reflective sail, radiation pressure acts across the surface area of the sail, giving thrust.

A photon sail has an area of 32 m^2 and exhibits perfect reflectance. Determine the magnitude of thrust if energy from the Sun reaches the sail perpendicularly with a total intensity of 1400 Wm^{-2}.

For each photon, employing de Broglie's wave-particle duality,

$$E = hf = \frac{hc}{\lambda} \implies E = pc$$

For perfect reflectance, the change in momentum for each photon is

$$\Delta p = p_f - p_i = 2p$$

From Newton's 2nd law,

$$\begin{aligned} F = \frac{\mathrm{d}p}{\mathrm{d}t} &= \frac{2p}{t} \cdot \text{no. of photons} \\ &= \frac{2}{t} \cdot \frac{\sum E}{c} \\ &= \frac{2}{c} \cdot \text{intensity} \cdot \text{area} \\ &= \frac{2}{3 \times 10^8} \cdot 1400 \cdot 32 \\ &= 2.99 \times 10^{-4}\ \text{N} \end{aligned}$$

19.3.1 Electron-Diffraction Experiment

In an electron-diffraction experiment, the apparatus consists of

- an electron gun, which produces a beam of electrons with controlled energy;
- a thin film of crystalline graphite as the target, acting as a diffraction medium for the electrons
- and a display, which shows a circular Fraunhofer diffraction pattern of glowing rings.

This entire apparatus is housed in a vacuum chamber, which prevents electron scattering by air molecules.

When electrons are accelerated through a potential V in the electron gun, its resultant de Broglie wavelegth is given by

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2meV}} \quad \left(\because \frac{1}{2}mv^2 = eV \right)$$

The graphite film acts like a diffraction grating for electrons, due to the spacing d between carbon atom layers. As such, the diffraction pattern follows a principle similar to the grating equation $d \sin\theta = n\lambda$ covered earlier in the topic of superposition.

For a higher V, the value of λ will be smaller, resulting in smaller angles θ and more closely spaced rings. Vice versa, for a lower V, the rings of the diffraction pattern will be spaced wider.

19.4 Energy Levels in Isolated Atoms

Atoms consist of a nucleus surrounded by electrons. These electrons occupy specific **discrete energy levels**. The ground state, denoted by $n = 1$, represents the lowest energy level, whereas higher integers of $n \in \mathbb{Z}^+$ represent higher energy levels.

Electrons can move between these energy levels by either absorbing or emitting energy.

Importantly, each element has a unique set of energy levels, resulting in a distinct spectral pattern. An **atomic energy level diagram** shows the possible paths back to the ground state for an atom. This uniqueness allows for the identification of elements based on the energy levels of the electrons, done through a process called spectroscopy.

The most studied atom is the hydrogen atom. The figure below shows the hydrogen atomic energy level diagram.

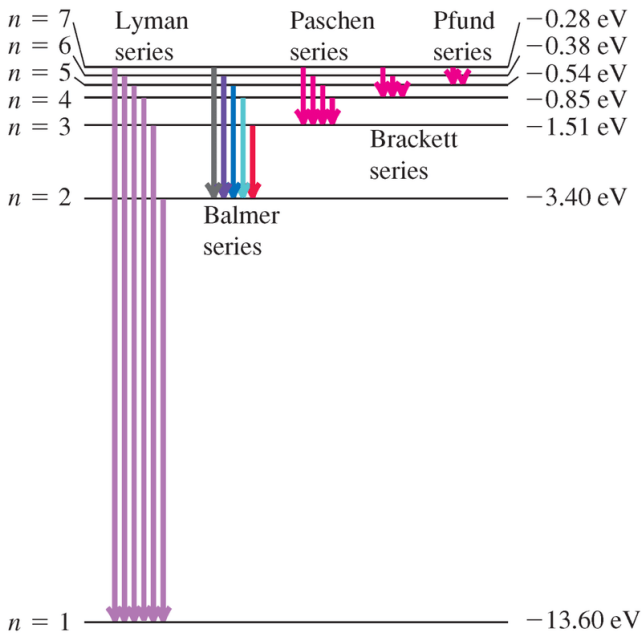


FIGURE 19.3. The atomic energy level diagram for the different transitions between energy states in a hydrogen atom.

Spectral series result from transitions between specific energy levels. For hydrogen atoms, the Lyman series involves transitions to $n = 1$, the Balmer series involving transitions to $n = 2$, followed by the Paschen, Brackett, and Pfund series in a progressive manner.

Some key features of the atomic energy level diagram are described below.

- The principal quantum n must be a positive integer. The ground state $n = 1$ represents the lowest energy level, where the atom is most stable. Electrons normally occupy the ground state.
- As $n \to \infty$, it represents an energy state where the electron is no longer bound to the atom (via ionisation), which is assigned an energy of 0. Consequently, all energy levels have negative values, with the more negative energy states corresponding to greater stability.
- Electrons can transition between energy levels by absorbing or emitting energy, not only as photons but also through other processes like collisions.

We now formally describe the processes of energy absorption and emission by electrons.

19.4.1 Excitation

When an electron jumps to a higher energy level, it absorbs energy in a process called excitation. However, electrons in these excited states are rather unstable, and can only maintain these levels for a short time.

Electrons are mainly excited either through **photon absorption** or **high speed collisions** with other particles. In the first case, the excitation of electrons will give rise to the absorption line spectra.

For photon absorption,

- cool gas is exposed to electromagnetic radiation, which causes an electron to absorb energy from a single photon;
- this photon energy must exactly match the energy difference between levels, governed by the equation $hf = \Delta E = E_f - E_i$;
- this precise requirement means that *only photons of particular frequencies* can excite electrons in a given atom, leading to the characteristic absoprtion spectra we observe.

For high-speed collisions,

- cool gas is bombarded with fast-moving particles (ions, atoms, or electrons); an electron absorbs part or all of the collidng particle's kinetic energy;
- the kinetic energy K of the colliding particle must be at least equal to the energy level difference, described by the inequality $K \geq |\Delta E|$;
- this allows for a range of particle energies to cause excitation, as any excess energy can be distributed as kinetic energy among the colliding particles after the interaction.

19.4.2 De-Excitation

Almost immediately after excitation, the electrons falls to a lower energy level, releasing energy via de-excitation. The energy change is emitted through photons that possess similar energies, giving rise to the emission line spectra.

This can happen in two ways, either through **direct transitions** or **cascading transitions**.

- In direct transition, the electron drops *directly* from excited to ground state in a single step, emitting one photon with energy equal to the full difference between these levels.
- In cascading transitions, the electron descends through a series of intermediate energy levels before reaching the ground state; as such, it emits a separate photon for each transition, with each photon's energy corresponding to the energy difference per step.

19.4.3 Emission Line Spectra

Definition 19.9: Emission Line Spectra

The emission line spectra consists of discrete bright lines of different colours on a dark background.

When a gas is heated, some of its molecules undergo excitation and transition to higher energy levels. They remain in this state briefly before being de-excited.

When the electrons in this hot gas are de-excited, photons of energies corresponding to the difference between the energy levels will be emitted.

Due to the existence of discrete energy levels in isolated atoms, these emitted photons have specific energies which correspond to specific wavelengths. These photons are diffracted at specific angles due to their wavelengths, giving rise to a set of characteristic spectral lines when viewed through through a spectroscope.

The set of spectral lines emitted by an element is **unique** to that element.

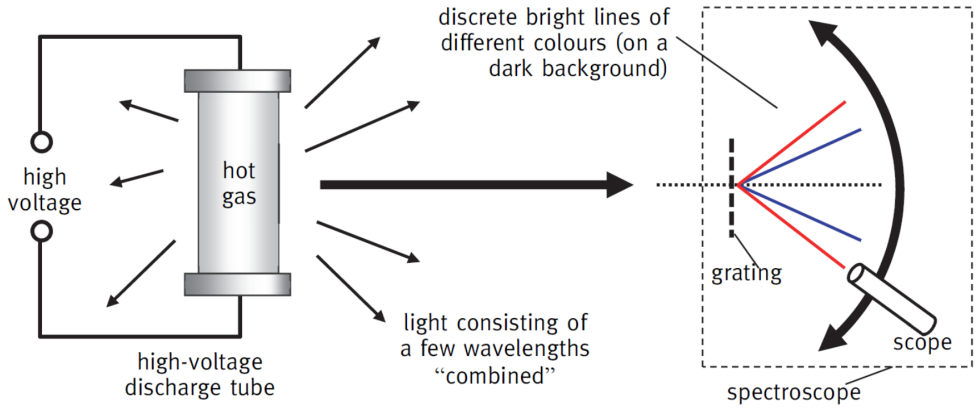


FIGURE 19.4. When a gas is heated, commonly using a high-voltage circuit, it is able to emit discrete wavelengths of light.

19.4.4 Absorption Line Spectra

Definition 19.10: Absorption Line Spectra

The **absorption line spectra** consists of continuous spectrum crossed by dark lines.

The absorption line spectrum is produced by passing a continuous spectrum of white light through a cool gas, as shown in Figure 19.5.

As white light passes through the gas, the electrons of the gas absorb photons of certain frequencies and get excited to higher energy levels.

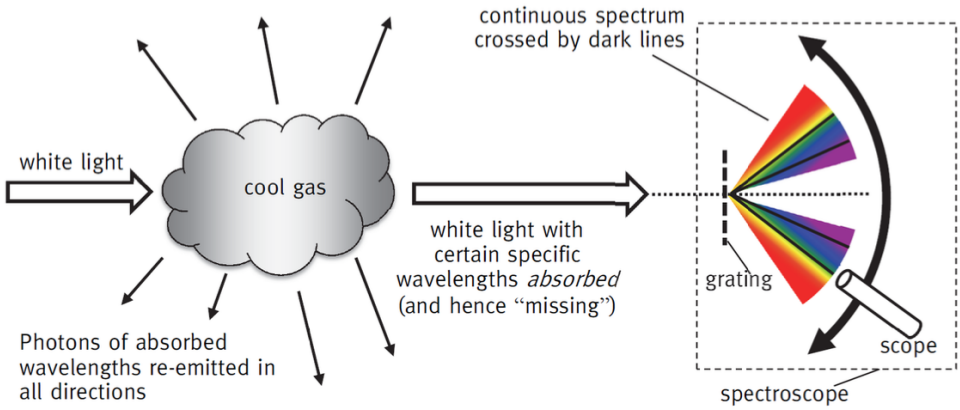


FIGURE 19.5. When white light is shone through a cool gas, some photons undergo interactions with the gas molecules, resulting in missing wavelengths.

The photons that are absorbed must have energy exactly equal to the energy difference between the two energy levels of the gas atoms. Due to the existence of discrete energy levels in isolated atoms, the absorbed photons have specific energies which correspond to specific wavelengths.

When the electrons are de-excited, the photons are re-emitted in random directions, causing them to be missing. Thus, dark lines corresponding to the absorption spectrum are observed when viewed through a detector placed on the other side of the gas.

The set of absorption spectral lines absorbed by an element is unique to that element.

Furthermore, this emission of light by the excited gas also creates its own emission spectrum. If observed from the side of the gas sample rather than in line with the original light source, one would see bright spectral lines characteristic of the gas.

19.4.5 X-Rays

X-rays are extremely high-energy photons. Since X-rays are ionising radiation, they can be rather dangerous. To produce X-rays, we rely on X-ray tubes, which operate on high voltage in order to generate X-ray photons.

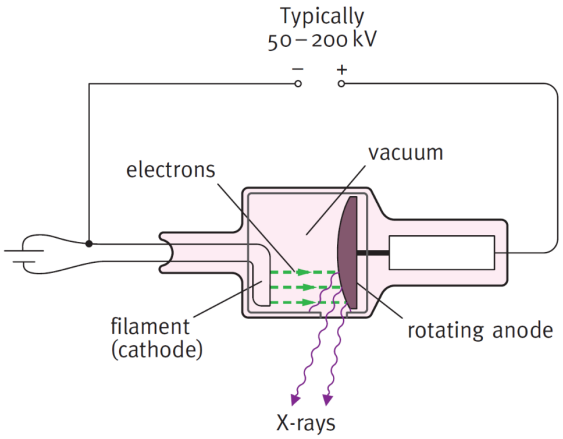


FIGURE 19.6. X-rays being emitted as high-speed electrons undergo deceleration upon collision with a target.

The filament is heated by passing a current through it, and emits electrons. The emitted electron gains kinetic energy given by

$$E = e\Delta V$$

where ΔV is the potential difference in which the electron is accelerated across.

When these electrons collide with a metal target, they either stop or slow down, and emit X-ray photons, which have a range of energies. A spectrum diagram shows a **broad continuous spectrum** and a few sharp peaks, termed **characteristic lines**.

An example graph of the X-ray spectrum is shown in Figure 19.7.

λ_{min} is the X-ray photon with the *most photon energy*. It is when all the energy of one electron is given up in one collision and is converted into a single X-ray photon.

The continuous background spectrum, also known as the Bremsstrahlung radiation, is the electromagnetic radiation (in the form of photons) produced whenever an electron is decelerated.

The characteristic lines originate from the same mechanism as emission spectra, involving electron transitions in-between the electron energy levels within an atom of the target metal.

- Inner shell electrons can be ejected from the metal's atom by the fired electrons, thus creating an electronic vacancy.
- Outer shell electrons then transit to lower energy levels in order to fill this vacancy. Doing so, they emit photons corresponding to the difference in electron energy levels, creating characteristic lines.

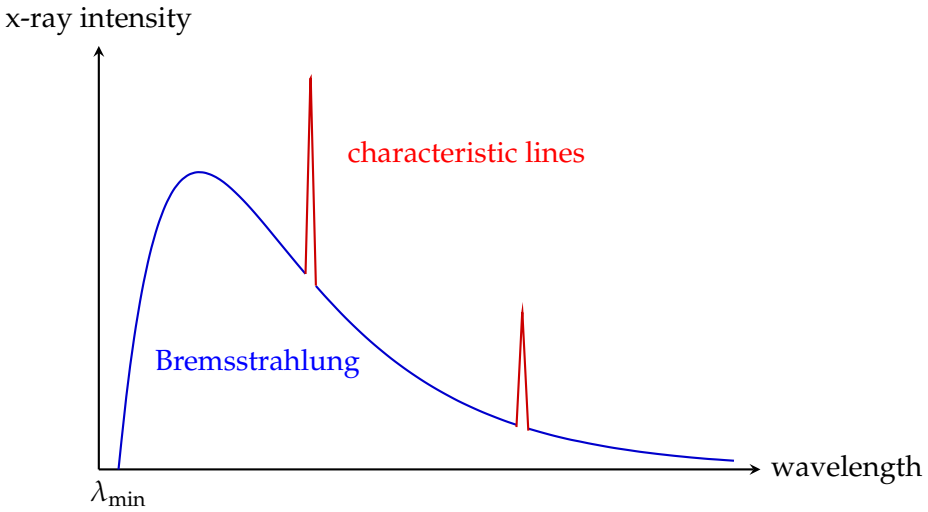


FIGURE 19.7. A graph relating the intensity of X-ray photons with their wavelength. Bulk of the photons follow the continuous Bremsstrahlung spectrum, with some characteristic lines.

- Thus, metals can be uniquely identified via this method too.

19.5 Heisenberg's Uncertainty Principle

Definition 19.11: Heisenberg's Position-Momentum Uncertainty Principle

The **Heisenberg position-momentum uncertainty principle** states that for a particle, the product of the uncertainties in position Δx and momentum Δp in the same direction **cannot be smaller than the Planck constant**.

$$\Delta x \Delta p \geq h$$

20 Nuclear Physics

> The unleashed power of the atom has changed everything [...] we thus drift toward unparalleled catastrophe.
>
> — ALBERT EINSTEIN

20.1 Rutherford Scattering

Rutherford's gold-foil experiment, used to test the plum-pudding model, consists of a narrow beam of alpha particles (helium nuclei) directed normally onto a sheet of very thin gold foil (around 1 micrometer thick), all within a vacuum environment.

- The gold foil must be thin to reduce the possibility of multiple deflections of alpha particles with gold nuclei, so that only single interactions between them can be studied.
- The vacuum chamber is needed to ensure that the deflection of alpha particles is entirely due to its interaction with gold atoms, and not surrounding air molecules.

An illustration of the set-up is shown in Figure 20.1.

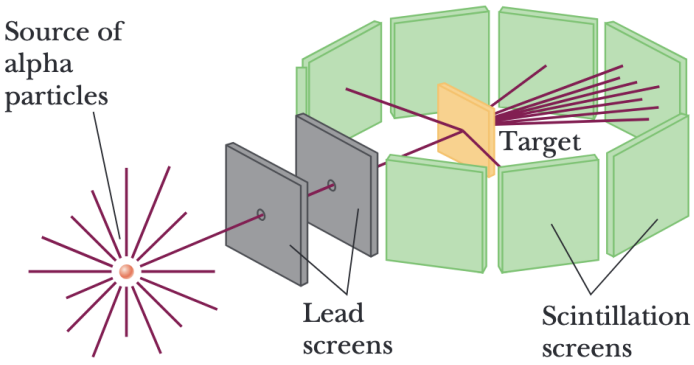


FIGURE 20.1. A schematic for Rutherford's gold foil experiment.

Surrounding the thin gold foil is a zinc sulfide screen, which produce a scintillation when struck by alpha particles. This allows for the detection of angle at which the alpha particles are deflected.

It was seen that while most alpha particles passed through with minimal deflection, a tiny proportion had significant deflection.

The number of scintillations observed at each scattering angle is given in Figure 20.2.

There are two main observations and inferences arising from Rutherford's gold foil experiment.

1. Most of the α-particles were deviated through **small angles**. This suggests that the size of the nucleus is small compared to the atom, as majority of the α-particles passed through the atom far enough to experience negligible repulsion.

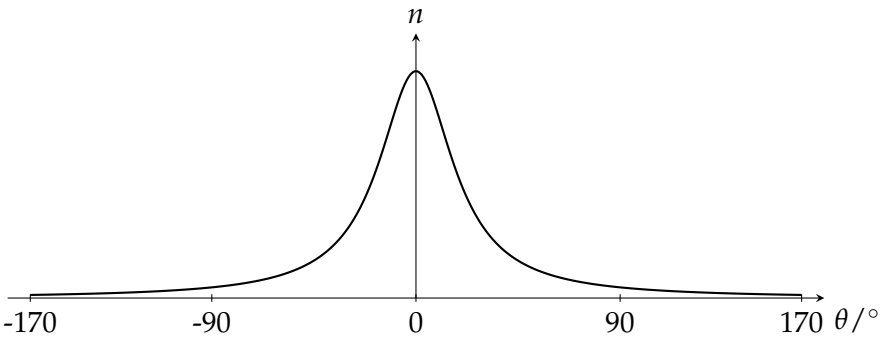


FIGURE 20.2. A graph showing the spread of scintillations as the scattering angle varies.

2. A smaller but significant percentage of the α-particles deviated through **large angles** greater than 90 degrees. This suggests that the mass of the atom is concentrated in the nucleus, and that the nucleus is positively charged.

Example 20.1.

In Rutherford's α-scattering experiment, an α-particle ($+2e$ charge) of energy 7.7 MeV is directed head-on towards a gold nucleus ($+79e$ charge). Calculate the distance of closest approach for the α-particle.

At the distance of closest approach, the α-particle is momentarily at rest. All kinetic energy is converted into electric potential energy.

By conservation of energy,

$$\Delta K + \Delta U = 0 \implies 7.7 \times 1.60 \times 10^{-19} = \frac{79 \times 2 \times (1.60 \times 10^{-19})^2}{4\pi\varepsilon_0 d}$$
$$\implies \boxed{d = 2.95 \times 10^{-14}\ \text{m}}$$

20.2 Notation

20.2.1 Atomic Structure

Definition 20.1: Nucleons

Nucleons are the collective **protons and nucleons** in a nucleus.

Definition 20.2: Nucleon Number

The nucleon number A (also mass number) of a nucleus is the **total number of nucleons** in the nucleus.

Definition 20.3: Proton Number

The proton number Z (also atomic number) of a nucleus is the **total number of protons** in the nucleus.

Definition 20.4: Unified Atomic Mass Constant

The unified atomic mass constant u is defined as one-twelfth of the mass of a carbon-12 atom.

$$1u = 1.66 \times 10^{-27}\ \text{kg}$$

All atoms are neutral. For atoms, the proton number of its nucleus and the number of electrons orbiting the nucleus must be the same.

The table below summarises the properties of the common subatomic particles.

Particle	Proton	Neutron	Electron
Charge (C)	$+1.60 \times 10^{-19}$	0	-1.60×10^{-19}
Rest mass (kg)	1.6726×10^{-27}	1.6749×10^{-27}	9.11×10^{-31}
Rest mass (u)	1.007	1.008	0.0055

20.2.2 Isotopes

Definition 20.5: Isotopes

Isotopes are atoms of the same element with the same number of protons but a **different number of neutrons** in the nucleus.

Isotopes of an element are chemically similar to each other, but since their nuclei have different masses, they exhibit different physical characteristics.

20.2.3 Nuclides

Definition 20.6: Nuclides

A nuclide is a specific combination of protons and neutrons in a nucleus. This is represented as

$$^{A}_{Z}X$$

where X is the chemical symbol of the element, A is the nucleon number, and Z is the proton number.

The neutron number is given by $A - Z$.

In a nuclear reaction, there is a transformation of at least one nuclide to another. This can be represented by a nuclear equation, for example,

$$^{14}_{7}\text{N} + ^{4}_{2}\text{He} \longrightarrow ^{17}_{8}\text{O} + ^{1}_{1}\text{H}$$

20.2.4 Conservation Laws

This is a general problem-solving technique: nuclear reactions obey 4 important conservation laws, which are the conservation of

1. nucleon number,
2. proton number,
3. mass-energy,
4. and linear momentum.

20.3 Mass Defect & Binding Energy

Definition 20.7: Mass-Energy Equivalence Relation

The mass-energy equivalence relation states that

$$e = mc^2$$

where e is energy, m is mass, and $c = 3.0 \times 10^8$ is the speed of light.

Definition 20.8: Mass Defect

The mass defect of a nucleus, Δm, is the **difference** between the total mass of the individual, free nucleons and the mass of the nucleus as a whole.

$$\Delta m = (Zm_p + (A - Z)m_n) - m_{\text{nucleus}}$$

where m_p, m_n, and m_{nucleus} are the masses of the total no. of protons, total no. of neutrons, and the nucleus respectively.

The mass defect comes about when we first notice a peculiar fact — that the mass of a single nucleus is somehow always less than the total mass of its constituent nucleons.

- Observing mass-energy equivalence, we see that since mass is a measure of energy content, any change in mass is accompanied by a corresponding change in energy, and vice versa.
- We term this change as the **mass defect**.

Definition 20.9: Nuclear Binding Energy

The **nuclear binding energy** E_b of a nucleus is

- the energy **required** to completely **separate** the nucleons in the nucleus to infinity; or
- the energy **released** when nucleons **come together** to form a nucleus from infinity.

Mathematically, this is given by

$$E_b = \Delta m c^2$$

where Δm is the mass defect of the nucleus.

Imagine the process in which free nucleons bind together to form a nucleus. Since there is a mass defect, the difference in mass must be released as some form of energy. In breaking up a nucleus, energy must be provided as well.

- This amount of energy is known as the **binding energy of the nucleus**, which is the same amount of energy being released or absorbed during nuclear reactions.
- This energy input is converted into part of the mass of the free nucleons, via $E = mc^2$.

Definition 20.10: Exothermic & Endothermic Reactions

For an **exothermic reaction**,

- there is a net *release* of energy;
- the total mass of reactants is *more than* that of products;
- and the total binding energy of reactants is *less than* that of products.

For an **endothermic reaction**,

- there is a net *absorption* of energy;
- the total mass of reactants is *less than* that of products;
- and the total binding energy of reactants is *more than* that of products.

Example 20.2.

When lithium ^{7}Li is bombarded by a proton, two alpha particles are produced. Find

1. the net energy released during the reaction, and
2. the difference in total binding energies of the reactants and products.

The masses are $m_H = 1.007825u$, $m_{\text{He}} = 4.002603u$ and $m_{\text{Li}} = 7.016004u$.

The reaction is

$$^1_1\text{H} + ^7_3\text{Li} = ^4_2\text{He} + ^4_2\text{He}$$

The masses of the reactants and products are given by

$$m_{\text{reactants}} = 1.007825u + 7.016004u = 8.023829u m_{\text{products}} = 2(4.002603u) = 8.005206u$$

The **energy released** is thus

$$\begin{aligned} E = \Delta mc^2 &= (8.023829u - 8.005206u) \cdot c^2 \\ &= 0.018623 \times 1.66 \times 10^{-27} \times (3.0 \times 10^8)^2 \\ &= \boxed{+2.78 \times 10^{-12}\ \text{J}} \end{aligned}$$

Now, for the binding energy of the reactants,

$$\begin{aligned} E_b \text{ of reactants} &= \text{mass defect} \cdot c^2 \\ &= [(4 \cdot 1.007 + 3 \cdot 1.008) - 8.023829]u \cdot c^2 \\ &= 0.036171u \cdot c^2 \end{aligned}$$

For the binding energy of the products,

$$\begin{aligned} E_b \text{ of products} &= \text{mass defect} \cdot c^2 \\ &= [(4 \cdot 1.007 + 3 \cdot 1.008) - 8.005206]u \cdot c^2 \\ &= 0.054794u \cdot c^2 \end{aligned}$$

Therefore, the difference in binding energy is

$$\Delta E_b = (0.054794 - 0.036171) \times 1.66 \times 10^{-27} \times (3 \times 10^8)^2 \implies \boxed{\Delta E_b = 2.78 \times 10^{12}\ \text{J}}$$

Thus, we see that the difference in binding energy of the reactants and products is exactly equal to the energy released during the nuclear reaction.

20.4 Binding Energy Per Nucleon

Definition 20.11: Binding Energy Per Nucleon

The binding energy per nucleon of a nucleus is the binding energy of the nucleus divided by the number of nucleons in the nucleus.

It is also the average energy per nucleon required to completely separate the nucleons to infinity. **The higher the binding energy per nucleon, the more stable the nucleus is**.

The binding energy per nucleon is not held constant across all nuclei, as seen in Figure 20.3. Some important details are that

- the maximum binding energy per nucleon is 8.8MeV for the iron-56 nuclide;
- hydrogen-1 does not appear on the graph as it has no binding energy;
- and the largest nucleus in the graph is uranium-235, with a binding energy of 7.6MeV.

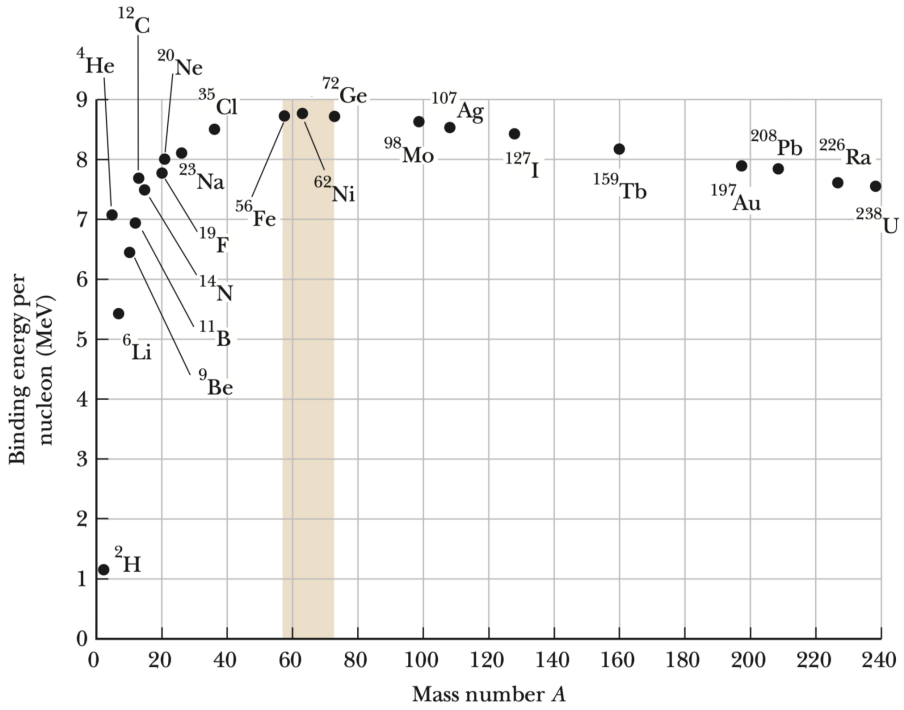


FIGURE 20.3. Binding energy per nucleon for nuclei of increasing mass. The tan line shows the region where binding energy per nucleon is the greatest.

20.5 Nuclear Reactions

Definition 20.12: Nuclear Reaction

A nuclear reaction is a process during which two nuclei, or one nucleus and an external subatomic particle, collide with each other to produce different nuclei.

It is important to understand that **nuclear reactions generally occur when the reactants have lower binding energy than the products.**

There are two important types of nuclear reactions, namely **nuclear fusion** and **nuclear fission**.

Energy is released when lighter nuclei undergo fusion or when heavier nuclei undergo fission, as these processes result in products that have a higher binding energy per nucleon.

20.5.1 Nuclear Fusion

Definition 20.13: Nuclear Fusion

Nuclear fusion is the fusing of two nuclei of low nucleon numbers to produce a larger nucleus with the release of energy.

20.5.2 Nuclear Fission

Definition 20.14: Nuclear Fission

Nuclear fission is the splitting of a heavy nucleus into two lighter nuclei of approximately the same mass with the release of energy and neutrons.

Nuclear fission is usually instigated by a neutron.

Nuclear fission can sometimes cause a chain reaction, where a series of similar reactions occur in which one of the products of a reaction causes further reactions to occur.

20.6 Decay Processes

Definition 20.15: Radioactive Decay

Radioactive decay is the **spontaneous** and **random** disintegration of the unstable nucleus in the form of the emission of alpha particles, beta particles, and gamma ray photons.

Spontaneous means that the decay is not affected by external or environmental factors, and **random** means that the nucleus has a constant probability of decay per unit time.

20.6.1 Alpha Decay

Definition 20.16: Alpha Particles

Alpha particles contain two protons and two neutrons, being the same as a helium nucleus ${}^{4}_{2}\mathrm{He}$. It has a charge of $+2e$ and mass $4u$, and may be emitted from a radioactive nucleus.

Definition 20.17: Alpha Decay

Alpha decay is the spontaneous decay of a nucleus with the emission of an alpha particle.

$$ {}^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X} \longrightarrow {}^{\mathrm{A}-4}_{\mathrm{Z}-2}\mathrm{Y} + {}^{4}_{2}\mathrm{He} $$

where X is the parent nucleus and Y is the daughter nucleus.

Since X is unstable compared to Y, it releases energy, causing its products to have less energy and thus less mass. Therefore, since the mass of the reactants is larger than that of the products, alpha decay is exothermic.

20.6.2 Beta Decay

Definition 20.18: Beta Particles

Beta particles are high-speed electrons emitted from within the nucleus of an atom.

Definition 20.19: Beta Decay

Beta decay is the spontaneous decay of a nucleus with the emission of a beta particle.

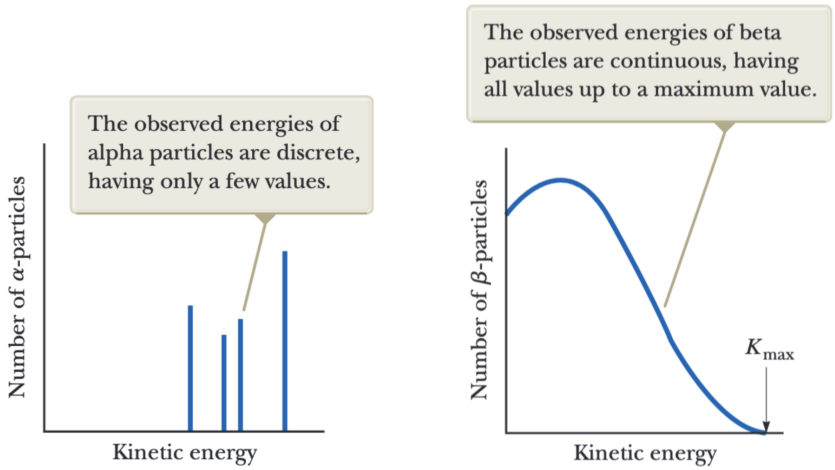
$$ {}^{A}_{Z}X \longrightarrow {}^{A}_{Z+1}Y + {}^{0}_{-1}e + {}^{\bar{0}} $$

where X and Y are the parent and daughter nucleus, and $\bar{\nu}$ is an antineutrino.

An Antineutrino?

It is not important to understand what an antineutrino is. What is important is how people discovered it was there in the first place.

In 1911, scientists Meitner and Hahn discovered that the energy of beta particles followed a **continuous distribution**, whereas the energy of alpha particles only took on **a few discrete values**. The kinetic energy of the system after the decay is equal to the decrease in rest energy of the system, and because all decaying nuclei in their experiment had the same initial mass, that means that the decrease in rest energy was always the same.



Therefore, the sole consideration of the beta particle creates a violation of the conservation laws for energy and momentum, for it is impossible that the emitted particles possess a range of kinetic energies instead of sticking to discrete values.

It was hence theorised that part of the energy released was carried by another particle, called the **antineutrino**, which was later verified. It has **zero electric charge** and its mass is a **tiny fraction of the electron's mass**.

20.6.3 Gamma Decay

Definition 20.20: Gamma Radiation

Gamma radiation is short-wavelength electromagnetic radiation.

Definition 20.21: Gamma Decay

Gamma decay is the spontaneous decay of a nucleus with the emission of gamma radiation.

$$ {}^{A}_{Z}X^{*} \longrightarrow {}^{A}_{Z}X + \text{fl} $$

Frequently, some of the energy released in the alpha and beta decay is in the form of gamma ray photons. Alpha and beta decay usually leave the daughter nucleus in an excited state, denoted by an asterisk. The daughter nucleus then emits energy in the form of a gamma ray photon to reach its ground state.

20.7 Radioactivity

Radiation detectors, upon detecting alpha, beta, and gamma radiation, obtain an electrical signal which is counted.

Definition 20.22: Count Rate

The count rate C is the number of counts per unit time received by a radiation detector.

20.7.1 Background Radiation

Background radiation is the radiation detected by a radiation counter when no radioactive source is nearby.

The background count rate must be subtracted from a measured count rate in order to obtain a more reliable measurement.

20.7.2 Activity

Definition 20.23: Activity

The activity (also decay rate) of a radioactive nuclide is the **number of decays per unit time**.

$$ A = -\frac{\mathrm{d}N}{\mathrm{d}t} = \lambda N $$

where the negative sign indicates a decrease in the **number of undecayed particles** N, and λ is the **decay constant**.

Definition 20.24: Decay Constant

The decay constant λ is the probability of decay of a nucleus per unit time.

Using ideas of differential equations, it can be proved that A, N, and C can be modelled as a function of time t, such that

$$ A = A_0 e^{-\lambda t}, \quad N = N_0 e^{-\lambda t}, \quad C = C_0 e^{-\lambda t} $$

where A_0, N_0, C_0 are the **initial** activity, number of undecayed particles, and count rate respectively.

20.7.3 Half-Life

Definition 20.25: Half-Life

The **half-life** $t_{\frac{1}{2}}$ of a radioactive nuclide is the average time taken for the initial number of nuclei of that particular nuclide to reduce to half its original value.

$$t_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$$

Proof. From $N = N_0 e^{-\lambda t}$, we obtain

$$\ln N = \ln N_0 - \lambda t$$

When $t = t_{\frac{1}{2}}$, $N = \frac{1}{2} N_0$, therefore

$$\ln\left(\frac{1}{2}N_0\right) = \ln N_0 - \lambda t_{\frac{1}{2}} \implies \ln\frac{1}{2} = -\lambda t_{\frac{1}{2}}$$
$$\implies t_{\frac{1}{2}} = \frac{\ln 2}{\lambda}$$

□

20.7.4 Effects of Ionising Radiation

Ionising radiation refers to alpha, beta, and gamma radiation. It can have direct and indirect effects on living tissues and cells.

Direct effects occur when the radiation interacts directly with the DNA or other cellular molecules. The energy from the radiation may cause breaks in DNA strands, leading to mutations or cell death.

Indirect effects occur when the radiation interacts with water molecules in the cell, creating highly reactive molecules. These reactive molecules can then damage the DNA and other cellular molecules.

Short-term effects of ionising radiation on living tissues include skin burns, nausea, vomiting, and diarrhea. **Long-term effects** include an increased risk of cancer and other diseases.

Part VII

H3 Physics

21 Inertial Frames

21.1 Frames of Reference

Frames of reference are *perspectives of motion*. There exists a stationary reference frame in which stationary observers undergo no movement relative to Earth's rotation. For observers that moving at a constant speed, they are within an *inertial reference frame*, where the frame's velocity may be different from the stationary frame, but exhibits zero acceleration.

Definition 21.1: Frame of Reference

A **frame of reference** is a set of coordinates and a time standard that allows an observer to measure the position and motion of objects.

A frame of reference can be uniquely defined by the **four-vector**

$$(x, y, z, t)$$

where x, y, z uniquely define the 3-dimensional coordinates of a body, and t refers to the time coordinate.

Definition 21.2: Galilean Relativity

The **principle of Galilean relativity** (also **Einstein's first postulate**) states that the laws of mechanics must be the same in all inertial frames of reference.

For inertial frames of reference, a body that is not acted on by an external force will continue in its state of constant motion or rest.

21.1.1 Galilean Transformations

Let S denote an original inertial frame, and S' denote an inertial frame moving at a constant velocity with respect to S. The Galilean transformation relates coordinates measured in one inertial reference frame $S \mapsto (x, y, z, t)$ to that measured in other inertial frame $S' \mapsto (x', y', z', t')$.

Proposition 21.1: Galilean Transformation Equations

The generalised form of the Galilean transformation equations, which provides us with the following two equations, where $\vec{r}$ is the position in frame S and $\vec{u}$ is the relative velocity between S and S'.

$$\vec{r}' = \vec{r} - \vec{u}t$$
$$t' = t$$

For example, if the S' frame is moving at a constant velocity u solely applied to the x-direction, the S' observer will perceive the stationary body in frame S to be moving in the opposite direction. Therefore, we obtain

$$(x', y', z', t') = (x - ut, y, z, t)$$

Across all frames of reference, time is invariant, therefore $t' = t$.

We may wish to consider how the *velocity* of particles change in the perspective of different inertial frames. Taking the derivative of the Galilean transformation equation,

$$\frac{\mathrm{d}\vec{r}'}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(\vec{r} - \vec{v}t) \implies \vec{v}' = \vec{v} - \vec{u}$$

This aligns with our natural intuition, particularly when we observe objects outside the window of a moving vehicle. The apparent velocity of an object in the new inertial frame is simply the subtraction of the frames' relative velocity from said object's original velocity.

21.2 Center of Mass

There is a special point in a mechanical system, termed the **center of mass** (also **barycenter**), that moves as if all of the mass of the system were concentrated at that one point. Mathematically speaking, it is the point where the weighted position vectors relative to this point all sum to zero.

Consequently, the system moves as if an external force were applied to a single particle of mass M (where M is the total mass of the system) located the center of mass.

Proposition 21.2: COM for Discrete Particles

For a system of discrete particles, the center of mass is defined to be

$$x_{\mathrm{CM}} = \frac{1}{M}\sum_i m_i x_i \qquad y_{\mathrm{CM}} = \frac{1}{M}\sum_i m_i y_i \qquad z_{\mathrm{CM}} = \frac{1}{M}\sum_i m_i z_i$$

Proof. As described earlier, for all discrete masses m_i and positions r_i, the coordinates r_{CM} of the center of mass satisfy

$$\sum_i m_i(r_i - r_{\mathrm{CM}}) = 0$$

Solving for r_{CM}, this yields

$$\begin{aligned}\sum_i m_i(r_i - r_{\mathrm{CM}}) = 0 &\implies r_{\mathrm{CM}}\sum_i m_i = \sum_i m_i r_i \\ &\implies r_{\mathrm{CM}} = \sum_i \frac{m_i r_i}{m_i} = \frac{1}{M}\sum_i m_i r_i\end{aligned}$$

□

Proposition 21.3: COM for Extended Objects

For an extended object, the center of mass is defined to be

$$x_{\mathrm{CM}} = \frac{1}{M}\int x\,\mathrm{d}m \qquad y_{\mathrm{CM}} = \frac{1}{M}\int y\,\mathrm{d}m \qquad z_{\mathrm{CM}} = \frac{1}{M}\int z\,\mathrm{d}m$$

Proof. For an extended object with a total mass M, we can model it as a system containing a large number of infinitesimally small mass elements. For the x-axis, this gives us

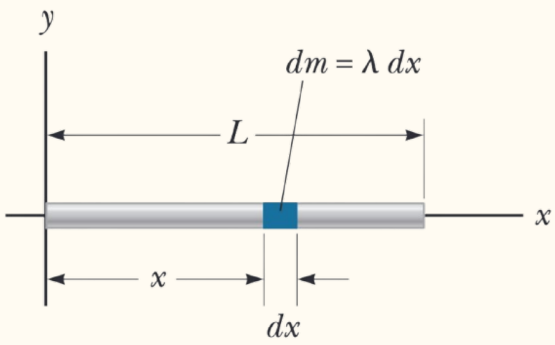
$$x_{\mathrm{CM}} = \lim_{\Delta m_i \to 0}\frac{1}{M}\sum_i x_i \Delta m_i \implies x_{\mathrm{CM}} = \frac{1}{M}\int x\,\mathrm{d}m$$

The same calculation holds for all three dimensional axes. □

Note that the center of mass does not need to be within the object itself.

Example 21.1. Serway & Jewett (9th Edition) (Ex. 9.11)

Find the center of mass of a uniform rod with mass M and length L.



Let the linear mass density be $\lambda = M/L$. If the rod is divided into elements $\mathrm{d}x$,

$$\mathrm{d}m = \lambda \, \mathrm{d}x$$

Finding the center of mass,

$$x_{\mathrm{CM}} = \frac{1}{M}\int x \, \mathrm{d}m = \frac{1}{M}\int_0^L x\lambda \, \mathrm{d}x$$

$$= \frac{\lambda}{M}\left[\frac{x^2}{2}\right]_0^L = \frac{\lambda L^2}{2M} = \frac{M}{L} \times \frac{L^2}{2M}$$

$$= \boxed{\frac{1}{2}L}$$

Example 21.2.

Suppose a rod of length L is non-uniform such that its mass per unit length varies linearly with x according to the expression $\lambda = kx$, where k is a constant.

Show that the center of mass of the rod is $\frac{2}{3}L$ away from the lighter end.

Expressing the infinitesimal mass element for a length element $\mathrm{d}x$,

$$\mathrm{d}m = \lambda \, \mathrm{d}x = kx \, \mathrm{d}x$$

Since the rod has non-uniform mass, we integrate $\mathrm{d}m$ throughout to obtain M.

$$M = \int \mathrm{d}m = \int_0^L kx \, \mathrm{d}x = \frac{kL^2}{2}$$

Finding the center of mass,

$$x_{\text{CM}} = \frac{1}{M}\int x\ \mathrm{d}m = \frac{k}{M}\int_0^L x^2\ \mathrm{d}x$$

$$= \frac{k}{M} \times \left[\frac{x^3}{3}\right]_0^L = \frac{2k}{kL^2} \times \frac{L^3}{3}$$

$$= \boxed{\frac{2}{3}L}$$

Example 21.3.

Find the center of mass of a flat, semi-circular object of radius R and uniform density.

Since the semicircle is symmetrical about the y-axis, $x_{\text{CM}} = 0$.

We divide the semicircle into horizontal strips of length $\mathrm{d}y$ and width $2\sqrt{R^2 - y^2}$ each. Let the area mass density be denoted by $\sigma = M/A$.

$$\mathrm{d}m = \sigma\ \mathrm{d}A = \sigma \times x \times \mathrm{d}y$$
$$= \sigma \times 2\sqrt{R^2 - y^2} \times \mathrm{d}y$$

Finding y_{CM},

$$y_{\text{CM}} = \frac{1}{M}\int_0^R y\ \mathrm{d}m = \frac{1}{M}\int_0^R y\,\sigma\, 2\sqrt{R^2 - y^2}\ \mathrm{d}y$$

$$= -\frac{2\sigma}{3M}\int_0^R -3y\sqrt{R^2 - y^2}\ \mathrm{d}y$$

$$= -\frac{2\sigma}{3M}\left[(R^2 - y^2)^{(3/2)}\right]_0^R$$

$$= \frac{2\sigma}{3M}(R^3)$$

Substituting $\sigma = M/A = M/(\frac{1}{2}\pi R^2)$,

$$y_{\text{CM}} = \frac{4}{3\pi R^2}(R^3) = \boxed{\frac{4R}{3\pi}}$$

21.2.1 Velocity & Momentum

Where the total mass M of the system remains constant, we can describe the motion of the system in terms of the velocity and acceleration of the center of mass of the system.

$$\vec{v}_{\text{CM}} = \frac{\mathrm{d}\vec{r}_{\text{CM}}}{\mathrm{d}t} = \frac{1}{M}\sum_i m_i \vec{v}_i$$

The total momentum of the system is equal to the total mass multiplied by the velocity of the center of mass.

$$\vec{p}_{\text{tot}} = \sum_i \vec{p}_i = \sum_i m_i \vec{v}_i = M\vec{v}_{\text{CM}}$$

This states that the total momentum *is equal to* the total mass times the velocity of the center of the mass.

21.2.2 Force & Acceleration

The acceleration of the center of mass is given by

$$\vec{a}_{\text{CM}} = \frac{\mathrm{d}\vec{v}_{\text{CM}}}{\mathrm{d}t} = \frac{1}{M}\sum_i m_i \vec{a}_i \implies M\vec{a}_{\text{CM}} = \sum_i \vec{F}_i$$

Summing over all internal forces, they cancel in pairs, and the net force on the system is caused only by external forces.

$$\sum \vec{F}_{\text{ext}} = M\vec{a}_{\text{CM}} = \frac{\mathrm{d}\vec{p}}{\mathrm{d}t}$$

When a system of particles is acted on by external forces, the center of mass moves just as though all the mass were concentrated at that point and it were acted on by a net force equal to the sum of external forces on the system.

If the net external force acting on the system is zero, then we see that

$$\frac{\mathrm{d}\vec{p}}{\mathrm{d}t} = 0 \implies M\vec{v}_{\text{CM}} = \vec{p} = \text{constant}$$

21.2.3 Center of Mass Frame

The **center of momentum frame** (also **zero momentum frame**) is a useful tool in solving two-body collision problems. This frame is chosen such that the total momentum is zero and the center of mass is at rest,, when the inertial frame travels at the same velocity as the center of mass.

Consider two objects of masses m_1, m_2 and u_1, u_2 respectively in the stationary frame. In the center of momentum frame,

$$u_1' = u_1 - v_{\text{CM}} \qquad u_2' = u_2 - v_{\text{CM}}$$

Let u' and v' denote the initial and final velocities respectively. Since $\sum p = 0$ for the center of mass frame, this quality implies that

$$m_1 u_1' + m_2 u_2' = m_1 v_1' + m_2 v_2' = 0$$

In both the pre-collision and post-collision states, we can express the speed of the second object in terms of the speed of the first object.

$$m_1 u_1' + m_2 u_2' = 0 \implies u_2' = -\left(\frac{m_1}{m_2}\right) u_1'$$

$$m_1 v_1' + m_2 v_2' = 0 \implies v_2' = -\left(\frac{m_1}{m_2}\right) v_1'$$

For elastic collisions, kinetic energy is also conserved. Therefore,

$$\frac{1}{2}m_1{u_1'}^2 + \frac{1}{2}m_2{u_2'}^2 = \frac{1}{2}m_1{v_1'}^2 + \frac{1}{2}m_2{v_2'}^2$$

Substituting our expressions of u_2 and v_2 into this equivalence, we obtain

$$\begin{aligned}\frac{1}{2}m_1{u_1'}^2 + \frac{1}{2}m_2\left(\frac{m_1}{m_2}\right)^2{u_1'}^2 = \frac{1}{2}m_1{v_1'}^2 + \frac{1}{2}m_2\left(\frac{m_1}{m_2}\right)^2{v_1'}^2 &\implies \left(m_1 + \frac{{m_1}^2}{m_2}\right){u_1}^2 = \left(m_1 + \frac{{m_1}^2}{m_2}\right){v_1}^2 \\ &\implies {u_1}^2 = {v_1}^2 \\ &\implies u_1 = \pm v_1\end{aligned}$$

This relation holds true for the second object as well. We have thus shown that the magnitudes of the initial and final velocities of each object are *equal*.

Considering that the two objects must rebound after colliding, we are left with one solution only, that is when **the final velocity is equal and opposite to the initial velocity**.

$$\vec{v}' = -\vec{u}'$$

22 Rotational Motion

A summary of the analogues between rotational and translational motion is shown below.

Rotational Motion About a Fixed Axis	**Translational Motion**
Angular speed $\omega = d\theta/dt$	Translational speed $v = dx/dt$
Angular acceleration $\alpha = d\omega/dt$	Translational acceleration $a = dv/dt$
Net torque $\Sigma\tau_{\text{ext}} = I\alpha$	Net force $\Sigma F = ma$
If α = constant $\begin{cases} \omega_f = \omega_i + \alpha t \\ \theta_f = \theta_i + \omega_i t + \frac{1}{2}\alpha t^2 \\ \omega_f^2 = \omega_i^2 + 2\alpha(\theta_f - \theta_i) \end{cases}$	If a = constant $\begin{cases} v_f = v_i + at \\ x_f = x_i + v_i t + \frac{1}{2}at^2 \\ v_f^2 = v_i^2 + 2a(x_f - x_i) \end{cases}$
Work $W = \int_{\theta_i}^{\theta_f} \tau \, d\theta$	Work $W = \int_{x_i}^{x_f} F_x \, dx$
Rotational kinetic energy $K_R = \frac{1}{2}I\omega^2$	Kinetic energy $K = \frac{1}{2}mv^2$
Power $P = \tau\omega$	Power $P = Fv$
Angular momentum $L = I\omega$	Linear momentum $p = mv$
Net torque $\Sigma\tau = dL/dt$	Net force $\Sigma F = dp/dt$

22.1 Angular Quantities

For rotational dynamics, we focus on **rigid objects**: non-deformable objects, where relative locations of the object's particles remain constant.

For an object rotating through a circular path with radius r, the arc length s and the angle θ it moves through is related by

$$s = r\theta \implies \theta = \frac{s}{r}$$

Definition 22.1: Angular Displacement

The angular displacement θ of an object is the change in the angular position of an object with respect to a certain point.

$$\Delta\theta = \theta_f - \theta_i$$

Definition 22.2: Angular Speed

The angular speed ω of an object is the rate of change of its angular displacement with respect to time.

$$\omega = \frac{\mathrm{d}\theta}{\mathrm{d}t} \implies \langle\omega\rangle = \frac{\Delta\theta}{\Delta t}$$

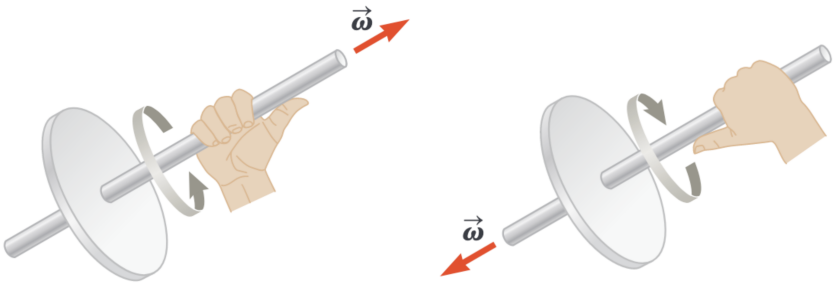
Definition 22.3: Angular Acceleration

The angular acceleration α of an object is the rate of change of its angular velocity with respect to time.

$$\alpha = \frac{\mathrm{d}\omega}{\mathrm{d}t} \implies \langle \alpha \rangle = \frac{\Delta\omega}{\Delta t}$$

Thus far, ω and a merely represent the magnitudes of the angular velocity and angular acceleration vectors $\vec{\omega}$ and $\vec{\alpha}$.

For rotation about a fixed axis, the only direction that uniquely specifies the rotational motion is the **direction along the axis of rotation**. The direction of both $\vec{\omega}$ and $\vec{\alpha}$ (since the direction of $\vec{\alpha}$ directly follows from that of $\vec{\omega}$) can be found using the right-hand rule.



22.1.1 Relation to Translational Quantities

For a rotating rigid object, translational velocity is always tangential to the circular path (and is hence called **tangential velocity**).

Since velocity is given by the rate of change of displacement,

$$v = \frac{\mathrm{d}s}{\mathrm{d}t} = r\frac{\mathrm{d}\theta}{\mathrm{d}t} \implies v = r\omega$$

The tangential speed of a point on a rotating rigid object is equal to the perpendicular distance of that point from the axis of rotation multiplied by the angular speed.

Hence, though every point on the rigid object has the **same angular speed**, they may **differ in tangential speed**, simply because r is not the same for all points on the object.

Tangential speed *increases* as one moves *outward* from the centre of rotation — the outer end of a rotating CD moves faster than a point near the centre.

We can relate the angular acceleration of the rotating rigid object to the **tangential acceleration** of any point by taking the time derivative of v.

$$a_t = \frac{\mathrm{d}v}{\mathrm{d}t} = r\frac{\mathrm{d}\omega}{\mathrm{d}t} \implies a_t = r\alpha$$

The expression for centripetal acceleration is standard.

$$a_c = \frac{v^2}{r} = v\omega = r\omega^2$$

The **total acceleration vector** at any point is thus

$$\begin{aligned}\vec{a} = \vec{a}_t + \vec{a}_c &\implies a = \sqrt{{a_t}^2 + {a_c}^2} \\ &\implies a = \sqrt{r^2\alpha^2 + r^2\omega^4} \\ &\implies a = r\sqrt{\alpha^2 + \omega^4}\end{aligned}$$

22.1.2 Rotational Kinematics

We consider the special case of **a rigid object under constant angular acceleration**.

Proposition 22.1: Rotational Kinematic Equations of Motion

For a rigid object under constant angular acceleration, the following equations hold true.

$$\begin{aligned}\omega_f &= \omega_i + \alpha t \\ \theta_f &= \theta_i + \omega_i t + \frac{1}{2}\alpha t^2 \\ {\omega_f}^2 &= {\omega_i}^2 = 2\alpha(\theta) \\ \omega_f &= \omega_i + \frac{1}{2}(\omega_i + \omega_f)t\end{aligned}$$

Proof. Recalling the equation for instantaneous angular acceleration and integrating from $t_i = 0$ to $t_f = t$, we can obtain an equation that determines the *angular speed* ω_f of the object *at any later time* t.

$$\begin{aligned}\alpha = \frac{\mathrm{d}\omega}{\mathrm{d}t} &\implies \mathrm{d}\omega = \alpha\ \mathrm{d}t \\ &\implies \int_0^t 1\ \mathrm{d}\omega = \int_0^t \alpha\ \mathrm{d}t \\ &\implies w_f = w_i + \alpha t\end{aligned}$$

Substituting this expression into our previous derivation for instantaneous ω, we obtain an expression relating the *angular displacement at any time* t in relation to initial displacement, velocity, and acceleration.

$$\begin{aligned}\omega = \frac{\mathrm{d}\theta}{\mathrm{d}t} &\implies \omega_i + \alpha t = \frac{\mathrm{d}\theta}{\mathrm{d}t} \\ &\implies \int_0^t (\omega_i + \alpha t)\ \mathrm{d}t = \int_0^t \frac{\mathrm{d}\theta}{\mathrm{d}t}\ \mathrm{d}t \\ &\implies \omega_i t + \frac{1}{2}\alpha t^2 = \theta_f - \theta_i \\ &\implies \theta_f = \theta_i + \omega_i t + \frac{1}{2}\alpha t^2\end{aligned}$$

Making t the subject in the first equation and substituting it into the second equation to elimi-

nate t, we obtain

$$\theta_f = \theta_i + \omega_i t + \frac{1}{2}\alpha t^2 \implies \theta_f = \theta_i + \omega_i \left(\frac{\omega_f - \omega_i}{\alpha}\right) + \frac{1}{2}\alpha \left(\frac{\omega_f - \omega_i}{\alpha}\right)^2$$

$$\implies 2\alpha(\theta_f - \theta_i) + 2\omega_i(\omega_f - \omega_i) + (\omega_f - \omega_i)^2$$

$$\implies 2\alpha(\theta_f - \theta_i) = {\omega_f}^2 - {\omega_i}^2$$

$$\implies {\omega_f}^2 = {\omega_i}^2 + 2\alpha(\theta_f - \theta_i)$$

Eliminating α instead,

$$\theta_f = \theta_i + \omega_i t + \frac{1}{2}\alpha t^2 \implies \theta_f = \theta_i + \omega_i t + \frac{1}{2}\left(\frac{\omega_f - \omega_i}{t}\right) t^2$$

$$\implies \theta_f = \theta_i + \frac{1}{2}(\omega_i + \omega_f)t$$

□

This set of equations is *analogous* to the *kinematic equations of motion* for translational motion.

22.2 Moment of Inertia

Definition 22.4: Moment of Inertia

The moment of inertia is a quantity of an object that is directly proportional to the masses of its particles and their distances from the axis of rotation squared.

For a *single particle*, its moment of inertia I is given as

$$I = mr^2$$

where m is the mass of the particle, and r is the distance from the particle to its axis of rotation.

An body's moment of inertia is **analogous to mass** for translational motion, for it is the *resistance* to changes in rotational motion.

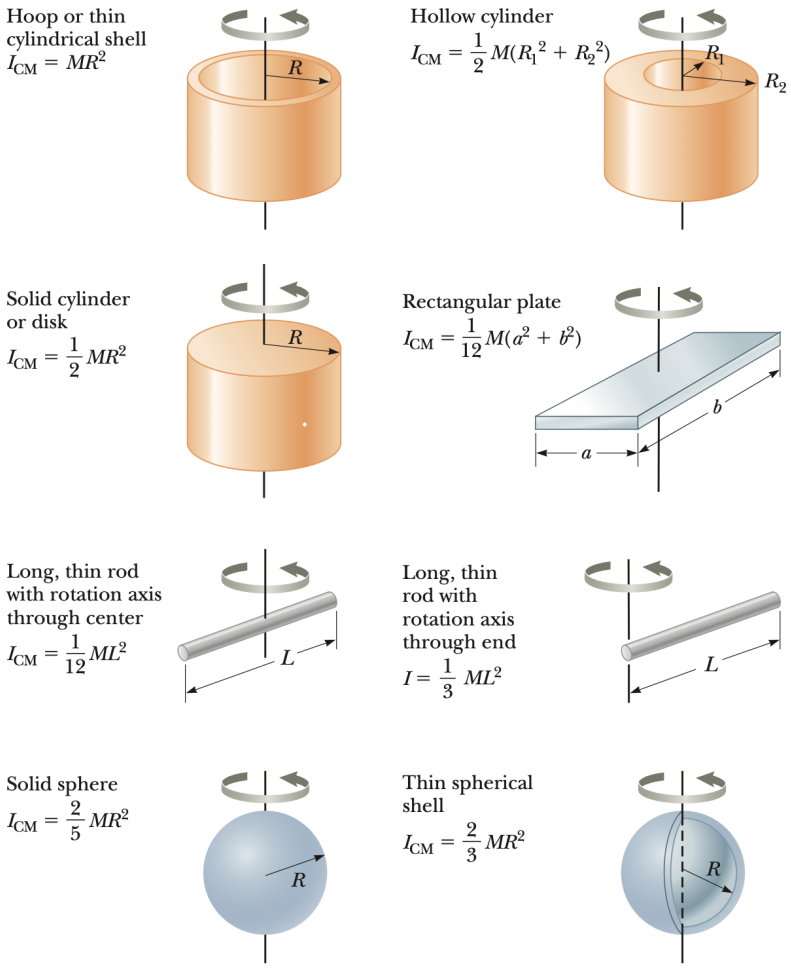
The calculation of the moment of inertia for extended objects becomes a bit more complicated. The figure below shows the moments of inertia for certain homogenous rigid objects with varying geometries.

22.2.1 Calculation

The moment of inertia of a system of discrete particles is simply the sum of the moments of inertia of all its particles.

For a continuous rigid object, we imagine the object being divided into many small elements of mass Δm_i, then take the limit.

$$I = \lim_{\Delta m_i \to 0} \sum_i {r_i}^2 \Delta m_i = \boxed{\int r^2 \, \mathrm{d}m}$$



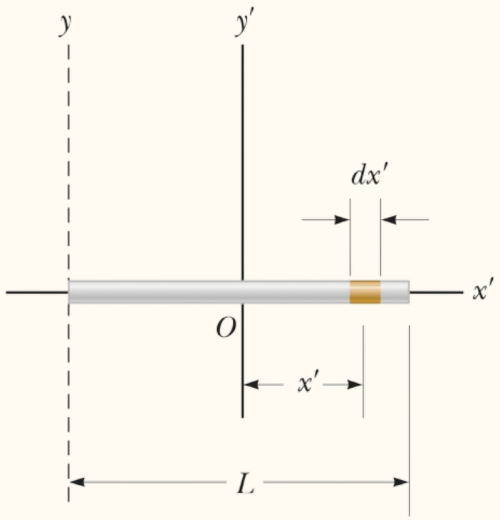
It is usually easier to calculate moments of inertia in terms of volume. Expressing density in volumetric mass form as $\rho = m/V$, we see that $\mathrm{d}m = \rho\ \mathrm{d}V$, hence

$$I = \int \rho r^2\ \mathrm{d}V$$

If the object is *homogenous*, ρ is *constant*, and the integral can be evaluated for a known geometry.

Example 22.1. Serway & Jewett (9th Edition) (Ex. 10.7)

Calculate the moment of inertia of a uniform thin rod of length L and mass M about an axis perpendicular to the rod and passing through its center of mass.



The shaded length $\mathrm{d}x$ has a mass $\mathrm{d}m$ equal to the mass per unit length λ multiplied by $\mathrm{d}x$.

$$\mathrm{d}m = \lambda \, \mathrm{d}x = \frac{M}{L} \, \mathrm{d}x$$

Substituting,

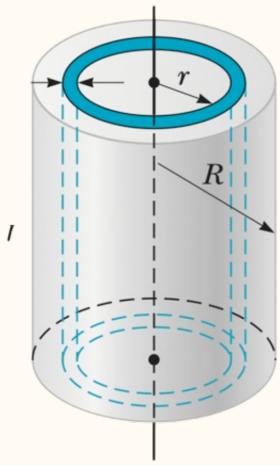
$$I = \int r^2 \, \mathrm{d}m = \int_{-L/2}^{L/2} (x)^2 \frac{M}{L} \, \mathrm{d}x = \frac{M}{L} \int_{-L/2}^{L/2} (x)^2 \, \mathrm{d}x$$

$$= \frac{M}{L} \left[\frac{(x)^3}{3} \right]_{-L/2}^{L/2}$$

$$= \boxed{\frac{1}{12} ML^2}$$

Example 22.2. Serway & Jewett (9th Edition) (Ex. 10.8)

A uniform solid cylinder has a radius R, mass M, and length L. Calculate its moment of inertia about its central axis.



Here, we divide the cylinder into many cylindrical shells, each having radius r, thickness $\mathrm{d}r$, and height L.

The density of the cylinder is ρ and the volume $\mathrm{d}V$ of each shell is given by

$$\mathrm{d}V = L\ \mathrm{d}A = L(2\pi r)\ \mathrm{d}r$$

Expressing $\mathrm{d}m$ in terms of $\mathrm{d}r$,

$$\mathrm{d}m = \rho\ \mathrm{d}V = \rho L(2\pi r)\ \mathrm{d}r$$

Substituting,

$$I = \int r^2\ \mathrm{d}m = \int r^2\ (pL(2\pi r)\ \mathrm{d}r) = 2\pi\rho L \int_0^R r^3\ \mathrm{d}r = \frac{1}{2}\pi\rho L R^4$$

Using the total volume to express density,

$$\rho = \frac{M}{V} = \frac{M}{\pi R^2 L}$$

Finishing off,

$$I = \frac{1}{2}\pi\left(\frac{M}{\pi R^2 L}\right) LR^4 = \boxed{\frac{1}{2}MR^2}$$

22.2.2 Parallel-Axis Theorem

Definition 22.5: Parallel-Axis Theorem

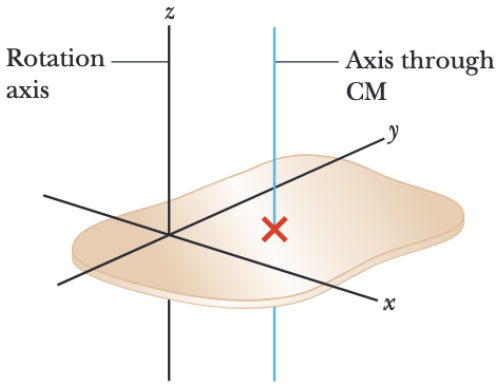
The **parallel-axis theorem** states that for a body of mass m with a moment of inertia I_{CM} when rotating about its center of mass, its moment of inertia I for when rotating about a parallel axis a distance d away from the center of mass axis is

$$I = I_{CM} + md^2$$

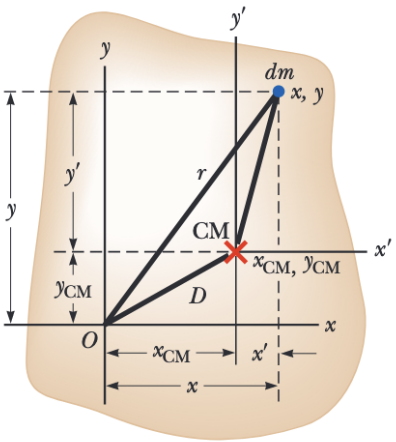
This is used to simplify calculations of moments of inertia for objects rotating around an arbitrary axis.

Proof. We note that the moment of inertia does not depend on the distribution of mass along the axis — as seen in the case of the cylinder, its moment of inertia is *independent of its length*.

As such, as long as the object is homogenous and symmetric, we can simply consider a cross-section of the object lying on a 2-dimensional plane, as shown below.



Now, we consider a mass element dm.



As this element is a distance $r = \sqrt{x^2 + y^2}$ from the axis of rotation (which passes through the

origin O), the moment of inertia of the entire object is given by

$$I = \int r^2 \, \mathrm{d}m = \int (x^2 + y^2) \, \mathrm{d}m$$

As a trick, we can relate the coordinates of $\mathrm{d}m$ to the object's centre of mass, where $x = x' + x_{\mathrm{CM}}$ and $y = y' + y_{\mathrm{CM}}$ respectively as shown by the figure above. Therefore,

$$\begin{aligned} I &= \int \left[(x' + x_{\mathrm{CM}})^2 + (y' + y_{\mathrm{CM}})^2 \right] \mathrm{d}m \\ &= \int \left[(x')^2 + (y')^2 \right] \mathrm{d}m + 2x_{\mathrm{CM}} \int x' \, \mathrm{d}m + 2y_{\mathrm{CM}} \int y' \, \mathrm{d}m + ({x_{\mathrm{CM}}}^2 + {y_{\mathrm{CM}}}^2) \int \mathrm{d}m \\ &= I_{\mathrm{CM}} + md^2 \end{aligned}$$

where m is the mass of the object and d is the perpendicular distance between the given axis and the axis through the centre of mass. □

22.3 Torque

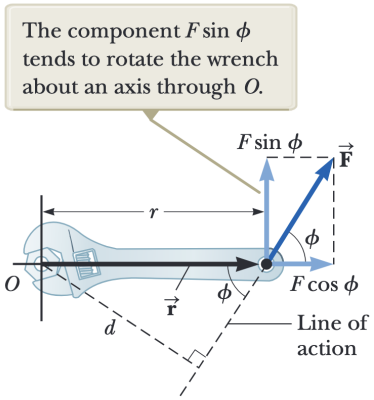
Definition 22.6: Torque

Torque, denoted by $\vec{\tau}$, is the cross product between a force $\vec{F}$ applied to a body and the body's perpendicular distance $\vec{r}$ to its axis of rotation.

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Torque acts as the rotational analogue to linear force.

When a force is exerted on a rigid object pivoted about an axis, the object tends to rotate about that axis. The tendency of a force to rotate an object about some axis is measured by torque $\vec{\tau}$.



Considering the wrench above, the force $\vec{F}$ acts at an angle ϕ to the length of the wrench $\vec{r}$. We define the **magnitude of the torque** associated with $\vec{F}$ around the axis O by the expression

$$\tau = |\vec{r} \times \vec{F}| = rF \sin \phi$$

22.3.1 Rigid Object Under Net Torque

Consider a particle of mass m rotating in a circle of radius r under the influence of a tangential net force $\sum \vec{F}_t$ and a radial net force for $\sum \vec{F}_r$. This tangential net force causes a tangential acceleration

$\vec{a}_t$, whereas the radial net force provides the centripetal acceleration for the circular motion.

By Newton's second law, examining the tangential acceleration,

$$\sum F_t = ma_t$$

The magnitude of the net torque due to $\sum \vec{F}_t$ on the particle about an axis perpendicular to the page through the centre of the circle is

$$\sum \tau = \sum F_t \, r = (ma_t) \, r$$

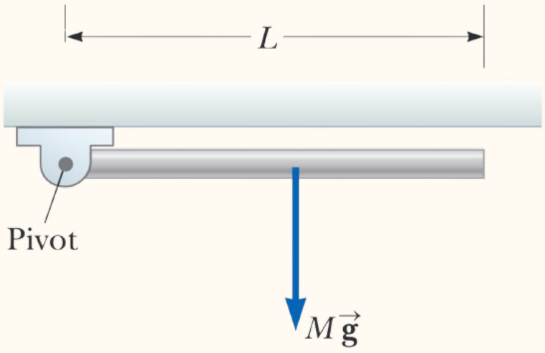
As mentioned earlier in the *relation between angular and translational quantities*, $a = r\alpha$, hence

$$\sum \tau = (ma_t) \, r = (mr\alpha) \, r = (mr^2) \, \alpha \implies \boxed{\sum \tau = I\alpha}$$

The above statement is analogous to $\vec{p} = m\vec{v}$.

Example 22.3. Serway & Jewett (9th Edition) (Ex. 10.4)

A uniform rod of length L and mass M is attached at one end to a frictionless pivot and is free to rotate about the pivot in the vertical plane.



The rod is released from rest in the horizontal position. What is the initial angular acceleration of the rod and the initial translational acceleration of its right end?

By taking moments about the pivot,

$$\sum \tau = F \cdot r = Mg\frac{L}{2}$$

Since $\sum \tau = I\alpha$,

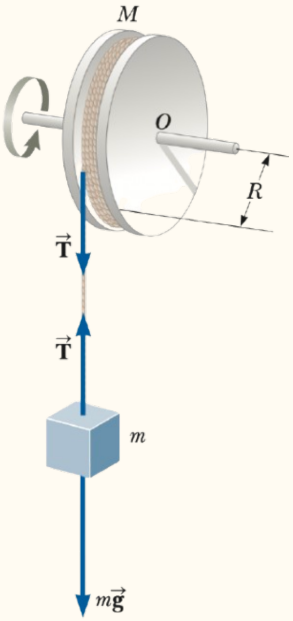
$$\alpha = \frac{\sum \tau}{I} = \frac{Mg(L/2)}{\frac{1}{3}ML^2} = \boxed{\frac{3g}{2L}}$$

$$a_t = L\alpha = \boxed{\frac{3}{2}g}$$

Example 22.4. Serway & Jewett (9th Edition) (Ex. 10.6)

A wheel of radius R, mass M and moment of inertia I is mounted on a frictionless, horizontal axle.

A light cord wrapped around the wheel supports an object of mass m. When the wheel is released, the object accelerates downward, the cord unwraps off the wheel, and the wheel rotates with an angular acceleration.



Find expressions for

1. the angular acceleration of the wheel,
2. the translational acceleration of the object, and
3. the tension in the cord.

From $\sum \tau = I\alpha$,

$$\alpha = \frac{\sum \tau}{I} = \frac{TR}{I}$$

Applying Newton's second law,

$$mg - T = ma \implies a = \frac{mg - T}{m}$$

Due to conservation of string, the translational acceleration of the object is equal to the

tangential acceleration of the cord.

$$a = R\alpha \implies \frac{mg - T}{m} = \frac{TR^2}{I}$$

$$\implies T = \boxed{\frac{mg}{1 + (mR^2/I)}}$$

$$\implies a = \boxed{\frac{g}{1 + (I/mR^2)}}$$

$$\implies \alpha = \frac{a}{R} = \boxed{\frac{g}{R + (I/mR)}}$$

22.4 Angular Momentum

Definition 22.7: Angular Momentum

The angular momentum $\vec{L}$ of a particle relative to an axis through the origin is defined by the cross product of the object's position vector $\vec{r}$ and its instantaneous linear momentum $\vec{p}$.

$$\vec{L} = \vec{r} \times \vec{p}$$

Analogous to Newton's 2nd law,

$$\sum \vec{\tau} = \frac{\mathrm{d}\vec{L}}{\mathrm{d}t}$$

Since $\vec{p} = m\vec{v}$, the magnitude of $\vec{L}$ is

$$L = mvr \sin\phi$$

where r is the distance of the particle from the origin, and ϕ is the angle between $\vec{r}$ and $\vec{p}$.

L is zero when $\vec{r}$ is parallel to $\vec{p}$, whereas $L = mvr$ when the vectors are perpendicular to each other.

Furthermore, for a rigid object rotating in the xy-plane about the z-axis with an angular speed ω, its angular momentum is

$$L_z = \sum_i L_i = \left(\sum_i m_i {r_i}^2\right)\omega = \boxed{I\omega}$$

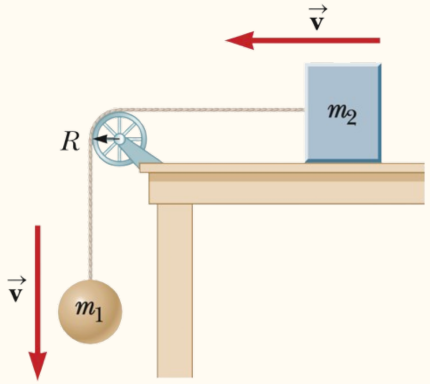
22.4.1 System of Particles

For a system that rotates about an axis, if there is a net external torque acting on the system, *the rate of change of angular momentum is equal to the torque*.

$$\boxed{\sum \vec{\tau} = \frac{\mathrm{d}\vec{L}_{\text{tot}}}{\mathrm{d}t} \iff \Delta\vec{L}_{\text{tot}} = \int \sum \vec{\tau}\ \mathrm{d}t}$$

Example 22.5. Serway & Jewett (9th Edition) (Ex 11.4)

A sphere of mass m_1 and a block of mass m_2 are connected by a light cord that passes over a pulley as shown.



The radius of the pulley is R, and the mass of the thin rim is M. The spokes of the pulley have negligible mass. The block slides on a frictionless, horizontal surface.

Find an expression for the linear acceleration of the two objects, using the concepts of angular momentum and torque.

Since all the objects move with equal speed, the total angular momentum of the sphere, block, and pulley is

$$L = (m_1 + m_2 + M)vR$$

Substituting the expression for total external torque,

$$\sum \tau = \frac{\mathrm{d}L}{\mathrm{d}t} \implies m_1 gR = \frac{\mathrm{d}}{\mathrm{d}t}[(m_1 + m_2 + M)vR]$$

$$\implies m_1 gR = (m_1 + m_2 + M)R\frac{\mathrm{d}v}{\mathrm{d}t}$$

$$\implies a = \boxed{\frac{m_1 g}{m_1 + m_2 + M}}$$

22.4.2 Conservation of Angular Momentum

Definition 22.8: Conservation of Angular Momentum

For an *isolated system*, the conservation of angular momentum states that the total angular momentum of a system is constant in both magnitude and direction if the net external torque acting on the system is zero.

By this principle,

$$\sum \vec{\tau} = 0 \implies \frac{\mathrm{d}\vec{L}_{\text{tot}}}{\mathrm{d}t} = 0$$

$$\implies \boxed{\Delta \vec{L}_{\text{tot}} = 0}$$

Hence, using the formula $L = I\omega$,

$$\vec{L}_i = \vec{L}_f \implies I_i\omega_i = I_f\omega_f = \text{constant}$$

22.5 Rotational Kinetic Energy

Definition 22.9: Rotational Kinetic Energy

The rotational kinetic energy of a system of particles of moment of inertia I and rotating with angular speed ω is

$$K_R = \frac{1}{2}I\omega^2$$

This is because it is the *sum of all the kinetic energies* of each individual particle.

$$K_R = \sum_i K_i = \sum_i \frac{1}{2}m_i{v_i}^2 = \frac{1}{2}\left(\sum_i m_i{r_i}^2\right)\omega^2$$

22.5.1 Rotational Work

Definition 22.10: Rotational Work

The **work done** by a force causing a rotational torque τ to move through an angle from θ_i to θ_f is

$$W = \int_{\theta_i}^{\theta_f} \tau \ \mathrm{d}\theta$$

Proof. To derive this, when an external force $\vec{F}$ is applied at a point on a rigid object, causing it to rotate through an infinitesimal distance $\mathrm{d}s = r\ \mathrm{d}\theta$, we see that

$$dW = \vec{F} \cdot \mathrm{d}\vec{s} = (F\sin\phi)r\ \mathrm{d}\theta = \tau\ \mathrm{d}\theta$$

Therefore,

$$\boxed{W = \int_{\theta_i}^{\theta_f} \tau \ \mathrm{d}\theta}$$

□

22.5.2 Work-Energy Theorem

Definition 22.11: Rotational Work-Energy Theorem

The **work-energy theorem** also holds **for rotational dynamics**, which states that the *net work done* by external forces in *rotating a symmetrical rigid object* about a fixed axis equals to the *change in the object's rotational energy*.

Proof. Examining net torque,

$$\sum \tau = I\alpha = I\frac{d\omega}{dt} = I\frac{d\omega}{d\theta}\frac{d\theta}{dt} = I\frac{d\omega}{d\theta}\omega$$

Note that from above, $dW = \sum \tau \, d\theta$. Hence,

$$\sum \tau \, d\theta = dW = I\omega \, d\omega \implies W = \int_{\omega_i}^{\omega_f} I\omega \, d\omega = \frac{1}{2}I{\omega_f}^2 - \frac{1}{2}I{\omega_i}^2$$

□

22.5.3 Power

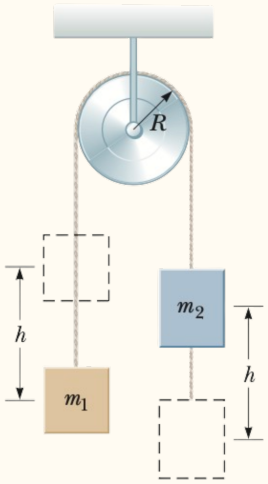
The **power** delivered by such a force is

$$P = \frac{dW}{dt} = \boxed{\tau\omega}$$

$$\langle P \rangle = \frac{\Delta W}{\Delta t}$$

Example 22.6. Serway & Jewett (9th Edition) (Ex. 10.12)

Two blocks having different masses m_1 and m_2 are connected by a string passing over a pulley.



The pulley has a radius R and moment of inertia I about its axis of rotation. The system is released from rest.

Using this information, find

1. the translational speeds of the blocks after block 2 descends through a distance h, and
2. the angular speed of the pulley at this time.

Using the isolated system model, where v_f is the final translational speed of both blocks, and ω_f is the final angular speed of the pulley,

$$\Delta K = -\Delta U \implies \frac{1}{2}m_1{v_f}^2 + \frac{1}{2}m_2{v_f}^2 + \frac{1}{2}I{\omega_f}^2 = -(m_1gh - m_2gh)$$

Substituting $v = R\omega$,

$$\frac{1}{2}m_1{v_f}^2 + \frac{1}{2}m_2{v_f}^2 + \frac{1}{2}I\frac{{v_f}^2}{R^2} = m_2gh - m_1gh \implies \frac{1}{2}\left(m_1 + m_2 + \frac{I}{R^2}\right){v_f}^2 = (m_2 - m_1)gh$$

$$\implies v_f = \boxed{\sqrt{\frac{2(m_2 - m_1)gh}{m_1 + m_2 + I/R^2}}}$$

$$\implies \omega_f = \boxed{\frac{1}{R}\sqrt{\frac{2(m_2 - m_1)gh}{m_1 + m_2 + I/R^2}}}$$

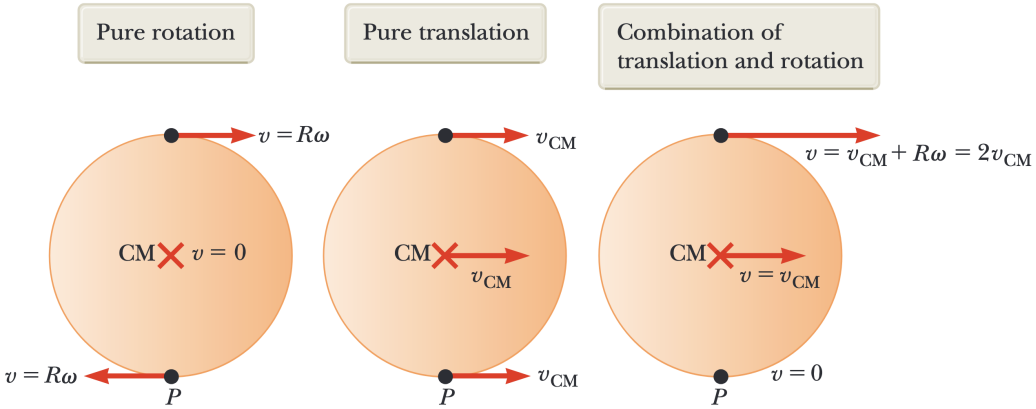
22.6 Rolling Motion

For a uniform cylinder of radius R undergoing **pure rolling motion** (i.e. no slipping), its centre of mass will always follow a straight-line trajectory.

$$v_{CM} = R\omega_{CM} \implies a_{CM} = R\alpha_{CM}$$

This holds only for cylinders and spheres undergoing pure rolling motion. More practically, **rolling friction** may cause mechanical energy to transform to internal energy. This rolling friction needs to be considered in most questions.

It is important to consider the combination between the rotational motion of the object and the translational motion of the object.



When we consider the combination, we can model the rolling motion of a body as being equivalent to pure rotation about a fixed pivot point P (where the body touches the ground). As such, the *total kinetic energy* of the rolling body is given as

$$K = \frac{1}{2} I_p \omega^2$$

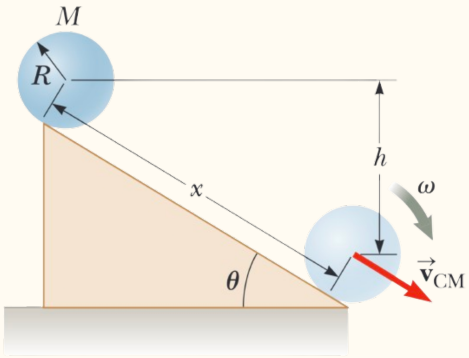
Applying the parallel-axis theorem and the pure rolling equation $v_{CM} = R\omega_{CM}$,

$$K = \frac{1}{2} I_p \omega^2 \implies K = \frac{1}{2} I_{CM} \omega^2 + \frac{1}{2} (MR^2) \omega^2$$

$$\implies \boxed{K = \frac{1}{2} I_{CM} \omega^2 + \frac{1}{2} M {v_{CM}}^2}$$

Example 22.7. Serway & Jewett (9th Edition) (Ex. 10.13) modified

A uniform cylinder of radius R and mass M rolls from rest without slipping down an inclined plane which makes an angle θ with the horizontal. The center of mass of the cylinder has dropped a vertical distance h when it reaches the bottom of the inclined plane.



Ignoring rolling friction, what is the speed of the center of the mass of the cylinder when it reaches the bottom of the inclined in terms of h and the acceleration of free fall g only?

Since the cylinder undergoes pure rolling motion, the equation $v_{CM} = R\omega_{CM}$ applies. The final kinetic energy of the cylinder is

$$K_f = \frac{1}{2} I_{CM} \omega^2 + \frac{1}{2} M {v_{CM}}^2 = \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \left(\frac{v}{R} \right)^2 + \frac{1}{2} M v^2$$

Due to the conservation of energy,

$$\Delta K + \Delta U = 0 \implies Mgh = \frac{1}{2} \left(\frac{1}{2} MR^2 \right) \left(\frac{v}{R} \right)^2 + \frac{1}{2} M v^2$$

$$\implies gh = \frac{1}{4} v^2 + \frac{1}{2} v^2$$

$$\implies v^2 = \frac{4}{3} gh$$

$$\implies \boxed{v = \sqrt{\frac{4}{3} gh}}$$

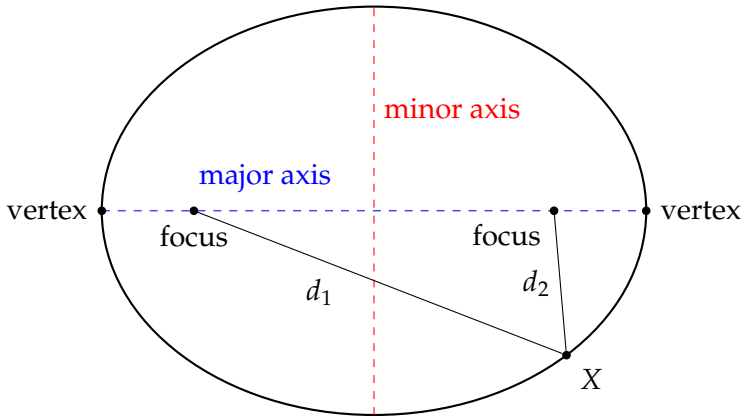
23 Planetary & Satellite Motion

> The bright sun was extinguish'd, and the stars
> Did wander darkling in the eternal space,
> Rayless, and pathless, and the icy earth
> Swung blind and blackening in the moonless air.
>
> — LORD BYRON, *Darkness*

23.1 Elliptical Orbits

23.1.1 Ellipse

An **ellipse** is mathematically defined by choosing two points, each called a **focus**, and drawing a curve through which the sum of distances from each focus d_1 and d_2 is a constant.



The longest distance through the center between points on the ellipse is the **major axis** (with the line from one center to a vertex being a **semi-major axis**), whereas the shortest distance through the center is a **minor axis**.

It is common to denote the length of the semi-major axis as a, and the length of the minor axis as b. Furthermore, the distance of each focus from the centre is denoted as c.

23.1.2 Periapsis & Apoapsis

The **periapsis** is the point of closest approach, and the **apoapsis** is the point of furthest approach. The periapsis and apoapsis are vertices on the elliptical orbit, the arrangement depending which focus the body is orbiting around.

The distances of the periapsis and apoapsis from a focus are denoted as r_p and r_a respectively. The sum of these two distances is equal to the length of the major axis.

$$2a = r_p + r_a$$

23.1.3 Eccentricity

The **eccentricity** of an ellipse is defined as

$$e = \frac{c}{a} = \sqrt{1 - \left(\frac{b}{a}\right)^2}$$

Eccentricity describes shape. For a circle, the focus is at the origin, hence $c = 0 \implies e = 0$. The *longer and thinner* an ellipse is, the *greater* e is.

Expressions for r_p and r_a utilise eccentricity.

$$r_p = a(1-e),\ r_a = a(1+e)$$

23.1.4 Semi-Latus Rectum

The length of a chord through any focus, perpendicular to the major axis, is called the **latus rectum**. One half of it, which is the vertical distance from one focus to the boundary of the ellipse, is the **semi-latus rectum** ℓ.

$$\ell = \frac{b^2}{a} = a(1-e^2)$$

23.2 Vis-Viva Equation

As mentioned later in our discussion of Kepler's laws, a **Keplerian orbit** is an orbit that is elliptical, parabolic, or hyperbolic. In this syllabus, we are merely concerned with elliptical orbits.

This introduces all elliptical orbits, not ones that are just circular. While Newton's formulation for gravity still holds for this broader class of orbits, certain assumptions and formulas that are true for circular orbits do not hold true for all elliptical orbits.

One notable case arises when we are finding the velocity of an orbiting object. An expression for velocity can be easily found for a circular orbit, by considering the centripetal acceleration offered by gravitational force.

$$\frac{GMm}{r^2} = m\frac{v^2}{r} \implies v = \sqrt{\frac{GM}{r}}$$

However, this is not as straightforward for elliptical orbits. Herein, we introduce a formula that gives the velocity of an object in all Keplerian orbits.

Proposition 23.1: Vis-Viva Equation

For any object in a Keplerian orbit around another object of mass M, with separation r and semi-major axis a, its velocity v is given by

$$v = \sqrt{GM\left(\frac{2}{r} - \frac{1}{a}\right)}$$

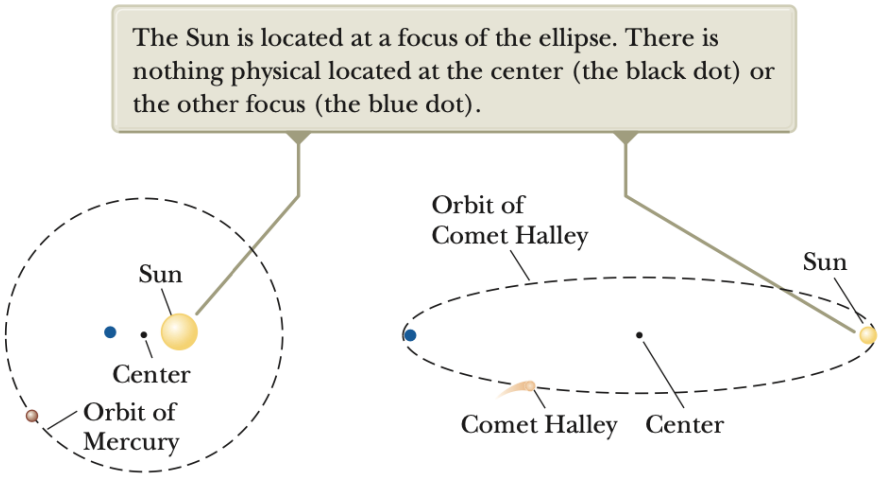
This is known as the **vis-viva equation**.

23.3 Kepler's Laws

23.3.1 Kepler's First Law

Definition 23.1: Kepler's First Law

Kepler's first law states that the orbit of a planet is an ellipse with the Sun at one of the two foci.



The aphelion is the point where the planet is farthest away from the Sun, and the perihelion is the point where the planet is nearest to the Sun.

23.3.2 Kepler's Second Law

Definition 23.2: Kepler's Second Law

Kepler's second law states that the radius vector drawn from the Sun to a moving planet sweeps out equal areas in equal time intervals.

Proof. This is a result of the conservation of angular momentum. For a planet of mass M moving around the Sun with *no net torque* (since gravitational force $\vec{F}$ is parallel to $\vec{r}$),

$$\begin{aligned} \Delta\vec{L} = 0 &\implies \vec{L} = \vec{r} \times \vec{p} = \text{constant} \\ &\implies L = M|\vec{r} \times \vec{v}| = \text{constant} \end{aligned}$$

Considering a geometric interpretation, in a time interval dt, the radius vector $\vec{r}$ sweeps out a triangle of area dA, given as

$$\mathrm{d}A = \frac{1}{2}|\vec{r} \times \mathrm{d}\vec{r}| = \frac{1}{2}|\vec{r} \times \vec{v}|\,dt$$

Substituting the earlier expression,

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{L}{2M} = \text{constant} \implies \text{equal areas in equal time intervals}$$

□

23.3.3 Kepler's Third Law

Definition 23.3: Kepler's Third Law

Kepler's third law states that *square of a planet's period of revolution* is directly proportional to the *cube of the semi-major axis* of its orbit around the Sun.

$$T^2 = \frac{4\pi^2}{GM}a^3$$

Proof. We begin with Kepler's second law. For a orbital period of T, the total area swept out in one orbit is

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{L}{2m} \implies A = \frac{L}{2m}T$$

An ellipse of semi-major and semi-minor axes a and b has an area described by

$$A = \pi ab = \pi a^2\sqrt{1-e^2}$$

Substituting this area,

$$\begin{aligned}
\pi a^2\sqrt{1-e^2} = \frac{L}{2m}T &\implies T = \frac{2m}{L}\left(\pi a^2\sqrt{1-e^2}\right) \\
&\implies T^2 = \frac{4m^2}{L^2}(\pi^2 a^4)(1-e^2) \\
&\implies T^2 = \frac{4m^2}{L^2}(\ell)(\pi^2 a^3)
\end{aligned}$$

The last step above introduces the semi-latus rectum ℓ of the elliptical orbit, which we can express using the equation $\ell = a(1-e^2)$.

At the periapsis with distance $r_p = a(1-e)$, using the vis-viva equation,

$$\begin{aligned}
v^2 = GM\left(\frac{2}{r_p} - \frac{1}{a}\right) &= GM\left(\frac{2}{a(1-e)} - \frac{1}{a}\right) \\
&= GM\left(\frac{1+e}{a(1-e)}\right)
\end{aligned}$$

Since angular momentum $L = mvr$,

$$\begin{aligned}
L^2 = (mvr_p)^2 &\implies \frac{L^2}{m^2} = GM\left(\frac{1+e}{a(1-e^2)}\right)(a(1-e))^2 \\
&\implies \frac{L^2}{GMm^2} = a(1-e^2) = \ell
\end{aligned}$$

Substituting the expression $\ell = L^2/(GMm^2)$,

$$\begin{aligned}
T^2 = \frac{4m^2}{L^2}\left(\frac{L^2}{GMm^2}\right)(\pi^2 a^3) = \frac{4\pi^2}{GM}a^3 \\
\implies T^2 \propto a^3
\end{aligned}$$

Thus, we conclude that the ratio of the square of the orbital period to the cube of the semi-major axis is always constant, with the constant of proportionality being $4\pi^2/(GM)$. □

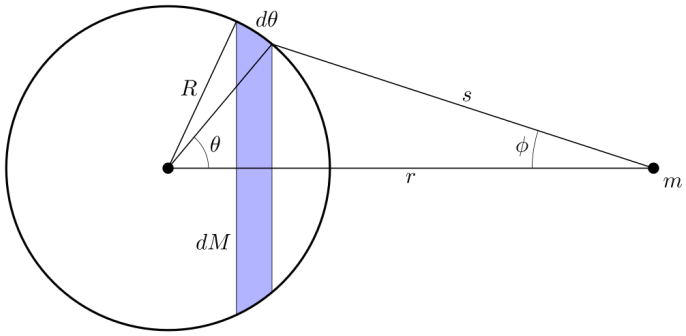
23.4 Newton's Shell Theorem

Definition 23.4: Shell Theorem

The shell theorem comprises of two corollaries, stating that

1. a spherically symmetric body affects an external point mass as though all the mass of the spherical body were *concentrated at its center*;
2. if the body is a spherically symmetric shell, *no net gravitational force is exerted* by the shell *on any object inside*, regardless of the object's location within the shell.

Proof. Consider a point mass m at a distance r from the centre of a spherical shell of mass M and radius R.



Consider a thin ring of mass corresponding to an angle θ between r and any point on the shell. Let s denote the distance of m and the ring.

Letting $\mu = \dfrac{M}{4\pi R^2}$ be the surface mass density of the shell, we can find the mass of a thin ring dm as shown.

$$\begin{aligned} \mathrm{d}M &= \mu(2\pi R\sin\theta)(R\ \mathrm{d}\theta) \\ &= \left(\frac{M}{4\pi R^2}\right)(2\pi R^2\sin\theta)\ \mathrm{d}\theta \\ &= \left(\frac{M}{2}\sin\theta\right)\mathrm{d}\theta \end{aligned}$$

By cosine rule,

$$s^2 = R^2 + r^2 - 2Rr\cos\theta$$

Differentiating with respect to θ, we obtain

$$2s\left(\frac{\mathrm{d}s}{\mathrm{d}\theta}\right) = 2Rr\sin\theta \implies \sin\theta = \frac{s}{Rr}\left(\frac{\mathrm{d}s}{\mathrm{d}\theta}\right)$$

The gravitational potential energy between the ring of mass $\mathrm{d}M$ and mass m is

$$\begin{aligned}\mathrm{d}U &= -\frac{Gm}{s}\,\mathrm{d}M \\ &= -\frac{Gm}{s}\left(\frac{M}{2}\sin\theta\right)\mathrm{d}\theta \\ &= -\frac{Gm}{s}\left(\frac{M}{2}\cdot\frac{s}{Rr}\left(\frac{\mathrm{d}s}{\mathrm{d}\theta}\right)\right)\mathrm{d}\theta \\ &= -\frac{GMm}{2Rr}\,\mathrm{d}s\end{aligned}$$

To prove the **first corollary**, we consider the region outside the shell, $r > R$.

$$U = \int_{r-R}^{r+R} -\frac{GMm}{2Rr}\,\mathrm{d}s = -\frac{GMm}{2Rr}[s]_{r-R}^{r+R} = \boxed{-\frac{GMm}{r}}$$

Therefore,

$$F = -\frac{\mathrm{d}U}{\mathrm{d}r} = -\frac{GMm}{r^2}$$

To prove the **second corollary**, we consider the region inside the shell, $r < R$.

$$U = \int_{R-r}^{r+R} -\frac{GMm}{2Rr}\,\mathrm{d}s = -\frac{GMm}{2Rr}[s]_{R-r}^{r+R} = \boxed{-\frac{GMm}{R}}$$

Therefore,

$$F = -\frac{\mathrm{d}U}{\mathrm{d}r} = 0$$

□

23.5 Effective Radial Potential

Definition 23.5: Effective Radial Potential

The effective radial potential energy of a mass m is defined as the sum of the r-dependent terms of its energy.

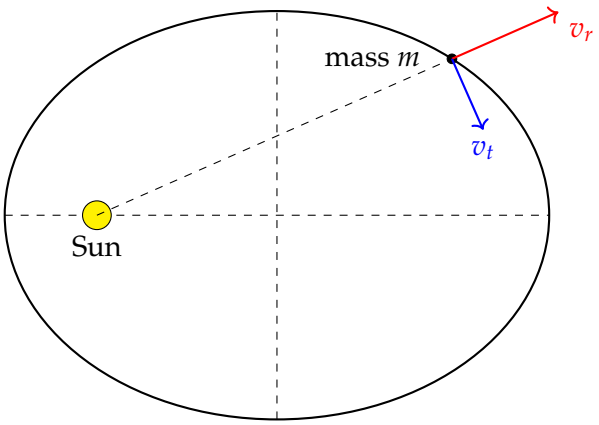
$$U_{\text{eff}} = \frac{L^2}{2mr^2} - \frac{GMm}{r}$$

where the first term is the transverse kinetic energy, and the second term is the **gravitational potential energy**.

The effective radial potential of the mass m may be used to determine the *orbits* of planets, and perform semi-classical atomic calculations.

23.5.1 Derivation

Consider a mass m orbiting around the Sun of mass M. The velocity of the mass m can be resolved into the radial component v_r and the tranverse component v_t, as shown below.



Thus, the kinetic energy of the mass is given as

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m({v_r}^2 + {v_t}^2)$$

From the angular momentum of the mass,

$$L = m\vec{v} \times \vec{r} = mv_t r \implies v_t = \frac{L}{mr}$$
$$\implies \frac{1}{2}m{v_t}^2 = \frac{L^2}{2mr^2}$$

Considering the Hamiltonian of the system,

$$E = K + U = \frac{1}{2}m{v_r}^2 + \frac{L^2}{2mr^2} - \frac{GMm}{r}$$

Therefore, taking the terms that depend on r, we obtain the effective radial potential.

$$U_{\text{eff}} = \frac{L^2}{2mr^2} - \frac{GMm}{r}$$

23.5.2 Turning Points

A **turning point** is a finite radial value $r = r^*$ such that

$$U_{\text{eff}} = E \implies \frac{1}{2}m{v_r}^2 = 0$$
$$\implies v_r = 0$$

This implies that a turning point exists in a path of mass m where it has no radial component to its velocity.

Example 23.1.

Determine the roots of $U_{\text{eff}} = E$. Hence, or otherwise, show that the length of the planet's semi-major axis is given by

$$a = -\frac{GMm}{2(E)}$$

From $U_{\text{eff}} = \sum E$,

$$\frac{L^2}{2m}\left(\frac{1}{r}\right)^2 - GMm\left(\frac{1}{r}\right) - E = 0$$

Using the quadratic formula,

$$\frac{1}{r} = \frac{GMm \pm \sqrt{(GMm)^2 - 4(L^2/2m)(-E)}}{L^2/m}$$

$$= \frac{1 \pm \sqrt{1 + 2L^2(E)/m(GMm)^2}}{L^2/(GMm)m}$$

$$\implies \frac{L^2}{GMm^2}\frac{1}{r} = 1 \pm \sqrt{1 + \frac{2L^2(E)}{m(GMm)^2}}$$

Finding the length of the semi-major axis, we employ Vieta's formula. Recall that r_p and r_a represent the distances of the periapsis and apoapsis respectively.

$$\frac{1}{r_a} + \frac{1}{r_p} = \frac{r_a + r_p}{r_a r_p} = \frac{GMm}{L^2/2m}$$

$$r_a r_p = \left(\frac{1}{r_a} \cdot \frac{1}{r_p}\right)^{-1} = -\frac{L^2/2m}{E}$$

Since $a = \frac{1}{2}(r_a + r_p)$,

$$a = \frac{1}{2} \cdot \frac{GMm}{L^2/2m} \cdot -\frac{L^2/2m}{E}$$

$$a = \boxed{-\frac{GMm}{2E}}$$

23.5.3 Bounded States

The graph below shows the relationship of U_{eff} against r.

Circular Orbits

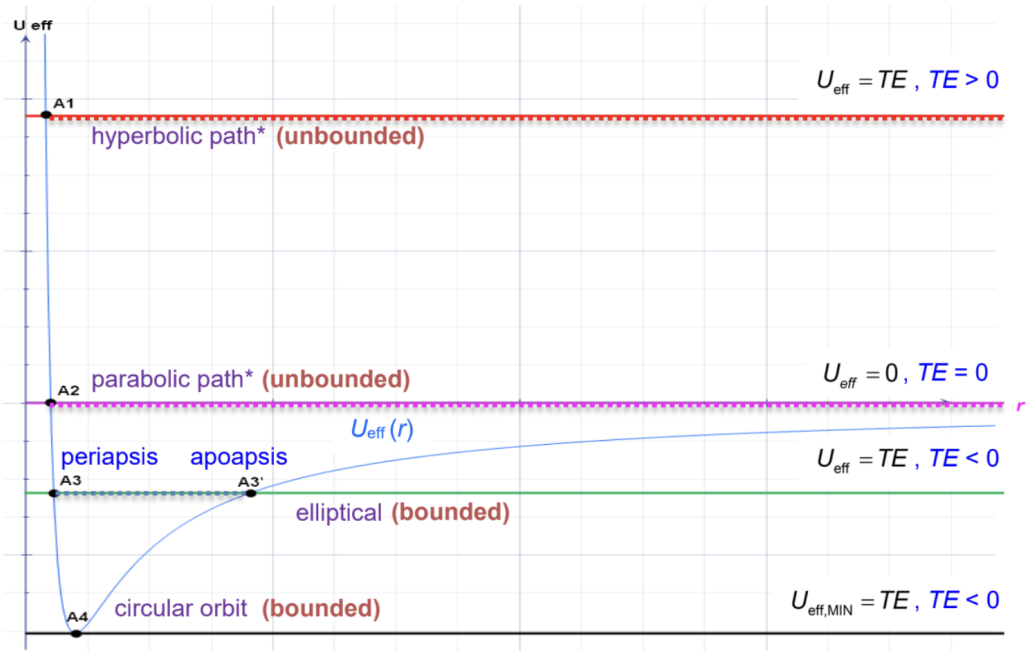
The lowest energy state $U_{\text{eff, min}}$ corresponds to the *minimum* of the effective radial potential.

Setting the derivative of U_{eff} to zero,

$$\frac{\mathrm{d}U_{\text{eff}}}{\mathrm{d}t} = -\frac{L^2}{mr^3} + \frac{GMm}{r^2} = 0 \implies r = \frac{L^2}{GMm^2}$$

For a circular orbit, the distance between m and M does not change.

$$v_t = \frac{L}{mr} = \frac{GMm}{mv_t r} = \sqrt{\frac{GM}{r}}$$



The speed calculated above is the speed *required for a circular orbit*. Let this speed be v_c.

Elliptical Orbits

Recall that the **escape velocity** is given as

$$v_e = \sqrt{\frac{2GM}{r}}$$

The speed of the elliptical orbit can take on two states, depending on whether it has surpassed the speed required for a circular orbit.

$$0 < v_t < v_c \text{ or } v_c < v_t < v_e$$

23.6 Hohmann Orbital Transfer

A Hohmann orbital transfer is a two-impulse elliptical transfer between two co-planar circular orbits.

The transfer itself consists of a periapsis at the inner orbit, before moving into an elliptical orbit, and ending off with an apoapsis at the outer orbit.

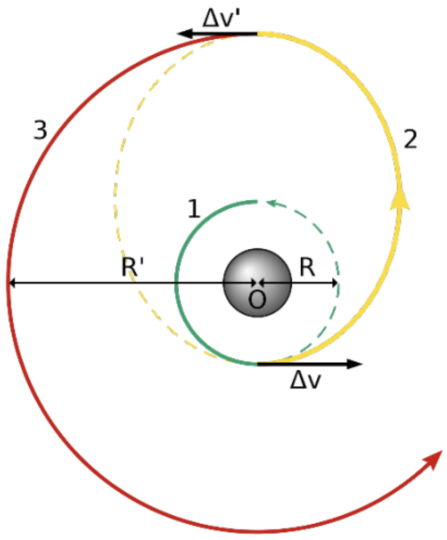
One assumption is that *velocity change occurs instantaneously* for the spacecraft. In the following, we examine the efficiency of the Hohmann orbital transfer.

If $\vec{v}_i$ is the velocity of the initial orbit and $\vec{v}_f$ is the velocity of the final orbit, the change in velocity $\Delta\vec{v}$ required to perform the orbital transfer is

$$\Delta\vec{v} = \vec{v}_f - \vec{v}_i$$

Using the law of cosines,

$$v_f{}^2 = v_i{}^2 + \Delta v^2 - 2(v_i)(\Delta v)\sin\beta$$



where β is the angle between $\vec{v}_i$ and $\Delta\vec{v}$.

The specific energy charge (energy change per unit mass) is given by

$$\frac{\Delta E}{m} = \frac{1}{2}\Delta v^2 + (v_i)(\Delta v)\cos\beta$$

Note that this specific energy change is greatest when $\beta = 0$ and v_i is at its maximum. In other words, the most energy-efficient directions of impulse would be collinear with the tangential velocity and apply it at the elliptical orbit's periapsis, where speed is greatest.

Example 23.2.

A communication satellite was carried by the Space Shuttle into low earth orbit at an altitude of 322 km and is to be transferred to a geostationary orbit at an alitude of 35 860 km using a Hohmann transfer.

Determine the magnitude of the total change in velocity after the Hohmann transfer is complete given the radius of Earth is 6 378 km.

For the low earth orbit,

$$\begin{aligned} v_1 = \sqrt{\frac{GM}{r_1}} &= R_{\text{earth}}\sqrt{\frac{g}{r_1}} \\ &= 7718 \text{ m s}^{-1} \end{aligned}$$

For the geostationary orbit,

$$\begin{aligned} v_2 = \sqrt{\frac{GM}{r_2}} &= R_{\text{earth}}\sqrt{\frac{g}{r_2}} \\ &= 3073 \text{ m s}^{-1} \end{aligned}$$

For the Hohmann transfer orbit, using the conservation of energy, at the periapsis,

$$K + U = \frac{-GMm}{2a} \implies \frac{1}{2}m{v_p}^2 - \frac{GMm}{r_1} = -\frac{GMm}{(r_1 + r_2)}$$

$$\implies -\frac{gR_{\text{earth}}^2}{r_1 + r_2} = \frac{1}{2}{v_p}^2 - \frac{gR_{\text{earth}}^2}{r_1}$$

$$\implies \frac{1}{2}{v_p}^2 = gR_{\text{earth}}^2\left(\frac{r_2}{r_1(r_1 + r_2)}\right)$$

$$\implies v_p = R_{\text{earth}}\sqrt{2g\left(\frac{r_2}{r_1(r_1 + r_2)}\right)}$$

$$\implies v_p = 10140 \text{ m s}^{-1}$$

Similarly, at the apoapsis,

$$K + U = \frac{-GMm}{2a} \implies \frac{1}{2}m{v_p}^2 - \frac{GMm}{r_2} = -\frac{GMm}{(r_1 + r_2)}$$

$$\implies v_a = R_{\text{earth}}\sqrt{2g\left(\frac{r_1}{r_2(r_1 + r_2)}\right)}$$

$$\implies v_a = 1608 \text{ m s}^{-1}$$

Therefore, the magnitude of the total change in velocity

$$\Delta v = (v_p - v_1) + (v_2 - v_a) = (10140 - 7718) + (3073 - 1608) \implies \boxed{\Delta v = 3890 \text{ m s}^{-1}}$$

24 Electric & Magnetic Fields

24.1 Properties of Electric Charges

There are three main properties of electric charges.

1. Charge comes in two varieties, termed **positive** (protons) and **negative** (electrons). Charges of the same sign **repel** one another and charges with opposite signs **attract** one another.
2. Charge is **always conserved** in an isolated system. Charge is neither created nor destroyed, but rather transferred from one object to another.
    - The **global** conservation of charge states that the total charge of the universe is fixed for all time.
    - The **local** conservation of charge states that for a charge to be transferred from one location to another, it must have passed along some continuous path (i.e. charges cannot suddenly teleport)
3. Charge is **quantised**, coming only in discrete lumps.
    - For an electric charge q, we can write $q = \pm Ne,\ N \in \mathbb{Z}$.
    - The electron has a charge $-e$ and the proton has a charge $+e$. The neutron carries no charge.

24.1.1 Conductors

Electrical conductors are materials in which some of the electrons are free electrons that are not bound to atoms and can move relatively freely through the material. This includes copper, aluminium, and silver.

Likewise, **electrical insulators** are materials in which all electrons are bound to atoms and cannot move freely through the material. This includes glass, rubber, and dry wood.

Semiconductors are between insulators and conductors, with its electrical properties able to be changed over many orders of magnitude by the addition of controlled amounts of certain atoms to the materials. This includes silicon and germanium.

24.1.2 Charging by Induction

Conductors are charged by a process known as **induction**.

Consider a neutral conducting sphere insulated from the ground. Since the charge is zero, there are an *equal numhber of electrons and protons* in the sphere.

- When a *negatively-charged* rubber rod is brought near the sphere, electrons in the sphere are repulsed and move to the opposite side of the sphere.
- This migration leaves the side of the sphere near the rod with an *effective positive charge* (purely due to the absence of electrons, for *protons do not move*).
- If the sphere has a *ground*, the electrons in the conductor are so strongly repelled that they move into the Earth.

- In this scenario, positive charges are *induced* on the sphere. For charging by induction, the position of the grounding wire does not matter — the induced charges will always remain on the sphere.

Charging an object by induction requires no contact. This is in contrast to charging by rubbing (by conduction).

24.2 Electric Field

To begin, it may be important to re-emphasise upon the notion of a net force.

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \cdots + \vec{F}_n$$

We consider the case of **electrostatics** where all source charges are stationary (but the test charge may be moving). We use the term **point charge** to refer to a charged particle of negligible size.

24.2.1 Coulomb's Law

Definition 24.1: Coulomb's Law (H3)

Coulomb's law describes the **electric force** between two charges.

$$\vec{F} = \frac{1}{4\pi\varepsilon_o}\frac{q_1 q_2}{r^2}\hat{\mathbf{r}}$$

where

- $\vec{F}$ is the electric force (N)
- q_1 is the charge of the first point charge (C)
- q_2 is the charge of the second point charge (C)
- r is the separation between the two charges (m)
- $\hat{\mathbf{r}}$ is the separation unit vector from q_1 to q_2, purely to provide direction
  - $\vec{r} = \vec{x} - \vec{x'}$, where $\vec{x}$ is the location of q_1 and $\vec{x'}$ is the location of q_2
- ε_0 is the permittivity of free space, with a value of $8.85 \times 10^{-12}\ \text{C}^2\,\text{kg}^{-1}\,\text{m}^{-3}\,\text{s}^2$

If $q_1 q_2$ is positive, the electric force on one particle is directed away from the other particle. If the product is negative, the electric force on both particles point towards each other.

24.2.2 Electric Field Strength

If we have *several* point charges $q_1, q_2, \cdots, q_n$ at distances $r_1, r_2, \cdots, r_n$ from a source charge Q, the total force on Q is

$$\begin{aligned}\sum \vec{F} &= \vec{F}_1 + \vec{F}_2 + \cdots + \vec{F}_n \\ &= \frac{1}{4\pi\varepsilon_0}\left(\frac{q_1 Q}{{r_1}^2}\hat{\mathbf{r}}_1 + \frac{q_2 Q}{{r_2}^2}\hat{\mathbf{r}}_2 + \cdots + \frac{q_n Q}{{r_n}^2}\hat{\mathbf{r}}_3\right) \\ &= \frac{Q}{4\pi\varepsilon_0}\left(\frac{q_1}{{r_1}^2}\hat{\mathbf{r}}_1 + \frac{q_2}{{r_2}^2}\hat{\mathbf{r}}_2 + \cdots + \frac{q_n}{{r_n}^2}\hat{\mathbf{r}}_n\right)\end{aligned}$$

This can be expressed as

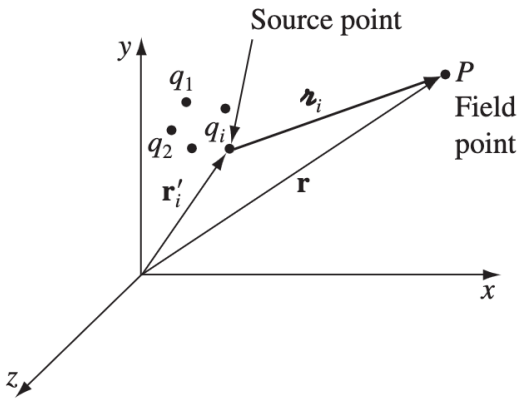
$$\vec{F}_{\text{net}} = q\vec{E}$$

Definition 24.2: Electric Field Strength (H3)

The **electric field strength** $\vec{E}$ at a point in the field is defined as the electric force per unit positive charge experienced by a charge placed at that point.

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i^2}\hat{\mathbf{r}}_\mathbf{i}$$

$\vec{E}$ is a function of the position vector $\vec{r}$, which characterising the distance between the source point q_i and a field point Q. This is because the separation vectors $\vec{r_i}$ depend on the location of the **field point** Q.



Physically, $\vec{E}(\vec{r})$ is the force per unit charge that would be exerted on a positive test charge, if you were to place one at P.

24.2.3 Electric Field Lines

Electric field patterns are represented by **electric field lines**, related to the electric field in a region of space in the following manner.

1. The *electric field vector* $\vec{E}$ is tangential and parallel to the electric field line at each point. The direction of the line is that of the *force on a positive charge* placed in the field.
2. The number of lines per unit area is proportional to the magnitude of the electric field in that region — field lines are close together for stronger fields, and far apart for weaker fields.

Field lines begin on positive charges and end on negative charges — they cannot terminate in midair.

For two charges of equal magnitude, they cancel out. The number of field lines leaving the positive charge must be equal to the number of field lines terminating at the negative charge.

This is similar to what is described in the H2 syllabus.

24.2.4 Motion of Charged Particles

When a particle of charge q and mass m is placed in an electric field $\vec{\boldsymbol{E}}$, the electric force exerted on the charge is $q\vec{\boldsymbol{E}}$.

Given that this force is the net force,

$$\vec{\boldsymbol{F}}_{\text{net}} = q\vec{\boldsymbol{E}} = m\vec{\boldsymbol{a}}$$

therefore giving the particle's acceleration to be

$$\vec{\boldsymbol{a}} = \frac{q\vec{\boldsymbol{E}}}{m}$$

24.2.5 Continuous Charge Distribution

The previous formula for electric field strength holds true *only if* the source of the field is a collection of **discrete point chrages**.

If the charge is distributed continuously over some region, then the sum becomes an integral. This integration is a vector operation and must be treated appropriately.

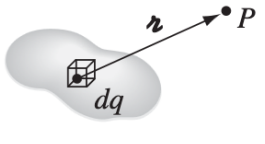
$$\vec{\boldsymbol{E}}(\vec{\boldsymbol{x}}) = \frac{1}{4\pi\varepsilon_0}\int \frac{1}{r}\hat{\mathbf{r}}\,\mathrm{d}q$$

This calculation can be extended for a line charge, area charge, and volume charge. We can use the concept of **charge density**. If a charge Q is uniformly distributed along a line of length l / area A / volume V, the linear / surface / volume charge density λ/ σ / ρ (C/m or C/m^2 or C/m^3) is

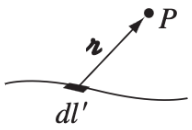
$$\lambda = \frac{Q}{l} \qquad \sigma = \frac{Q}{A} \qquad \rho = \frac{Q}{V}$$

If the charge is non-uniformly distributed over a line, surface, or volume, the amounts of charge $\mathrm{d}q$ in a small volume is equivalent to the following infinitesimal values.

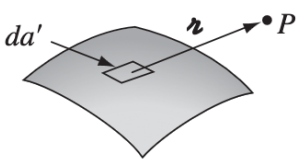
$$\mathrm{d}q \to \lambda\,dl \sim \sigma\,\mathrm{d}A \sim \rho\,dV$$



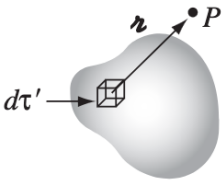
(a) Continuous distribution



(b) Line charge, λ



(c) Surface charge, σ



(d) Volume charge, ρ

Consequently, specifically, for a line charge,

$$\vec{E}(\vec{x}) = \frac{1}{4\pi\varepsilon_0}\int \frac{\lambda(\vec{x'})}{r^2}\hat{\mathbf{r}}\,\mathrm{d}l$$

For a surface charge,

$$\vec{E}(\vec{x}) = \frac{1}{4\pi\varepsilon_0}\int \frac{\sigma(\vec{x'})}{r^2}\hat{\mathbf{r}}\,\mathrm{d}A$$

And, rather importantly, for a volume charge,

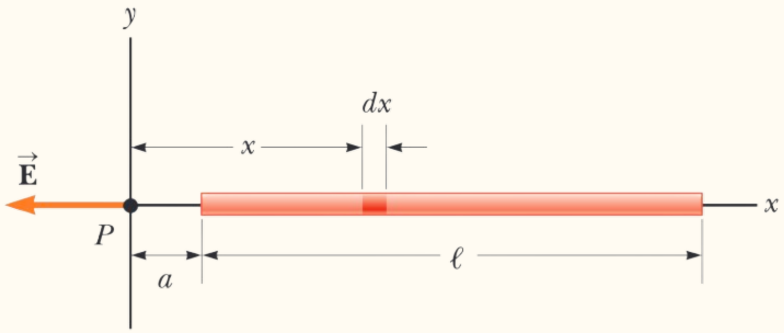
$$\vec{E}(\vec{x}) = \frac{1}{4\pi\varepsilon_0}\int \frac{\rho(\vec{x'})}{r^2}\hat{\mathbf{r}}\,\mathrm{d}V$$

where $\vec{r}$ is the vector from $\mathrm{d}q$ (and naturally from $dl, \mathrm{d}A$ and dV) to the field point $\mathbf{x}$.

Uniformly Charged Rod

Example 24.1. Serway & Jewett (9th Edition) (Ex. 23.7)

A rod of length l has a uniform positive charge per unit length λ and a total charge Q. We wish to calculate the electric field at a point P that is located along the long axis of the rod and a distance a from one end.



Consider an infinitesimal element of the rod with length $\mathrm{d}x$. The charge on this segment is $\mathrm{d}q$, which can be expanded to form $\lambda\,\mathrm{d}x$. Thus, the electric field contribution from this segment is given by

$$\mathrm{d}\vec{E} = -\frac{1}{4\pi\varepsilon_0}\frac{\mathrm{d}q}{x^2}\hat{\mathbf{x}} = -\frac{1}{4\pi\varepsilon_0}\frac{\lambda\,\mathrm{d}x}{x^2}\hat{\mathbf{x}}$$

(The negative sign in this equation is due to the fact that the electric field points rightward, where we take leftwards to be positive).

Integrating, we obtain the total electric field

$$\begin{aligned}\vec{E} = \int \mathrm{d}\vec{E} &= -\frac{1}{4\pi\varepsilon_0}\int_a^{a+l}\frac{\lambda\,\mathrm{d}x}{x^2}\hat{\mathbf{x}} \\ &= -\frac{\lambda}{4\pi\varepsilon_0}\left[-\frac{1}{x}\right]_a^{a+l}\hat{\mathbf{x}}\end{aligned}$$

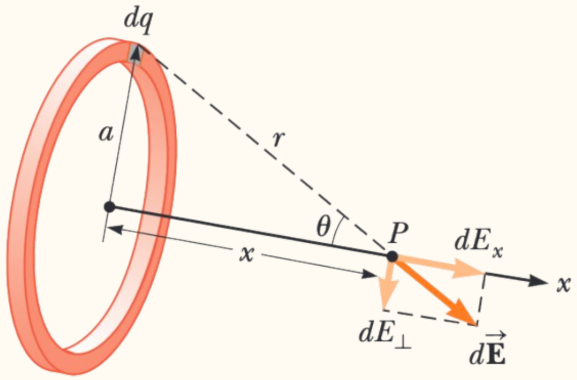
Since $\lambda = \frac{Q}{l}$,

$$\vec{E} = -\frac{1}{4\pi\varepsilon_0}\frac{Q}{l}\left(\frac{1}{a} - \frac{1}{a+l}\right)\hat{\mathbf{x}} = \boxed{-\frac{1}{4\pi\varepsilon_0}\frac{Q}{a(a+l)}\hat{\mathbf{x}}}$$

Uniformly Charged Ring

Example 24.2. Serway & Jewett (9th Edition) (Ex. 23.8)

A ring of radius a carries a uniformly distributed positive total charge Q. Calculate the electric field due to the ring at a point P lying a distance x from its center along the central axis perpendicular to the plane of the ring.



Consider an infinitesimal element of the ring with length dx. We notice that for each element, there is another element directly opposite to it on the ring. With reference to P, the vertical component of each element will be cancelled out, leading to the overall electric field solely possessing a horizontal component.

The charge on this segment is $\mathrm{d}q$.

Thus, the electric field contribution from the segment is

$$\mathrm{d}\vec{E} = \mathrm{d}E_x\hat{\mathbf{x}} = |\,\mathrm{d}\vec{E}\,|\cos\theta\,\hat{\mathbf{x}} = \frac{1}{4\pi\varepsilon_0}\frac{\mathrm{d}q}{r^2}\cos\theta\,\hat{\mathbf{x}}$$

Integrating, we obtain the total electric field.

$$\begin{aligned}\vec{E} = \int \mathrm{d}\vec{E} &= \frac{1}{4\pi\varepsilon_0}\int \frac{\mathrm{d}q}{r^2}\cos\theta\,\hat{\mathbf{x}} \\ &= \frac{1}{4\pi\varepsilon_0}\int \frac{\mathrm{d}q}{\sqrt{x^2+a^2}^2}\frac{x}{\sqrt{x^2+a^2}}\hat{\mathbf{x}} \\ &= \frac{1}{4\pi\varepsilon_0}\int \frac{x\,\mathrm{d}q}{(x^2+a^2)^{\frac{3}{2}}}\hat{\mathbf{x}}\end{aligned}$$

Here, we realise that x and a are constants provided in the question. Hence, we can easily

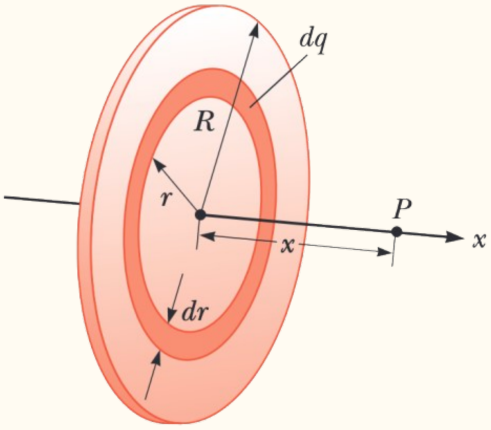
take them out of the integral.

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\frac{x}{(x^2+a^2)^{\frac{3}{2}}}\int \mathrm{d}q\,\hat{\mathbf{x}} = \boxed{\frac{1}{4\pi\varepsilon_0}\frac{Qx}{(x^2+a^2)^{\frac{3}{2}}}\hat{\mathbf{x}}}$$

Uniformly Charged Disc

Example 24.3. Serway & Jewett (9th Edition) (Ex. 23.9)

A disc of radius R has a uniform surface charge density σ. Calculate the electric field at a point P that lies along the central perpendicular axis of the disk and a distance x from the center of the disc.



We can approach this problem by considering the disk to be a set of concentric rings. This gives us the agency to use our previous result and sum the contributions of all rings making up the disk.

Each ring element has charge $\mathrm{d}q$, which gives

$$\mathrm{d}q = \sigma\ \mathrm{d}A = \sigma(2\pi r\ \mathrm{d}r)$$

Calculating the electric field contribution,

$$\mathrm{d}E_x = \frac{1}{4\pi\varepsilon_0}\frac{x}{(x^2+a^2)^{\frac{3}{2}}}(2\pi\sigma r\ \mathrm{d}r)$$

Integrating, we obtain the total electric field.

$$E_x = \int \mathrm{d}E_x = \int \frac{1}{4\pi\varepsilon_0} \frac{x}{(x^2+r^2)^{\frac{3}{2}}} (2\pi\sigma r \, \mathrm{d}r)$$

$$= \frac{\pi\sigma x}{4\pi\varepsilon_0} \int \frac{2r \, \mathrm{d}r}{(x^2+r^2)^{\frac{3}{2}}}$$

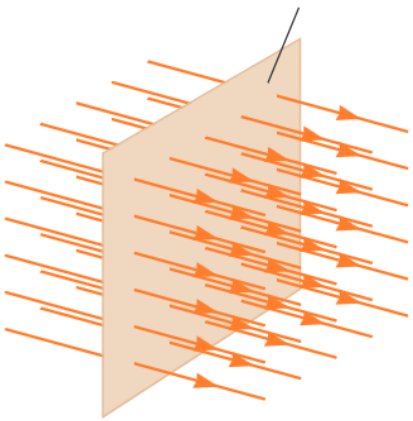
$$= \frac{\pi\sigma x}{4\pi\varepsilon_0} \left[\frac{(x^2+r^2)^{-\frac{1}{2}}}{-1/2} \right]_0^R$$

$$= \boxed{\frac{2\pi\sigma}{4\pi\varepsilon_0} \left(1 - \frac{x}{(R^2+x^2)^{\frac{1}{2}}} \right)}$$

24.3 Electric Flux

In this section, we treat electric field lines more quantitatively.

Consider an electric field that is uniform in both magnitude and direction. The field lines penetrate a rectangular area of a plane perpendicular to the direction of the electric field, as illustrated below.



The field line density is proportional to the magnitude of the electric field. Hence, the total number of lines penetrating the surface is *proportional* to the product EA.

Definition 24.3: Electric Flux

This product of the magnitude of the electric field E and surface area A perpendicular to the field is called the electric flux, Φ_E, of units $\mathrm{N\,m^2\,C^{-1}}$.

$$\Phi_E = \vec{E} \cdot \vec{A} = EA\cos\theta$$

For more general situations, the electric field may vary over a large surface. This general surface can be divided into a large number of small elements, each of area ΔA_i, with corresponding

vector $\Delta\vec{A_i}$ whose magnitude represents the area of the ith element of the surface and direction perpendicular to the surface.

This gives us

$$\Phi_{E,i} = \vec{E_i} \cdot \Delta\vec{A_i}$$

Proposition 24.1: Integral Form of Electric Flux

Taking the limit to zero, the sum is replaced by an integral, giving us the **general definition of electric flux**.

$$\Phi_E = \oint \vec{E} \cdot \mathrm{d}\vec{A}$$

where $\oint$ denotes a **surface integral**. The value of Φ_E depends on both the field pattern and the surface.

Due to the definition of the area vector, the **net electric flux** through the surface is proportional to the **net number of lines leaving** the surface — this net number refers to the number of lines leaving the surface minus the number of lines entering the surface.

- If more lines are leaving then entering, net electric flux is **positive**.
- If more lines are entering then leaving, net electric flux is **negative**.

These conventions give us

$$\Phi_E = \oint \vec{E} \cdot \mathrm{d}\vec{A} = \oint E_n \, \mathrm{d}A$$

where E_n represents the component of the electric field normal to the surface.

24.4 Gauss's Law

The integrations involved with using Coulomb's law are cumbersome. To avoid this, we can apply Gauss's law, a mathematical alternative to Coulomb's law, to efficiently calculate the electric field of symmetrical objects.

24.4.1 Explanation

Gauss's law describes a general relationship between the net electric flux through a closed surface (or **Gaussian surface**) and the charge enclosed by the surface.

From the derivation of electric flux, we can infer that the flux through any *closed* surface is a measure of the total charge inside it — this is because the field lines that originate on a positive charge must either pass out through the surface, or else terminate on a negative charge inside.

Moreover, any charge *outside* the surface will *not* contribute to the total flux, since its field lines will pass in one side and out the other.

This is the qualitative essence of Gauss's law.

24.4.2 Derivation

Definition 24.4: Gauss's Law

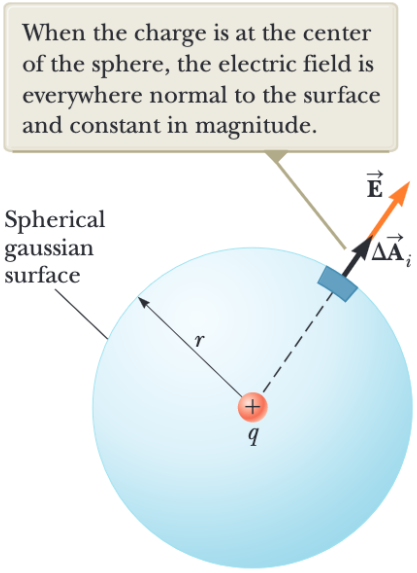
The mathematical form of Gauss's law is

$$\Phi_E = \oint \vec{E} \cdot \mathrm{d}\vec{A} = \frac{q_{\text{enc}}}{\varepsilon_0}$$

where $\vec{E}$ represents the **total** electric field at any point on the surface and q_{enc} represents the net charge enclosed within the surface.

Proof. Consider a sphere enclosing a positive charge $+q$ directly in the centre. The net flux through the spherical Gaussian surface is

$$\Phi_E = \oint \vec{E} \cdot \mathrm{d}\vec{A} = \oint E \, \mathrm{d}A \cos 0^\circ = E \oint \mathrm{d}A$$



E has moved outside of the integral because, by symmetry, E is constant over the surface. Recall that $E = \frac{1}{4\pi\varepsilon_o}\frac{q}{r^2}$. Due to the spherical surface, $\oint \mathrm{d}A = A = 4\pi r^2$. Hence, the net flux through the spherical Gaussian surface is

$$\Phi_E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}(4\pi r^2) = \frac{q}{\varepsilon_0}$$

This equation actually holds for *all Gaussian surfaces*. The net flux through *any* closed surface surrounding a point charge q is given by q/ε_0 and is *independent* of the shape of that surface. □

In practice, Gauss's law can be solved for $\vec{E}$ in highly symmetrical situations, such as charge distributions that possess spherical, cylindrical, or planar symmetry.

24.4.3 Applications

The following examples demonstrate ways of choosing the Gaussian surface over which the surface integral can be simplified.

In choosing the surface, take advantage of the symmetry of the charge distribution so that E can be removed from the integral. To do so, determine a surface that satisfies at least one of the following conditions.

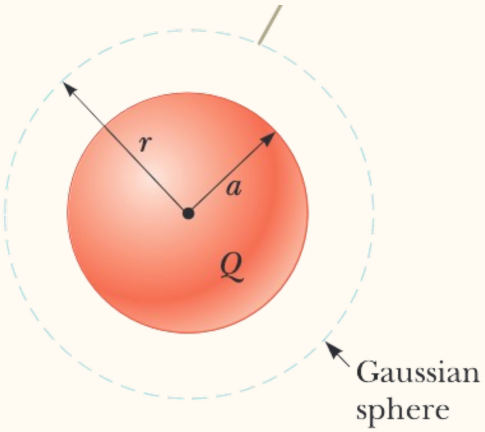
1. The value of the electric field is constant over the surface due to symmetry.
2. $\vec{\boldsymbol{E}}$ and $\mathrm{d}\vec{\boldsymbol{A}}$ are parallel, hence allowing $\vec{\boldsymbol{E}} \cdot \mathrm{d}\vec{\boldsymbol{A}} = E\ \mathrm{d}A$.
3. $\vec{\boldsymbol{E}}$ and $\mathrm{d}\vec{\boldsymbol{A}}$ are perpendicular, hence allowing $\vec{\boldsymbol{E}} \cdot \mathrm{d}\vec{\boldsymbol{A}} = 0$.
4. The electric field is zero over the surface.

Spherically Symmetric Charge Distributions

Example 24.4. Serway & Jewett (9th Edition) (Ex. 24.3)

Consider a solid sphere of radius a with uniform charge density ρ carrying a total positive charge Q. Calculate the magnitude of the electric field at a point P for cases outside and inside the sphere.

For *points outside the sphere*, we can draw another Gaussian surface, with r being the distance from the enclosed charge to the point P.



Because the charge is distributed uniformly throughout the sphere, the charge distribution has spherical symmetry and we can apply Gauss's law.

Replacing $\vec{\boldsymbol{E}} \cdot d\mathbf{A}$ in Gauss's law with $E\ \mathrm{d}A$,

$$\Phi_E = \oint \vec{\boldsymbol{E}} \cdot \mathrm{d}\vec{\boldsymbol{A}} = \oint E\ \mathrm{d}A = \frac{Q}{\varepsilon_0}$$

By symmetry, E has the same value everywhere.

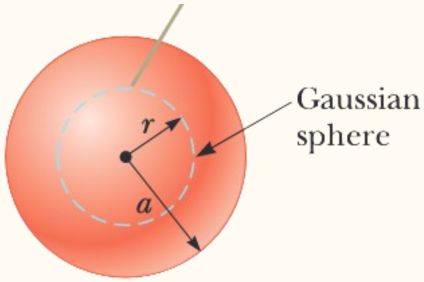
$$\Phi_E = \oint E\ \mathrm{d}A = E \oint \mathrm{d}A = E(4\pi r^2) = \frac{q_{\text{enc}}}{\varepsilon_0}$$

Hence, we can solve for E.

$$\therefore \boxed{E = \frac{q_{\text{enc}}}{4\pi\varepsilon_0 r^2} \text{ (where } r > a\text{)}}$$

This field is identical to that for a point charge. Therefore, the electric field due to a uniformly charged sphere in the region external to the sphere is *equivalent* to that of a point charge located at the centre of the sphere.

This calculation differs for *points inside the sphere*.



For a spherical Gaussian surface having a radius $r < a$ that encloses a volume of V', the enclosed charge q_{enc} is smaller than the total charge Q.

$$q_{\text{enc}} = \rho V' = \rho\left(\frac{4}{3}\pi r^3\right)$$

Since the surface is still spherically symmetrical,

$$\Phi_E = \oint \vec{\boldsymbol{E}} \cdot \mathrm{d}\vec{\boldsymbol{A}} = E \oint \mathrm{d}A = E(4\pi r^2) = \frac{q_{\text{enc}}}{\varepsilon_0}$$

Solving for E and substituting q_{enc},

$$E = \frac{q_{\text{enc}}}{4\pi\varepsilon_0 r^2} = \frac{\rho(\frac{4}{3}\pi r^3)}{4\pi\varepsilon_0 r^2} = \frac{\rho}{3\varepsilon_0} r$$

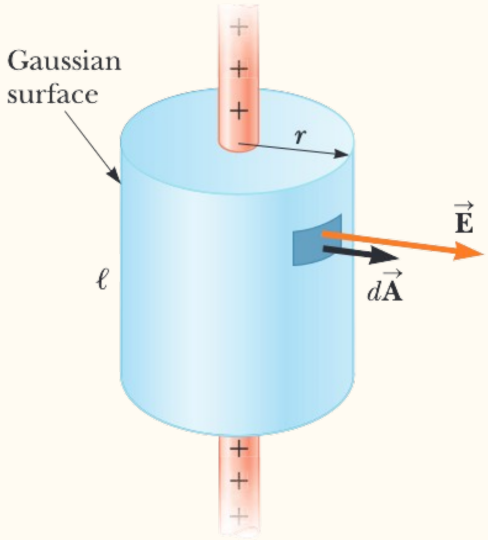
Since the total charge $Q = \rho \times \frac{4}{3}\pi a^3$,

$$\therefore E = \frac{Q/\frac{4}{3}\pi a^3}{3\varepsilon_0} r = \boxed{\frac{Q}{4\pi\varepsilon_0 a^3} r \text{ (where } r < a\text{)}}$$

Cylindrically Symmetric Charge Distribution

Example 24.5. Serway & Jewett (9th Edition) (Ex. 24.4)

Find the electric field a distance r from a line of positive charge of infinite length and constant charge per unit length λ.



Since the line of charge is infinitely long, the field is the same at all points equidistant from the line.

To reflect the symmetry of the charge distribution, we choose a cylindrical Gaussian surface of radius r and length l. For the curved part, $\vec{E}$ is constant and perpendicular to the surface.

For the flat ends of the cylinder, $\vec{E}$ is *perpendicular* to $d\vec{A}$, hence $\vec{E} \cdot \mathrm{d}\vec{A} = 0$ at those surfaces. This allows us to *focus on the curved surfaces only*.

$$\Phi_E = \oint \vec{E} \cdot \mathrm{d}\vec{A} = E \oint \mathrm{d}A = EA = \frac{q_{\text{enc}}}{\varepsilon_0} = \frac{\lambda l}{\varepsilon_0}$$

Substituting the area $A = 2\pi rl$,

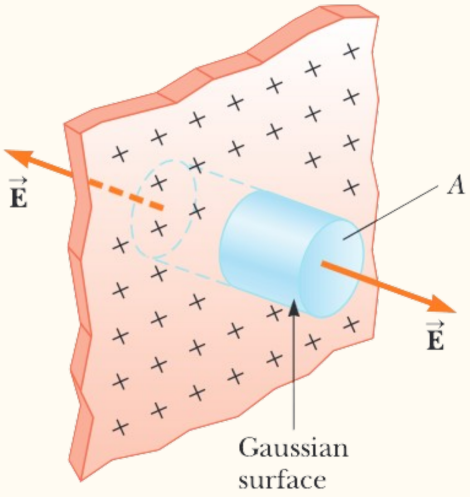
$$E(2\pi rl) = \frac{\lambda l}{\varepsilon_0} \implies \boxed{E = \frac{\lambda}{2\pi\varepsilon_0 r}}$$

Planarly Symmetric Charge Distribution

Example 24.6. Serway & Jewett (9th Edition) (Ex. 24.5)

Find the electric field due to an infinite plane of positive charge with uniform surface charge density σ.

By symmetry, $\vec{E}$ must be perpendicular to the plane at all points. Here, we can characterise a cylindrically-shaped Gaussian surface (or Gaussian "pillbox") whose axis is perpendicular to the plane and whose ends each have an area A and are equidistant from the plane.



$\vec{E}$ is parallel to the curved surface and thus causes no electric flux on the curved parts.

For the flat ends of the cylinder, $\vec{E}$ is constant and parallel to $d\vec{A}$.

The flux through each end of the cylinder is EA, hence the total flux through the entire Gaussian surface is double that, $\Phi_E = 2EA$.

$$\Phi_E = 2EA = \frac{q_{\text{enc}}}{\varepsilon_0} = \frac{\sigma A}{\varepsilon_0}$$

Solving for E, we obtain

$$\therefore \boxed{E = \frac{\sigma}{2\varepsilon_0}}$$

24.5 Conductors in Electrostatic Equilibrium

A good electrical conductor contains charges that are not bound to any atom and therefore are free to move about within the material.

When there is no net motion of charge within a conductor, the conductor is in **electrostatic equilibrium**. Such conductors have the following properties.

1. The $\vec{E}$ field is *zero everywhere* inside the conductor, whether the conductor is solid or hollow.
2. If the conductor is isolated and carries a charge, the charge resides on its surface.
3. The electric field at a point just outside a charged conductor is perpendicular to the surface of the conductor and has a magnitude of σ/ε_0, where σ is the surface charge density at that point.
4. The surface of a conductor is an *equipotential surface*, and the electric potential is *constant everywhere* inside the conductor and equal ot its value at the surface.
5. On an irregularly-shaped conductor, the surface charge density is greatest at locations where the radius of curvature of the surface is smallest.

24.6 Electric Potential

When a charge q is placed in an electric field $\vec{E}$, the particle experiences an electric force $\vec{F} = q\vec{E}$. This force is *conservative* because the force between charges described by Coulomb's law is conservative.

For an infinitesimal displacement $\mathrm{d}\vec{s}$ of a point charge q in an electric field, the work done within the charge-field system by the electric field on the charge is $W_{\text{int}} = \vec{F} \cdot \mathrm{d}\vec{s} = q\vec{E} \cdot \mathrm{d}\vec{s}$.

As learnt previously, the internal work done in a system is equal to the negative of the change in the potential energy of the system, or $W_{\text{int}} = -\Delta U$.

Therefore, as the charge q is displaced, the electric potential energy of the charge-field system is changed by an amount $dU = -W_{\text{int}} = -q\vec{E} \cdot \mathrm{d}\vec{s}$.

Definition 24.5: Electric Potential Energy (H3)

For a finite displacement of the charge from some point A in space to some other point B, the **change in electric potential energy** of the system is

$$\Delta U = -q \int_A^B \vec{E} \cdot \mathrm{d}\vec{s} = -\frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r}$$

The integration is performed along the path that q follows as it moves from A to B. As the electric force is conservative, the line integral does not depend on the specific path taken from A to B.

Definition 24.6: Electric Potential (H3)

Dividing the potential energy by the charge gives a physical quantity that depends only on the source charge distribution and has a value at every point at an electric field. This quantity is called the **electric potential**.

$$V = \frac{U}{q}$$

Both potential energy and potential are scalar quantities.

Definition 24.7: Potential Difference (H3)

The **potential difference** $\Delta V = V_B - V_A$ between two points A and B in an electric field is defined as the charge in electric potential energy of the system when a charge q is moved between the points, divided by the charge.

$$\Delta V = \frac{\Delta U}{q} = -\int_A^B \vec{E} \cdot \mathrm{d}\vec{s}$$

The potential difference is *not equivalent* to the difference in potential energy.

Potential difference between A and B exists solely due to a source charge — potential energy requires at least two charges.

Now, consider the situation where an *external agent* moves a charge from A to B with changing

the kinetic energy.

This agent performs work that changes the potential energy, where, by the work-energy theorem, $W = \Delta U$.

Definition 24.8: Work Done for Charges

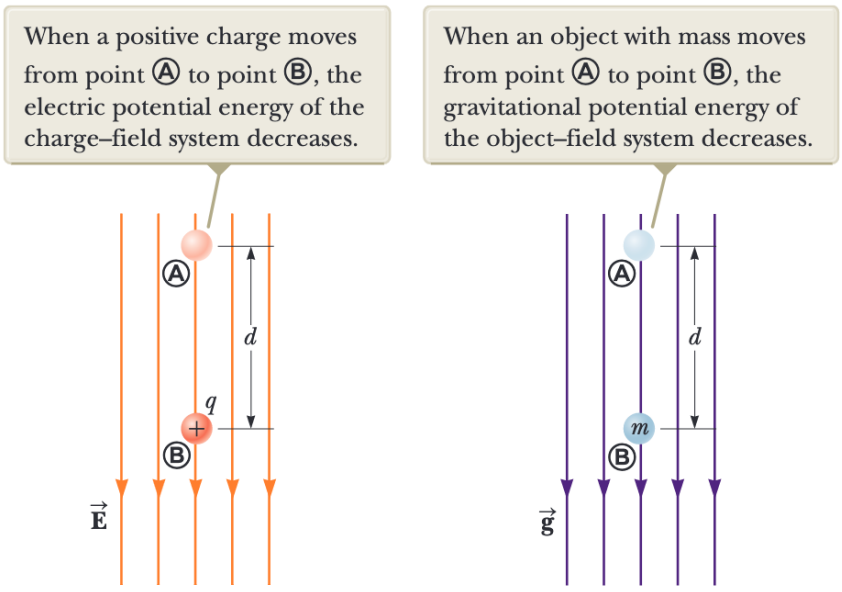
The **work done** by an external agent in moving a charge q through an electric field at constant velocity is thus

$$W = q\Delta V$$

24.6.1 Potential in Uniform Electric Field

From the previous equation, we can highlight an important point regarding units.

- Electrical potential and potential difference have S.I. unit volt (V) or joules per coulomb.
- However, as defined earlier, potential difference also has units of *electric field multiplied by distance*.
- Thus, the S.I. unit of electric field can be expressed in volts per meter $1\,\mathrm{N\,C^{-1}} = 1\,\mathrm{V\,m^{-1}}$.
- This shows that the electric field is a measure of the *rate of change of the electrical potential* with respect to position.



We can calculate the potential difference between points A and B separated by a distance d, where the displacement $\vec{s}$ points from A to B and is parallel to the field lines.

$$\Delta V = -\int_A^B \vec{E} \cdot \mathrm{d}\vec{s} = -\int_A^b E\ \mathrm{d}s(\cos 0) = -\int_A^B E\ \mathrm{d}s$$

Since E is constant, it can be removed from the integral sign, giving

$$\Delta V = -E\int_A^B \mathrm{d}s = -Ed$$

This shows that electric field lines *always point in the direction of decreasing electric potential*.

Suppose a charge q moves from A to B. The change in potential energy is given by

$$\Delta U = q\Delta V = -qEd$$

This shows if q is positive, then ΔU is negative, and vice versa. In a system consisting of a positive charge and an electric field, the electric potential energy of the system *decreases* when the charge moves in the direction of the field.

24.6.2 Potential for Continuous Charges

The electric potential due to a *single point charge* q at any distance r from the charge is

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \implies U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$$

For *multiple discrete point charges*,

$$V = \frac{1}{4\pi\varepsilon_0}\sum_i \frac{q_i}{r_i}, \quad U = \frac{1}{4\pi\varepsilon_0}\sum_{i<j}\frac{q_i q_j}{r_{ij}}$$

For a *continuous charge distribution*,

$$V = \frac{1}{4\pi\varepsilon_0}\int \frac{\mathrm{d}q}{r}$$

and the total potential energy $\sum U$ is calculated by integrating dU for *every pair of charges*.

Note that the electric field can derived by taking the negative derivative of the electric potential.

$$E_r = -\frac{\mathrm{d}V}{\mathrm{d}r}$$

24.7 Electric Dipole

Definition 24.9: Electric Dipole

An **electric dipole** consists of two charges of equal magnitude and opposite sign separated by a certain distance. It is common to denote this distance as $2a$.

Definition 24.10: Electric Dipole Moment

The **electric dipole moment** of an electric dipole configuration is defined as the vector $\vec{\boldsymbol{p}}$ directed from the negative charge $-q$ to the positive charge $+q$ along the line joining the charges, which is defined to be

$$\vec{\boldsymbol{p}} \equiv 2\vec{\boldsymbol{a}}q$$

where $2\vec{\boldsymbol{a}}$ is the separation between the two electric charges.

Definition 24.11: Torque of Electric Dipole

For an electric dipole placed in an electric field $\vec{E}$ at an oblique angle θ with the field, the **net torque** created about the midpoint **of the dipole** is given as

$$\tau = \vec{p} \times \vec{E} = pE \sin\theta$$

Proof. Suppose an electric dipole with separation $2a$ is placed in a uniform electric field $\vec{E}$, whilst making an angle θ with the field. The $\vec{E}$ field is *external* to the dipole, and not the field that is created by the dipole.

Within this electric field, each charge experiences an equal and opposite force pointing in the direction of the external electric field. Though the net force acting on the dipole is zero, these two forces produce a net torque about the midpoint of the two charges.

The torque due to the positive charge has magnitude $Fa\sin\theta$. The torque due to the negative charge has magnitude $Fa\sin\theta$ as well. Therefore, the magnitude of the net torque about its midpoint is

$$\begin{aligned}\tau &= 2Fa\sin\theta = 2aqE\sin\theta \\ &= pE\sin\theta\end{aligned}$$

□

24.7.1 Potential Energy of Electric Dipoles

Definition 24.12: Potential Energy of Electric Dipole

For an electric dipole placed in an electric field $\vec{E}$ with electric dipole moment $\vec{p}$, the **potential energy possessed by said dipole** is given as

$$U_E = -\vec{p} \cdot \vec{E}$$

The potential energy of any electric dipole is associated with a *rotational configuration* of the system. Harkening back to our understanding of rotational mechanics, the work dW required to rotate the dipole through an angle $d\theta$ is $dW = \tau d\theta$.

Hence, for a rotation from θ_i to θ_f, the change in potential energy of the system is

$$\begin{aligned}\Delta U = U_f - U_i = \int_{\theta_i}^{\theta_f} \tau \, \mathrm{d}\theta &= \int_{\theta_i}^{\theta_f} pE\sin\theta \, \mathrm{d}\theta \\ &= pE\int_{\theta_i}^{\theta_f} \sin\theta \, \mathrm{d}\theta \\ &= pE[-\cos\theta]_{\theta_i}^{\theta_f} \\ &= pE(\cos\theta_i - \cos\theta_f)\end{aligned}$$

Choosing $\theta_i = \pi/2$ and $\theta_f = \theta$, this expression can be further simplified into

$$\Delta U = -pE\cos\theta$$

24.8 Magnetic Fields

Definition 24.13: Gauss's Law in Magnetism

Gauss's law in magnetism states that the net magnetic flux through any closed surface is always zero.

$$\oint \vec{\boldsymbol{B}} \cdot \mathrm{d}\vec{\boldsymbol{A}} = 0$$

This implies that magnetic monopoles do not exist.

24.8.1 Ampere's Law

Definition 24.14: Ampere's Law

Ampere's law states the the line integral of $\vec{\boldsymbol{B}} \cdot \mathrm{d}\vec{\boldsymbol{s}}$ around any closed path is equal to $\mu_0 I$, where I is the total current passing through any surface bounded by the closed linear path.

$$\oint \vec{\boldsymbol{B}} \cdot \mathrm{d}\vec{\boldsymbol{s}} = \mu_0 I$$

Long Wire

Example 24.7. Serway & Jewett (9th Edition) (Ex. 30.5)

A long straight wire of radius R carries a steady current I that is uniformly distributed through the cross-section of the wire. Calculate the magnetic field at a distance r from the centre of the wire in the regions inside the wire and outside the wire.

When **inside** the wire, $r < R$, the current I' within the closed loop 2 is such that

$$\frac{I'}{I} = \frac{J \times \pi r^2}{J \times \pi R^2} = \frac{r^2}{R^2} \quad \left(J = \frac{I}{A}\right)$$

Therefore, by Ampere's law,

$$\oint \vec{\boldsymbol{B}} \cdot \mathrm{d}\vec{\boldsymbol{s}} = B \oint ds = B(2\pi r) = \mu_0 I' = \mu_0 \left(\frac{r^2}{R^2} I\right)$$

$$\implies \boxed{B = \frac{\mu_0 I}{2\pi R^2} r}$$

When **outside** the wire, $r \geq R$.

$$\oint \vec{\boldsymbol{B}} \cdot \mathrm{d}\vec{\boldsymbol{s}} = B \oint ds = B(2\pi r) = \mu_0 I$$

$$\implies \boxed{B = \frac{\mu_0 I}{2\pi r}}$$

Toroid

Example 24.8. Serway & Jewett (9th Edition) (Ex. 30.6)

A toroid, consisting of wire wrapped around a non-conducting ring, is often used to create a near-uniform magnetic field in some enclosed area. For a toroid having N closely-spaced turns of wire, determine the magnetic field in the region occupied by the torus, a distance r from the centre.

Consider the circular loop 1 of radius r within the ring. By symmetry, the magnitude of the field B is constant in this circle and tangential to it ($\vec{\boldsymbol{B}} \cdot \mathrm{d}\vec{\boldsymbol{s}} = B\,ds$). Furthermore, the total current is NI.

Applying Ampere's law,

$$\oint \vec{\boldsymbol{B}} \cdot \mathrm{d}\vec{\boldsymbol{s}} = B \oint ds = B(2\pi r) = \mu_0 NI$$

$$\implies \boxed{B = \frac{\mu_0 NI}{2\pi r}}$$

Solenoid

Example 24.9.

A solenoid is a long wire wound in the form of a helix. Considering an ideal solenoid with N turns, find the magnitude of the uniform magnetic field created within the interior.

Consider the loop of length ℓ and width w. Applying Ampere's law, we obtain

$$\oint \vec{\boldsymbol{B}} \cdot \mathrm{d}\vec{\boldsymbol{s}} = \mu_0 I \implies B\ell = \mu_0 NI$$

$$\implies \boxed{B = \mu_0 \frac{N}{l} I}$$

24.9 Magnetic Dipoles

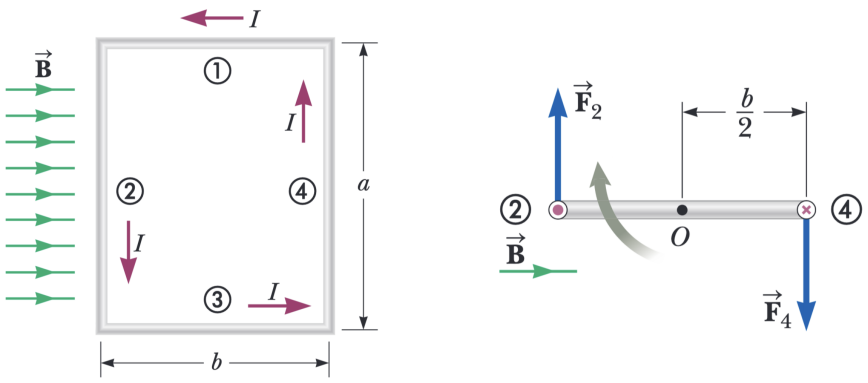
We begin our analysis of magnetic forces acting magnetic dipoles by first considering a current-carrying conductor.

Take a rectangular loop carrying a current I in the presence of a uniform magnetic field directed *parallel to the plane* of the loop.

Magnetic forces only act on the sides that are oriented perpendicular to the field (that is, the sides of length a). The magnitudes of said forces are

$$F_2 = F_4 = BI\ell = BIa$$

We also notice that these forces impose a net torque on the rectangular loop, rotating it about the



centre O. The magnitude of this torque is given by

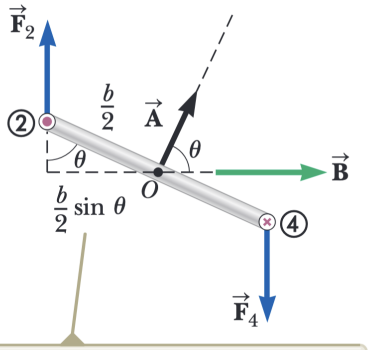
$$\begin{aligned}\tau = F_2 \cdot \frac{b}{2} + F_4 \cdot \frac{b}{2} &= BIa \cdot \frac{b}{2} + BIa + \frac{b}{2} \\ &= IabB \\ &= IAB\end{aligned}$$

where A is the area enclosed by the loop.

Where the loop is oriented at an oblique angle θ with respect to the direction of the magnetic field, the magnitude of the resultant torque about O is given by

$$\tau = IAB \sin\theta$$

as seen from the diagram below, where the perpendicular between O and the line of the magnetic force is $b/2 \cdot \sin\theta$ for each side of the loop.



When the normal to the loop makes an angle θ with the magnetic field, the moment arm for the torque is $(b/2)\sin\theta$.

Definition 24.15: Magnetic Dipole Moment

The product $I\vec{A}$ is defined to the magnetic dipole moment (also magnetic moment) of the loop.

$$\vec{\mu} = I\vec{A}$$

where $\vec{A}$ is the *area vector* of the loop.

For a coil with N loops of the same area, the magnetic moment of the coil is then

$$\vec{\mu} = NI\vec{A}$$

The S.I. unit of magnetic dipole moment is the ampere square meter, $\mathrm{A\,m^{-2}}$.

Definition 24.16: Torque on Current-Carrying Loop

To simplify our derivation of torque even further, the **torque exerted on a current-carrying loop** placed in a uniform magnetic field $\vec{B}$ is given by

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

where $\vec{\mu}$ is the magnetic moment of the loop.

Definition 24.17: Potential Energy of Magnetic Dipole

In a similar vein with electric dipoles, the **potential energy of a magnetic dipole** in a magnetic field is given by

$$U_B = -\vec{\mu} \cdot \vec{B}$$

which has the lowest energy when $\vec{\mu}$ is *parallel* to $\vec{B}$, and has the highest energy when $\vec{\mu}$ is *anti-parallel* to $\vec{B}$.

25 Capacitors & Inductors

25.1 Capacitors

Definition 25.1: Capacitor

A capacitor is formed when two conductor plates are arranged parallel to each other.

A potential difference ΔV exists if the two plates have equal and opposite charges.

Definition 25.2: Capacitance

The capacitance C of a capacitor is defined as the ratio of the magnitude of the charge on either conductor to the magnitude of potential difference between the conductors.

$$C = \frac{Q}{\Delta V}$$

The S.I. unit of capacitance is the farad (F), or coulombs per volt.

We often think of capacitance as the *amount of charge* that the capacitor can store per unit of potential difference. Capacitance is *always* a positive quantity.

25.1.1 Combinations

For a combination of **capacitors arranged in series**, they have the *same charge* Q with different potential differences. The equivalent capacitance is

$$C_{\text{eq}} = \left(\frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots \right)^{-1}$$

For a combination of **capacitors arranged in parallel**, they have the *same potential* V with different charges, depending on the capacitance of each capacitor. The equivalent capacitance is

$$C_{\text{eq}} = C_1 + C_2 + C_3 + \cdots$$

25.1.2 Energy in Capacitors

Definition 25.3: Energy in a Capacitor

The **work done in charging a capacitor** appears as electric potential energy U_E stored in the capacitor. This is given as

$$U_E = \frac{1}{2} C V^2 = \frac{1}{2} Q V = \frac{1}{2} \frac{Q^2}{C}$$

Proof. Note that U_E is equal to the work done W by a battery of potential difference V exerting a force F to put a charge Q on a capacitor with capacitance C.

Since the force exerted on the charge by the battery is equal and opposite to the electric force F_E exerted by the capacitor,

$$\begin{aligned} F = -F_E = -QE = Q\frac{\mathrm{d}V}{\mathrm{d}s} &\implies \int F \,\mathrm{d}s = \int Q\frac{\mathrm{d}V}{\mathrm{d}s}\,\mathrm{d}s \\ &\implies U_E = \int CV \,\mathrm{d}V \\ &\implies U_E = \frac{1}{2}CV^2 \end{aligned}$$

The other forms of U_E can be observed by using the relation $Q = CV$. □

25.2 Dielectrics & Ferromagnetism

25.2.1 Dielectric Materials

Dielectric materials help to **enhance capacitance** by allowing a greater stored charge at a given potential difference. Dielectrics are electrical insulators that prevents conducting plates from coming into direct contact, and can be polarised by an applied external electric field.

Dielectric breakdown can occur when the electric field is sufficiently strong, where a conducting path between the conducting plates is formed and charge jumps across the gap, damaging the capacitor.

25.2.2 Ferromagnetic Materials

Ferromagnetic materials help to **enhance inductance** by allowing the magnetic field within the inductor to align its domains.

The enhancement is non-linear. In other words, there exists an asymptotic threshold where the intrinsic magnetic dipole moments are almost aligned perfectly with the field, resulting in saturation magnetisation.

25.3 Inductance

Definition 25.4: Inductor

An inductor is a device that stores energy in a magnetic field when a current flows through it.

Its working principle is based on Faraday's law and Lenz's law. A current flowing through a conductor generates a magnetic field, consequently generating a magnetic flux linkage.

If the current **varies with time**, the magnetic flux through the loop varies and induces a **back electromotive force**.

This magnetic field **stores energy**. If opened, the inductor will quickly lose this energy via a spark at the opened switch or damped electromagnetic oscillations.

Definition 25.5: Self-Inductance

The inductance of an inductor can be defined by its **self-inductance**, which is defined as

- the ratio of the magnetic flux linkage in an electrical component to the current flowing through it; and also
- the ratio of the induced electromotive force in an electrical component to the rate of change of current causing it.

$$L = \frac{N\Phi}{I} = V \Big/ \frac{\mathrm{d}I}{\mathrm{d}t}$$

The unit of inductance is the **henry** (H), one tesla square-meter per ampere.

Now, consider the case where there are two inductors placed adjacent to each other at near proximity. The magnetic fields generated by both inductors would affect each other's magnetic flux linkage and consequently their inductance. This is a phenomenon known as **mutual inductance**.

We commonly consider two coils, with N_1 and N_2 turns carrying currents I_1 and I_2 respectively. Let us define $\Phi_{a \leftarrow b}$ to be the magnetic flux linkage through coil a caused by the magnetic field from coil b.

Definition 25.6: Mutual Inductance

The **mutual inductance** of two coils is defined as

- the ratio of the magnetic flux linkage in one coil to the current in the other coil; and also
- the ratio of the induced electromotive force in one coil to the rate of change of current in the other.

$$M = \frac{N_2\Phi_{2\leftarrow 1}}{I_1} = \frac{N_1\Phi_{1\leftarrow 2}}{I_2}$$

$$= -V_1 \Big/ \frac{\mathrm{d}I_2}{\mathrm{d}t} = -V_2 \Big/ \frac{\mathrm{d}I_1}{\mathrm{d}t}$$

25.3.1 Combinations

Inductors combine similarly to resistors..

For inductors connected in series,

$$V_{\text{eq}} = \sum_i V_i = \sum_i L\frac{\mathrm{d}I}{\mathrm{d}t} = L_{\text{eq}}\frac{\mathrm{d}I}{\mathrm{d}t} \implies L_{\text{eq}} = L_1 + L_2 + \cdots$$

For inductors connected in parallel,

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \sum_i \frac{\mathrm{d}I_i}{\mathrm{d}t} = \sum_i \frac{V}{L_i} \implies V = \frac{1}{\sum_i \frac{1}{L_i}}\frac{\mathrm{d}I}{\mathrm{d}t} \implies L_{\text{eq}} = \left(\frac{1}{L_1} + \frac{1}{L_2} + \cdots\right)^{-1}$$

25.3.2 Energy in Inductors

Definition 25.7: Energy in an Inductor

The **potential energy stored in an inductor** U is given by

$$U = \frac{1}{2}LI^2$$

where L is the inductance and I is the current flowing through the inductor.

Proof. We note that at any given time t, we consider the power P flowing into the magnetic field using its classical and electrical definitions.

$$P = IV \implies \frac{\mathrm{d}U}{\mathrm{d}t} = I\left(L\frac{\mathrm{d}I}{\mathrm{d}t}\right)$$

$$\implies \int \mathrm{d}U = \int IL\ \mathrm{d}I$$

When there is no current, there is no magnetic field, therefore the stored energy is zero.

Neglecting resistive losses,

$$\int \mathrm{d}U = \int IL\ \mathrm{d}I \implies U = \int_0^I IL\ \mathrm{d}I$$

$$\implies U = \frac{1}{2}LI^2$$

□

25.4 Linear Analog Filters

This section concerns itself with linear analog electronic filters, all connected in series, namely the RC circuit, RL circuit, LC circuit, and RLC circuit. These circuits consist of resistors (R), inductors (L), and capacitors (C), which all affect each other differently in the time domain.

Preliminary concepts include Kirchhoff's circuit laws, which are crucial in fundamental network analyses in this area.

Definition 25.8: Kirchhoff's Current Law

Kirchhoff's current law states that the sum of currents entering any junction in an electric circuit is always equal to the sum of currents leaving that junction.

In other words, at a junction,

$$\sum \text{Current}_{\text{in}} = \sum \text{Current}_{\text{out}}$$

Definition 25.9: Kirchhoff's Voltage Law

Kirchhoff's voltage law states that in any closed loop in an electric circuit, the total e.m.f. E supplied equals to the total potential difference in that loop.

In other words, for a closed loop within a circuit,

$$\sum \text{EMF} = \sum \text{Potential Drop}$$

Analysis of analog filters requires knowledge of solving up to second-order differential equations.

25.4.1 RC Circuit

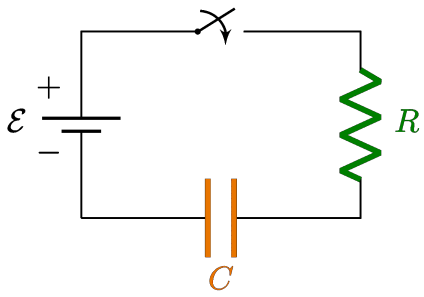
Definition 25.10: RC Circuit

A RC circuit is a circuit composed of resistors and capacitors. The **general solution to the RC series circuit** with a constant electromotive force is given as

$$Q = C\varepsilon + (Q_0 - C\varepsilon)e^{-t/\tau}, \quad \tau = RC$$

where Q is the charge at time t, R is resistance, C is capacitance, ε is the electromotive force of the battery, and Q_0 is the initial charge.

Proof. Consider the RC series circuit with a constant electromotive force.



When the switch is closed, by Kirchhoff's voltage law, we can form a differential equation in Q.

$$\varepsilon - iR - \frac{Q}{C} = 0 \implies \varepsilon - \frac{\mathrm{d}Q}{\mathrm{d}t}R - \frac{Q}{C} = 0$$

$$\implies \frac{\mathrm{d}Q}{\mathrm{d}t} + \frac{1}{RC}Q = \frac{\varepsilon}{R}$$

The integrating factor is given as

$$\text{I.F.} = e^{\int 1/RC \, \mathrm{d}t} = e^{t/RC}$$

Multiplying throughout,

$$e^{t/RC}\frac{dQ}{dt} + e^{t/RC}\frac{1}{RC}Q = e^{t/RC}\frac{\varepsilon}{R} \implies e^{t/RC}Q = \int \frac{\varepsilon}{R}e^{t/RC}\,dt$$
$$\implies e^{t/RC}Q = C\varepsilon e^{t/RC} + A$$
$$\implies Q = C\varepsilon + Ae^{-t/RC}$$

Letting initial charge be Q_0 at $t = 0$,

$$Q_0 = C\varepsilon + A \implies A = Q_0 - C\varepsilon$$

Therefore, the general solution to the RC series circuit with initial charge Q_0.

$$Q = C\varepsilon + (Q_0 - C\varepsilon)e^{-t/\tau}, \quad \tau = RC$$

□

Dividing throughout by C, we obtain the general solution for the voltage V_C across the capacitor.

$$V_C = \varepsilon + (V_0 - \varepsilon)e^{-t/\tau}$$

Some notes to take note of are that:

- if $V_C > \varepsilon$, the capacitor is charging; if $V_C < \varepsilon$, the capacitor is discharging;
- $\tau = RC$ is known as the **time constant** of the circuit;
- at $t = 0$, the capacitor acts as a short circuit; at $t \to \infty$, the capacitor acts like an open circuit, for there will be no current through the capacitor.

25.4.2 RL Circuit

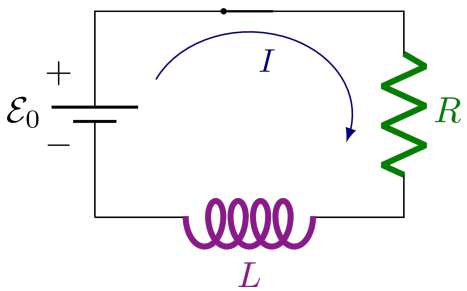
Definition 25.11: RL Circuit

A **RL circuit** is a circuit composed of resistors and inductors. The **general solution to the RL series circuit** with a constant electromotive force is given as

$$i = \frac{\varepsilon}{R} + \left(i_0 - \frac{\varepsilon}{R}\right)e^{-t/\tau}, \quad \tau = \frac{L}{R}$$

where i is the current at time t, R is resistance, L is inductance, ε is the electromotive force of the battery, and i_0 is the initial charge.

Proof. Consider the RL series circuit with a constant electromotive force.



When the switch is closed, by Kirchhoff's voltage law, we form a differential equation in i.

$$\varepsilon - iR - L\frac{\mathrm{d}i}{\mathrm{d}t} = 0 \implies \frac{\mathrm{d}i}{\mathrm{d}t} + \frac{R}{L}i = \frac{\varepsilon}{L}$$

The integrating factor is given as

$$\text{I.F.} = e^{\int R/L \, \mathrm{d}t} = e^{(R/L)t}$$

Multiplying throughout,

$$\begin{aligned} e^{(R/L)t}\frac{\mathrm{d}i}{\mathrm{d}t} + e^{(R/L)t}\frac{R}{L}i = e^{(R/L)t}\frac{\varepsilon}{L} &\implies e^{(R/L)t}i = \int \frac{\varepsilon}{L}e^{(R/L)t} \, \mathrm{d}t \\ &\implies e^{(R/L)t}i = \frac{\varepsilon}{R}e^{(R/L)t} + A \\ &\implies i = \frac{\varepsilon}{R} + Ae^{-(R/L)t} \end{aligned}$$

Letting initial current be i_0 at $t = 0$,

$$i_0 = \frac{\varepsilon}{R} + A \implies A = i_0 - \frac{\varepsilon}{R}$$

Therefore, the general solution to the RL series circuit with constant electromotive force is

$$i = \frac{\varepsilon}{R} + \left(i_0 - \frac{\varepsilon}{R}\right)e^{-t/\tau}, \quad \tau = \frac{L}{R}$$

□

Differentiating, the rate of change of current is given as

$$\frac{\mathrm{d}i}{\mathrm{d}t} = -\left(\frac{R}{L}\right)\left(i_0 - \frac{\varepsilon}{R}\right)e^{-t/\tau}, \quad \tau = \frac{L}{R}$$

25.4.3 LC Circuit

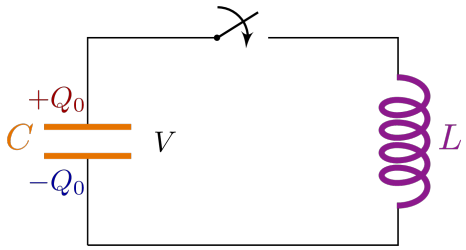
Definition 25.12: LC Circuit

A LC circuit is a circuit composed of inductors and capacitors. The **general solution to the LC series circuit** with a constant electromotive force is given as

$$Q = Q_{\max}\cos(\omega t + \phi), \quad \omega = \frac{1}{\sqrt{LC}}$$

where Q is the charge at time t, L is inductance, C is capacitance, ϕ is the phase angle, and $Q_{\max}$ is the maximum charge amount.

Proof. Consider the LC series circuit below, when the switch is closed.



We will take an energy-based approach to analysing this circuit. Considering the total energy contained within the system and taking its derivative,

$$U = \frac{Q^2}{2C} + \frac{1}{2}Li^2 \implies \frac{\mathrm{d}U}{\mathrm{d}t} = \frac{Q}{C}\frac{\mathrm{d}Q}{\mathrm{d}t} + Li\frac{\mathrm{d}i}{\mathrm{d}t}$$

Since the circuit is isolated, there is no change in the total energy. Therefore, we can set the time derivative of U to be zero.

$$0 = \frac{Q}{C}\frac{\mathrm{d}Q}{\mathrm{d}t} + L\frac{\mathrm{d}i}{\mathrm{d}t} \implies \frac{Q}{C}\frac{\mathrm{d}Q}{\mathrm{d}t} + L\frac{\mathrm{d}Q}{\mathrm{d}t}\frac{\mathrm{d}^2Q}{\mathrm{d}t^2} = 0$$

$$\implies \frac{\mathrm{d}^2Q}{\mathrm{d}t^2} = -\frac{1}{LC}Q$$

This forms a second-order differential equation in Q. For its characteristic equation,

$$m^2 = -\frac{1}{LC} \implies m = \pm\frac{1}{\sqrt{LC}}i$$

Since m takes on a complex conjugate pair without any real component, the general solution is

$$Q = A_0\cos\left(\frac{1}{\sqrt{LC}}t\right) + B_0\sin\left(\frac{1}{\sqrt{LC}}t\right)$$

$$= A_1\cos\left(\frac{1}{\sqrt{LC}}t + B_1\right)$$

It is obvious that A_1 must be equal to the maximum value of Q. B_1 is an arbitrary constant, dependent on the initial conditions of the circuit, that we denote to be ϕ, the phase angle of the sinusoidal function. Lastly, we note that the angular frequency ω is equal to $1/\sqrt{LC}$. Therefore, we obtain the general solution of the LC series circuit.

$$Q = Q_{\max}\cos(\omega t + \phi), \quad \omega = \frac{1}{\sqrt{LC}}$$

□

Differentiating Q gives us the general solution for the current in the LC circuit.

$$i = \frac{\mathrm{d}Q}{\mathrm{d}t} = -\omega Q_{\max}\sin(\omega t + \phi)$$
$$= -I_{\max}\sin(\omega t + \phi)$$

As aforementioned, the value of ϕ can be derived from the initial conditions of the LC circuit.

Particularly, when $i = 0$ at $t = 0$, substituting these values into the expression for i gives

$$0 = -I_{\max} \sin(\phi) \implies \phi = 0$$

Furthermore, notice that i and Q are $\pi/2$ out of phase with each other. In other words, when either i or Q is zero, the other must be at its maximum value.

25.4.4 RLC Circuit

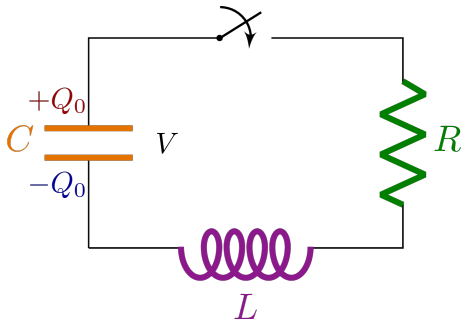
Definition 25.13: RLC Circuit

A RLC circuit is a circuit composed of resistors, inductors, and capacitors. The **general solution to the RLC circuit** is given as

$$Q = \begin{cases} Ae^{m_1 t} + Be^{m_2 t}, & R^2 > 4L/C \\ e^{mt}(A + Bt), & R^2 = 4L/C \\ De^{-(R/2L)t}\cos(\omega t + \phi),\ \omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}, & R^2 < 4L/C \end{cases}$$

where Q is the charge at time t, R is resistance, L is inductance, and C is capacitance.

Proof. Consider the RLC series circuit as shown below, when the switch is closed.



Similar to the LC series circuit, we begin analysing the total energy of this circuit. The central difference is that there is a resistor present, which incurs a power loss of i^2R over time.

$$U = \frac{Q^2}{2C} + \frac{1}{2}Li^2 \implies \frac{\mathrm{d}U}{\mathrm{d}t} = \frac{Q}{C}\frac{\mathrm{d}Q}{\mathrm{d}t} + Li\frac{\mathrm{d}i}{\mathrm{d}t} = -i^2R$$

$$\implies \frac{Q}{C}\frac{\mathrm{d}Q}{\mathrm{d}t} + L\frac{\mathrm{d}Q}{\mathrm{d}t}\frac{\mathrm{d}^2Q}{\mathrm{d}t^2} = -\left(\frac{\mathrm{d}Q}{\mathrm{d}t}\right)^2 R$$

$$\implies L\frac{\mathrm{d}^2Q}{\mathrm{d}t^2} + R\frac{\mathrm{d}Q}{\mathrm{d}t} + \frac{1}{C}Q = 0$$

Solving the characteristic equation of this second-order differential equation,

$$Lm^2 + Rm + \frac{1}{C} = 0 \implies m = \frac{-R \pm \sqrt{R^2 - 4L/C}}{2L}$$

This gives rise to three possible cases.

If $R^2 > 4L/C$, the electromagnetic oscillations are **overdamped**; the charge on the oscillator does not oscillate and decreases to zero **very slowly**. The general solution then takes the form

$$Q = Ae^{m_1 t} + Be^{m_2 t}$$

If $R^2 = 4L/C$, the electromagnetic oscillations are **critically damped**; the charge on the oscillator does not oscillate and decreases to zero **in the shortest possible time**. The general solution then takes the form

$$Q = e^{mt}(A + Bt)$$

If $R^2 < 4L/C$, the electromagnetic oscillations are **underdamped**; the charge on the capacitor oscillates at a specific angular frequency, where the charge decreases to zero **exponentially**.

Let us closely analyse the mathematical formulation of this general solution. Since the characteristic equation has complex roots,

$$Q = e^{\mathrm{Re(m)t}}(A\sin[\mathrm{Im}(m)t] + B\cos[\mathrm{Im}(m)t])$$

The angular frequency ω is equal to $\mathrm{Im}(m)$. Looking at the solutions of m,

$$\mathrm{Re}(m) = -\frac{R}{2L}\mathrm{Im}(m) = \frac{1}{2L}\sqrt{\frac{4L}{C} - R^2} = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

Therefore, the general solution is

$$Q = De^{-(R/2L)t}\cos(\omega t + \phi), \quad \omega = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}$$

where D and ϕ are both dependent on the initial conditions of the circuit. □

Apart from the angular frequency of the RLC oscillation, another important shorthand for the quantities within the general solutions is regarding the **neper frequency** or **attenuation** α.

$$\alpha = \frac{R}{2L}$$

As it is placed within the term of exponential decay, attentuation is primarily a measure of how fast the transient response of the RLC circuit will die over time.